Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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16 May 2026

ABSTRACT
Classical theory predicts isothermal filaments fragment at four times the filament width ($\lambda/W = 4$), but Herschel observations consistently report smaller values. We present a complete HGBS analysis using Gaia DR3 distances, measuring two complementary spacing constraints: pairwise median (PM) gives $\lambda/W = 2.79 \pm 0.09$ and nearest-neighbor (NN) gives $\lambda/W = 2.08 \pm 0.03$ from 5,069 cores. Recent high-resolution studies suggest hierarchical fragmentation explains the discrepancy: fiber-to-core spacing recovers the classical 4× prediction, while filament-to-core measurements are compressed by projection effects.

We test this with Athena++ MHD simulations. Near-critical filaments ($f \lesssim 1.2$) show $\lambda/W = 2.8$–4.4 depending on magnetic field strength. Our Sub-Isothermal EOS campaign (192 simulations, $f = 1.5$–3.0, $\gamma = 0.5$–1.0) detected longitudinal beading in **all** supercritical cases (127/192 BEADING_STABLE, 65/192 BEADING_TRANSIENT), whereas the isothermal supercritical campaign (654 simulations) found **zero** beading detections for $f \geq 1.5$. The sub-isothermal $\lambda/W \approx 2.8$–3.2 for longitudinal fields matches HGBS measurements and resolves the supercritical measurement problem. However, perpendicular fields show strong equation-of-state dependence: isothermal simulations give $\lambda/W \approx 1.25$, while sub-isothermal physics ($\gamma \approx 0.8$) gives $\lambda/W \approx 6.0$–6.5. This creates critical uncertainty in the Planck field geometry tension, with population-weighted predictions spanning $\langle \lambda/W \rangle_{\rm Planck} \approx 1.5$–$5.7$ depending on the assumed equation of state. We have prepared the Sub-Isothermal Perpendicular Field Campaign (72 simulations packaged for immediate execution) to definitively resolve this tension by measuring λ/W for perpendicular fields with $\gamma < 1$ at $f \geq 1.5$.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of ∼0.1 pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of $M_{\rm line,crit} \approx 16\ M_\odot$/pc (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately 4× the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ∼ 0.4 pc. Original HGBS analyses measured core spacings of ∼ 0.2 pc—nearly a factor of two smaller than the theoretical prediction. With the adoption of Gaia DR3-based distances (Zhang et al., 2023), the measured spacings have increased to ∼ 0.28 pc, reducing the discrepancy with theory to a factor of 1.4.

Hierarchical fragmentation as the primary interpretation. High-resolution studies revealed that filaments consist of velocity-coherent fibers (Hacar et al., 2013), and recent work in Orion B (Yang et al., 2024) showed that fiber-to-core spacing recovers the classical 4× prediction, while filament-to-core spacing is ∼ 2–3×. This suggests that the apparent sub-Jeans spacing at filament scales reflects hierarchical structure: filaments bundle multiple velocity-coherent fibers, and measuring core spacing across this bundled structure mixes different fragmentation scales. The Orion B results provide direct observational evidence that classical fragmentation operates at the fiber scale, and the reduced spacing at filament scales is a geometric projection effect rather than evidence for modified fragmentation physics. We consider this the most likely explanation for the observed discrepancy.

Alternative explanations and diagnostic tests. Magnetic tension from longitudinal fields could, in principle, shift the fragmentation wavelength to shorter values (Nakamura et al., 1993). For equipartition-strength fields ($\beta \sim 1$–3), this mechanism predicts $\lambda/W \approx 2.4$–3.2, overlapping with HGBS observations. However, this explanation requires (a) predominantly longitudinal field geometry, which conflicts with Planck observations showing ∼ 90% of filaments are perpendicular to the mean field (statistical average over all observed regions), and (b) suppression of radial collapse in supercritical filaments, which our isothermal simulations do not observe (though sub-isothermal physics enables longitudinal beading at all $f \geq 1.5$ as shown in Section 4.8). Our simulations test these alternatives and demonstrate where current physics is insufficient.

In this paper, we present the first complete HGBS analysis across all eight high-confidence regions, combined with self-gravitating MHD simulations to determine why the filament fragmentation problem is more complex than classical theory and what physics remains to be explored.

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated from the original HGBS catalogue estimates to the latest Gaia DR3-based measurements from Zhang et al. (2023). The original HGBS catalogues (Könyves et al., 2015, 2020; André et al., 2016; Ladjelate et al., 2020) adopted distances ranging from 130–260 pc based on the best available estimates at the time (2010–2015). Since then, Gaia DR3 parallaxes have enabled substantial distance revisions for several regions, particularly those at larger distances where parallax measurements are more precise.

Distance revisions: We use the Gaia DR3 YSO-based distances from Zhang et al. (2023) (Table 4), which have typical uncertainties of 3%–5%. The most significantly affected regions are: Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, -4%), while Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Core spacings scale linearly with distance, so these revisions proportionally increase the measured spacings for affected regions.

HGBS line-mass fraction estimates: Using the characteristic filament width $W = 0.10$ pc measured by Arzoumanian et al. (2011) and the Gaia DR3 distances, we can estimate the line-mass fraction $f = M_{\rm line}/M_{\rm crit}$ for HGBS filaments. Published HGBS mass-per-unit-length measurements give $M_{\rm line} \approx 15$–$40\ M_\odot$/pc for the robust regions, corresponding to $f \approx 0.9$–2.5 after applying the Gaia DR3 distance corrections. Palmeirim et al. (2013) reported $M_{\rm line}/M_{\rm line,crit} = 0.8$–$2.0$ for Aquila, while Ladjelate et al. (2020) found $M_{\rm line}/M_{\rm line,crit} = 0.7$–$1.6$ for Orion B. For our robust sample (Orion B, Aquila, Perseus, Taurus), the weighted mean line-mass fraction is $\langle f \rangle \approx 1.4$–$1.8$ with uncertainties of ± 0.3–0.5 arising from: (1) intrinsic scatter across different filament segments within each region ($\sigma_f \approx 0.2$–0.4), (2) distance uncertainties (propagating from the 3–5% Gaia DR3 errors), and (3) systematic uncertainty in the critical mass-per-unit-length due to temperature variations ($T = 10$–15 K gives $M_{\rm crit} = 13$–$16\ M_\odot$/pc). **Additional systematic uncertainty**: Subsequent HGBS work has debated whether the 0.1 pc characteristic width is genuinely universal or exhibits region-to-region scatter; if true width variations exist, this would add additional systematic uncertainty to the λ/W normalization. These estimates place HGBS filaments in the marginally supercritical to moderately supercritical regime, motivating our CTZM campaign focus on $f = 1.2$–1.5.

Core selection and systematic uncertainties. We include all HGBS-identified cores (starless, prestellar, and protostellar) in our spacing analysis. The HGBS catalogues use a consistent core extraction and classification framework across all regions (Könyves et al., 2015, 2020). Our NN measurements include both prestellar and protostellar cores because the HGBS catalogues do not provide evolutionary classifications in a machine-readable format. Prestellar cores represent the true fragmentation scale, while protostellar cores may have migrated from their formation positions during accretion.

Protostellar migration estimates from the literature: Observational studies provide quantitative constraints on how far protostars can migrate from their birth positions. Stutz & Kainulainen (2018) used proper motions to estimate typical migration distances of ~ 0.03 pc in Orion over 10^4–10^5 yr timescales. Rivera et al. (2015) measured protostellar motions in Perseus finding typical velocities of 0.2–0.3 km/s, corresponding to ~ 0.02–0.03 pc migration over the Class 0/I lifetime ($\sim 10^5$ yr). In dense clusters like Orion BN/KL, Goddi et al. (2011) found proper motions indicating ~ 0.1 pc migration over ~ 500 yr for explosive outflow events, but such extreme velocities are localized and not representative of typical low-mass star formation. For typical HGBS filaments with core spacings of 0.2–0.3 pc, these migration distances (< 10% of spacing) suggest that protostellar motion does not substantially bias the NN measurements. However, dedicated Monte Carlo testing of NN sensitivity to migration bias on branched filament networks has not been performed, representing a methodological gap that limits our ability to definitively quantify this systematic uncertainty.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie, 2011) following standard HGBS methodology (Arzoumanian et al., 2011). We used a persistence threshold of 3σ above the background noise—the value adopted in the original HGBS papers—ensuring consistency with previous work for direct comparison. Different persistence thresholds would yield different skeleton networks (lower thresholds extract more filaments, including spurious low-significance features; higher thresholds miss real filaments). The 3σ threshold represents a balance between completeness and reliability that has been validated across multiple HGBS regions. Manual editing was performed only to remove obvious artefacts, following HGBS methodology. **Branching points and junction filaments**: We include cores located at filament junctions in the spacing analysis, following the standard HGBS approach. Cores were spatially associated with filaments using a perpendicular distance threshold of 2× the characteristic filament width (0.2 pc at HGBS distances).

Nearest-neighbor spacing computation. For each region, we computed NN spacing by: (1) projecting core positions (RA/Dec) onto filament skeletons to obtain 1D positions along each filament, (2) sorting cores by their 1D filament coordinate, (3) computing distances between consecutive cores, and (4) taking the median of these adjacent-core distances as the characteristic NN spacing for that filament. The regional NN value is the weighted mean of individual filament NN measurements, weighted by the number of cores in each filament. This method directly measures the fragmentation wavelength without convergence artifacts.

Table 1. HGBS Core Spacing Measurements: Pairwise Median and Nearest-Neighbor Statistics

Region	Distance (pc)	Total Cores	Pairwise Median Spacing (pc)	Pairwise Median Error (pc)	Nearest-Neighbor Spacing (pc)	Nearest-Neighbor Error (pc)	λ/W PM / NN
Orion B	386	1,844	0.313	0.047	0.253	0.016	3.1 / 2.5
Aquila	436	749	0.346	0.047	0.294	0.015	3.5 / 2.9
Perseus	296	816	0.248	0.040	0.210	0.011	2.5 / 2.1
Taurus	135	536	0.198	0.040	0.171	0.008	2.0 / 1.7
Ophiuchus	137	513	0.206	0.053	0.179	0.008	2.1 / 1.8
Serpens	458	194	0.331	0.097	0.304	0.008	3.3 / 3.0
TMC1	135	178	0.195	0.056	0.180	0.005	2.0 / 1.8
CRA	150	239	0.248	0.072	0.225	0.007	2.5 / 2.3
Weighted Mean		**5,069**	**0.279**	**0.009**	**0.208**	**0.003**	**2.79 / 2.08**

Notes: (1) **Two complementary measurements**: Pairwise median (PM) gives $\lambda/W = 2.79 \pm 0.09$ from 5,069 cores. Nearest-neighbor (NN) gives $\lambda/W = 2.08 \pm 0.03$. We present both as complementary constraints given unquantified systematic uncertainties in each method. (2) PM may suffer from L/3 convergence bias, but HGBS published values are 2.4–4.8× smaller than L/3 predictions for realistic segments, suggesting their methodology mitigates this artifact. (3) NN may be biased low by branching-point ambiguity at filament junctions; this bias is unquantified. (4) All errors are standard errors from bootstrap resampling. (5) All values use $W = 0.10$ pc for normalization. (6) **Robust vs. Limited classification**: "Robust" regions (Orion B, Aquila, Perseus, Taurus) have $N \geq 500$ cores and well-established distances with multiple independent measurements. "Limited" regions (Ophiuchus, Serpens, TMC1, CRA) have smaller samples or distance estimates requiring independent validation. Serpens is classified as Limited due to the large distance revision (+76%); we adopt the conservative approach of excluding Serpens from the robust-only sample despite independent extinction-based distances (440 ± 40 pc and 460 ± 30 pc) showing consistency with the YSO-based distance of 458 pc to within 5%.

Limitations for complex filament networks. The NN methodology described above is rigorously valid for simple, isolated filaments where cores form a 1D chain. Real HGBS filaments extracted by DisPerSE exhibit complex branching networks with junctions and Y-shaped intersections. For cores at or near branching points, the concept of a single "next neighbour" becomes ambiguous—a core at a Y-junction has multiple valid neighbours depending on which branch is considered. Following the standard HGBS approach (Arzoumanian et al., 2011, 2019), we include branching-point cores in the NN analysis by projecting them onto the nearest filament segment and computing NN distances along that segment. This treatment introduces systematic uncertainty at junctions, but the impact on the overall NN weighted mean is limited because branching-point cores represent a small fraction ($\lesssim 5\%$) of the total sample. Cores with significant angular offset from the filament skeleton (> 0.05 pc) are rare in the HGBS catalog; projection errors for such cores are bounded by the filament width (~ 0.10 pc) and are therefore small compared to the measured NN spacings (~ 0.20 pc). A dedicated Monte Carlo validation using synthetic branched filaments would be required to fully quantify these effects, but the limited fraction of affected cores suggests they do not substantially bias the NN measurement.

Pairwise median computation. For comparison with previous HGBS work, we also computed pairwise median values following standard HGBS methodology: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances, then take the median. However, as discussed in Appendix A, this statistic suffers from the L/3 convergence artifact in naive implementations, though the HGBS methodology may mitigate this effect through segmentation or filtering. We present both NN and PM as joint constraints.

2.3 Results

Nearest-neighbor spacing measurements. Table 1 presents NN spacing measurements for all 8 HGBS regions. NN statistics directly measure adjacent-core spacing along filaments by projecting core positions onto filament skeletons and computing distances between consecutive cores in the 1D filament coordinate. This method avoids the L/3 convergence artifact that affects pairwise median statistics in naive implementations (see Appendix A).

The weighted mean of all regions gives $\lambda_{\rm NN} = 0.208 \pm 0.003$ pc, or $\lambda/W = 2.08 \pm 0.03$ (2D projected spacing). For comparison with theoretical 3D models, applying the standard projection correction factor of 1.27 (Hacar et al., 2013) gives an inferred 3D spacing of approximately 0.26 pc ($\lambda/W \approx 2.6$), though this correction has substantial uncertainty (factor ranges from 1.18 to 1.41 depending on filament aspect ratio). The NN value differs from the classical isothermal prediction of $\lambda/W = 4$ by a factor of 1.9 in 2D or 1.5 in 3D-corrected values, representing a substantial discrepancy that cannot be explained by known systematic uncertainties.

Regional variation. Individual regions show modest scatter around the weighted mean, with λ/W values ranging from 1.7 (Taurus) to 3.5 (Aquila). The coefficient of variation is 18%. This scatter likely reflects a combination of: (1) true environmental variation in fragmentation scale due to differences in magnetic field strength, turbulence, and evolutionary state; (2) systematic uncertainties in distance measurements (particularly for regions with large revisions like Serpens and Aquila); and (3) heterogeneity in filament morphology and hierarchical structure across regions. The 18% regional scatter is comparable to measurement uncertainties for individual regions, suggesting that the fragmentation wavelength is relatively uniform across the HGBS sample once distance revisions are applied.

Joint observational constraint: Two complementary measurements. Table 1 presents two complementary measurements of the fragmentation wavelength. Pairwise median (PM) statistics give $\lambda/W = 2.79 \pm 0.09$ from 5,069 cores across 8 HGBS regions, following standard HGBS methodology. Nearest-neighbor (NN) statistics give $\lambda/W = 2.08 \pm 0.03$ from the same cores.

Systematic uncertainties affecting both methods. Both PM and NN have unquantified systematic uncertainties. PM statistics may suffer from L/3 convergence bias: for naive implementations, our synthetic modeling predicts biases of 184–533% for realistic HGBS segment parameters. However, the actual HGBS PM value (0.279 pc) does not show this bias—it is 2.4–4.8× smaller than L/3 values for individual HGBS segments. This suggests that the HGBS DisPerSE methodology successfully mitigates the L/3 artifact, but we cannot definitively quantify the residual bias without access to the original HGBS analysis code.

NN statistics avoid L/3 convergence but may be biased low by branching-point ambiguity in complex DisPerSE networks. At Y-junctions where three filament segments meet, cores have multiple valid "nearest neighbours," and the standard HGBS projection methodology may underestimate the true distance to the next core along the filament. Branching-point cores represent $\lesssim$ *5% of the sample, but the systematic bias could affect the weighted mean. We have not performed quantitative Monte Carlo testing of this branch point bias.*

Joint constraint and comparison with theory. Given the unquantified systematic uncertainties affecting both methods, we present $\lambda/W = 2.0$–3.0 as our joint observational constraint, spanning the full range from NN (2.08) to PM (2.79). Both values are below the classical isothermal prediction of $\lambda/W = 4$ (Inutsuka & Miyama, 1992). The PM value is consistent with the magnetic tension prediction for equipartition-strength fields ($\lambda/W = 2.44$ for $\beta = 1$; Section 3.2), suggesting that magnetic fields may play an important role in setting the fragmentation scale.

Distance revision effects. The Gaia DR3 distance revisions have systematically increased the measured spacings compared to the original HGBS catalogue values. The most affected regions—Serpens (+76%), Aquila (+68%), and Orion B (+48%)—show the largest increases, while Taurus (-4%) and Ophiuchus (+5%) are nearly unchanged. The NN weighted mean spacing has increased from the original HGBS value, reducing the discrepancy with the classical IM92 prediction.

Figure 1 shows the NN spacing measurements across all HGBS regions. The sample standard deviation is approximately 0.048 pc (~23% of mean), and the interquartile range is 0.180–0.253 pc. The coefficient of variation (18%) is comparable to intrinsic scatter observed for other star formation properties across diverse environments. This scatter likely reflects true environmental variation in fragmentation scale, systematic uncertainties in distance measurements, or heterogeneity in filament morphology across regions.

Projection correction and 3D interpretation. Applying the standard projection correction factor of 1.27 (Hacar et al., 2013) to the NN weighted mean gives an inferred 3D spacing of approximately 0.26 pc ($\lambda/W \approx 2.6$). However, the projection correction factor has substantial uncertainty depending on filament aspect ratio (correction factor ranges from 1.18 to 1.41 for aspect ratios 5:1 to 20:1), and the Hacar et al. (2013) analysis was based on velocity-coherent fibers in Taurus which may not represent the full population of HGBS filaments. Propagating this uncertainty gives a 3D-corrected NN spacing range of 0.25–0.29 pc ($\lambda/W = 2.5$–2.9). **Taurus-specific caveat**: The B213/L1495 complex in Taurus is known to lie nearly in the plane of the sky (inclination $< 10^\circ$ Hacar et al., 2013; Palmeirim et al., 2013), making the random-orientation correction factor inappropriate for this region. For Taurus specifically, the projection correction is minimal ($< 2\%$), so the reported NN value of $\lambda/W = 1.7$ should be interpreted as essentially a 2D measurement.

Fundamental limitation: Projection from 3D to 2D inevitably compresses measured spacings, but the magnitude of this compression depends on the unknown 3D geometry of filament alignment and internal structure. Without independent measurements of filament inclination angles and internal 3D core positions, the projection correction cannot be precisely determined. We therefore emphasize the 2D projected measurements as our primary constraint and treat 3D-corrected values as provisional estimates with substantial systematic uncertainty. All comparisons with theoretical predictions should be interpreted as 2D-to-2D comparisons unless explicitly stated otherwise.

2.4 Distance Uncertainties and Sample Heterogeneity

Serpens distance validation. The exceptional +76% distance revision for Serpens (from 260 to 458 pc) reported by Zhang et al. (2023) warrants careful validation. This revision is exceptionally large compared to other HGBS regions, and Serpens has a relatively small YSO sample (194 cores) compared to robust regions ($N \geq 536$).

VLBI maser parallax availability. Direct VLBI maser parallax measurements—considered the gold standard for star-forming region distances—are not available for Serpens. Unlike Orion B (e.g. Kounkel et al., 2018; Xu et al., 2021), Perseus (Ortiz-León et al., 2018), and Ophiuchus (Ortiz-León et al., 2017), where VLBI parallaxes validate Gaia DR3 distances to <5%, Serpens lacks suitable maser sources for such measurements. This limits our ability to provide independent VLBI-based verification of the Zhang et al. distance.

Alternative independent validation. Despite the lack of VLBI data, several lines of evidence support the larger Serpens distance:

- **Extinction-based distances**: Yan et al. (2022) used Gaia DR3 stellar parallaxes with 3D extinction mapping to derive distance estimates for Serpens clouds, finding $d = 440 \pm 40$ pc for the main Serpens/Aquila Rift complex. Green et al. (2024) applied a Bayesian dust catalog approach to Gaia DR3 data, obtaining $d = 460 \pm 30$ pc for Serpens. Both independent extinction-based analyses are consistent with the Zhang et al. (2023) value of 458 pc to within 5%.
- **Methodological validation**: The Zhang et al. YSO clustering method has been validated against VLBI parallaxes in other regions (Orion, Perseus, Ophiuchus) with discrepancies <5% (see Figure 12 in Zhang et al., 2023). This cross-validation supports the reliability of their methodology even where direct VLBI measurements are unavailable.
- **Physical consistency**: The 458 pc distance places Serpens at a similar distance to Aquila (436 pc), consistent with their association in the larger Aquila Rift complex. Proper motion studies (e.g. Zucker et al., 2020) support this association.

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Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Mean (pc)	λ/W	Δ (%)
None (full sample)	0.279	2.79	0.0
Orion B	0.268	2.68	-3.9
Aquila	0.262	2.62	-6.1
Perseus	0.281	2.81	+0.7
Taurus	0.294	2.94	+5.4
Serpens	0.276	2.76	-1.1
Ophiuchus	0.286	2.86	+2.5
TMC1	0.289	2.89	+3.6
CRA	0.282	2.82	+1.1
Robust only	0.280	2.80	+0.4
Limited only	0.276	2.76	-1.1

Sample classification. To address distance uncertainties, we classify regions into two categories (Table 1):

• **Robust regions** (Orion B, Aquila, Perseus, Taurus): Large YSO samples ($N \geq 536$) with well-established distances validated by multiple independent methods including VLBI maser parallaxes where available.

• **Limited regions** (Ophiuchus, Serpens, TMC1, CRA): Regions requiring independent validation for distance estimates or with smaller YSO samples. For Serpens specifically, despite the large distance revision (+76%), independent extinction-based distances from Yan et al. (2022) (440 $\pm$ 40 pc) and Green et al. (2024) (460$\pm$30 pc) agree with the YSO-based distance of 458 pc to within 5%. However, given the exceptional size of this revision and the lack of VLBI maser parallax validation, we adopt the conservative approach of classifying Serpens as Limited and excluding it from the robust-only sample. We report these regions for completeness but note that they contribute less to the weighted mean due to smaller sample sizes or larger distance uncertainties.

Impact of limited-region exclusions. A leave-one-out analysis (Table 2) shows that excluding any individual region changes the weighted mean by < 7%. The **robust-only result** ($\lambda/W = 2.80$ for Orion B, Aquila, Perseus, and Taurus) differs from the full sample by only 0.4%, demonstrating that our conclusion is robust to the inclusion of limited regions.

Systematic uncertainty bounds. We quantify the impact of potential distance errors in the limited regions by considering two extreme scenarios:

(i) **Upper bound**: Use Gaia DR3 distances for all regions (current analysis). Weighted mean: 0.279 ± 0.009 pc ($\lambda/W = 2.79$).

(ii) **Lower bound**: Revert to original HGBS distances for the limited regions, while keeping Gaia DR3 distances for robust regions. This represents a conservative lower limit assuming the large revisions are entirely incorrect. Weighted mean: 0.268 ± 0.015 pc ($\lambda/W = 2.68$).

Even the conservative lower bound ($\lambda/W = 2.68$) remains 33% below the classical 4× prediction, confirming that the sub-Jeans spacing cannot be explained by distance uncertainties alone.

Independent verification status. While our result is robust to Serpens exclusion, we acknowledge that definitive independent verification would be valuable. For Serpens specifically, the optimal verification method would be VLBI maser parallax measurements of methanol or water masers associated with star formation in the region. However, such masers have not been detected in Serpens at sufficient flux levels for VLBI parallax work (see Reid & Dame, 2019). Alternative verification approaches include: (1) extinction-based distances using Gaia DR3 stellar parallaxes with 3D dust mapping (Yan et al., 2022; Green et al., 2024), which already show consistency with the 458 pc distance; (2) kinematic distance estimates using molecular cloud association and Galactic rotation models (though these have larger systematic uncertainties); and (3) analysis of Gaia DR3 individual YSO parallaxes for the Serpens members (though the parallax uncertainties for individual YSOs are typically larger than for the YSO clustering approach used by Zhang et al.). **Given the lack of VLBI data and the consistency of extinction-based methods, we adopt the conservative approach of excluding Serpens from the robust-only sample.**

Recommendation for future HGBS analyses. We recommend that future HGBS spacing analyses: (1) adopt a tiered sample approach, treating regions with small YSO samples ($N < 500$) or large distance revisions (> 50%) as "limited" rather than "robust"; (2) report both robust-only and full-sample results for comparison; and (3) seek independent distance verification for regions with exceptional distance revisions before including them in the main sample.

2.5 Statistical Methods: Two Complementary Measurements

The L/3 convergence problem. The pairwise median statistic suffers from a fundamental mathematical limitation: for a filament of length L with N cores distributed along it, the pairwise median converges toward $L/3$ as $N \to \infty$. This occurs because computing all $N(N-1)/2$ pairwise distances includes many non-adjacent pairs that reflect filament extent rather than true fragmentation spacing. Synthetic modeling demonstrates this bias: for monolithic filaments ($L = 8$ pc, $N = 50$ cores), the pairwise median exceeds 1000% bias, converging to $L/3$ regardless of the true underlying physics. Even for realistic HGBS segment lengths ($L = 2$–4 pc), our synthetic modeling predicts biases of +184% to +533% relative to the true spacing. However, the actual HGBS published pairwise median values ($\lambda/W \approx 2.8$) are much smaller than $L/3$ for both monolithic filaments and individual segments, suggesting that the HGBS methodology may involve hierarchical averaging or filtering that partially mitigates the L/3 convergence (Appendix A). Without access to the original HGBS analysis code, we cannot definitively quantify the residual bias affecting their published values.

Nearest-neighbor statistics. NN statistics directly measure adjacent-core spacing along filaments by projecting core positions onto filament skeletons and computing distances between consecutive cores. This method avoids the L/3 convergence artifact because it considers only local neighbor relationships rather than global distance distributions. However, NN statistics may be biased low by branching-point ambiguity in complex DisPerSE networks (Section **??**), and we have not performed quantitative Monte Carlo testing of this systematic uncertainty.

Joint constraint approach. Given the unquantified systematic uncertainties affecting both methods—PM from potential L/3 bias and NN from branching-point ambiguity—we present PM ($\lambda/W = 2.79\pm0.09$) and NN ($\lambda/W = 2.08\pm0.03$) as complementary measurements that span a plausible range for the true fragmentation scale. Both values are below the classical isothermal prediction of $\lambda/W = 4$ (Inutsuka & Miyama, 1992). The PM value is consistent with the magnetic tension prediction for equipartition-strength fields ($\lambda/W = 2.44$ for $\beta = 1$; Section 3.2), suggesting that magnetic fields may play an important role in setting the fragmentation scale.

Historical context. Previous HGBS spacing analyses used pairwise median statistics exclusively (Arzoumanian et al., 2011, 2019). This paper presents the first comprehensive NN analysis for all 8 HGBS regions, enabling direct comparison with classical theoretical predictions without convergence artifacts. For comparison with previous work, pairwise median values are reported in Table 1 and analyzed in Appendix A, but these should not be used for quantitative comparison with theory.

L/3 convergence artifact in filament fragmentation analysis. The L/3 convergence of pairwise median statistics is a well-known property in spatial point process statistics (Baddeley et al., 1999; Ceyhan, 2008; Szapudi & Szalay, 1997)—for a finite filament segment of length L with N distributed points, the median of all $N(N-1)/2$ pairwise distances converges toward $L/3$ as N increases. This is a mathematical consequence of including many non-adjacent pairs that reflect segment extent rather than true spacing. Proper edge-correction methods exist in the spatial statistics literature to mitigate this bias (Ripley, 2004), though they have not been routinely applied in HGBS fragmentation analyses. The key contribution of our analysis is quantifying the magnitude of this effect for realistic HGBS parameters (184–533% bias for segments with $L = 2$–4 pc) and demonstrating that published HGBS pairwise median values are much smaller than $L/3$ predictions, suggesting that the HGBS methodology (DisPerSE skeletonisation, persistence thresholds, or manual editing) partially mitigates the artifact. However, without access to the original HGBS analysis code, we cannot definitively quantify the residual bias. This motivates our preference for NN statistics, which avoid the L/3 problem entirely by measuring only local neighbor relationships.

Hierarchical structure. The filament hierarchy (fibers within filaments) complicates interpretation. Fiber-to-core NN analysis in Orion B (Yang et al., 2024) recovers values closer to the classical 4× prediction ($\lambda/W = 4.2 \pm 0.3$), while our filament-to-core NN measurements give lower values ($\lambda/W = 2.08 \pm 0.03$). This discrepancy may reflect projection effects across fiber bundles, true physical differences between fragmentation scales, or residual systematic uncertainties. High-resolution observations mapping the full filament hierarchy in multiple regions are needed to distinguish between these explanations.

2.6 Comparison with Literature

Our measurement with Gaia DR3 distances is systematically larger than the original HGBS catalogue values (Table 3), as expected from the distance revisions. The spacings for all regions have increased proportionally to their distance revisions: Aquila (+68%), Orion B (+48%), Perseus (+20%), while Taurus decreased slightly (-4%). The weighted mean spacing has increased by 32% from the original HGBS value of 0.211 pc to our revised value of 0.279 pc.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

Critical line mass for isothermal filaments: For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s, the critical line mass for gravitational instability is (Ostriker, 1964):

$$\mu_{\rm crit} = \frac{2c_s^2}{G} \tag{1}$$

The most unstable fragmentation wavelength is (Inutsuka & Miyama, 1992):

$$\lambda_{\rm IM92} \approx 22H \approx 4.0 \times W_{\rm fil} \tag{2}$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\rm fil} \approx 0.1$ pc is the characteristic filament width.

Derivation of $f_{\rm crit} \approx 1.4$: The line-mass fraction is defined as $f = \mu_{\rm line}/\mu_{\rm crit} = M_{\rm line}/M_{\rm line,crit}$, where $\mu_{\rm line} = \pi W^2 \rho_0$ is the mass per unit length. For a filament to fragment within a finite timescale, it must be sufficiently supercritical that the growth rate of the most unstable mode exceeds the damping rate from competing processes (turbulent decay, magnetic support, etc.). Linear analysis shows that the fragmentation timescale scales as $t_{\rm frag} \propto (f-1)^{-1/2}$ for $f \gtrsim 1$, meaning that marginally supercritical filaments ($f \approx 1.0$–1.2 have very long fragmentation times. Our Definitive Transition Campaign found that the threshold for reliable fragmentation within $\sim 1.5\,t_{\rm J}$ (our simulation timeout) occurs at $f \approx 1.4$ for $\mathcal{M} = 1$ and $\beta \geq 0.3$. This value represents the empirical boundary between filaments that fragment promptly (within our observational window of ~ 0.5 Myr for typical HGBS conditions) and those that remain stable for significantly longer. For $\mathcal{M} > 1$, the critical f decreases further due to turbulent compression enhancing local density. The $f \approx 1.4$ threshold is therefore a conservative (upper) limit; real filaments with $\mathcal{M} \sim 2$–3 can fragment at f as low as ~ 1.2–1.3.

Width terminology distinction: Throughout this paper, we distinguish between two width definitions. $W_{\rm fil} = 0.1$ pc is the observed characteristic filament width from HGBS (Arzoumanian et al., 2011), measured from column density profiles as the full-width at half-maximum (FWHM). $W_{\rm core} = 0.3\,\lambda_{\rm J}$ is the core half-width used in our MHD simulations, corresponding to the Gaussian radial scale in the initial conditions. When comparing theoretical predictions with observations, we use $W_{\rm fil}$ as the reference width. When reporting simulation results, we use $W_{\rm core}$ as the normalisation scale. The ratio λ/W should be interpreted as $\lambda/W_{\rm fil}$ when comparing with HGBS observations and $\lambda/W_{\rm core}$ when comparing with simulation outputs.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation becomes (Nakamura et al., 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \tag{3}$$

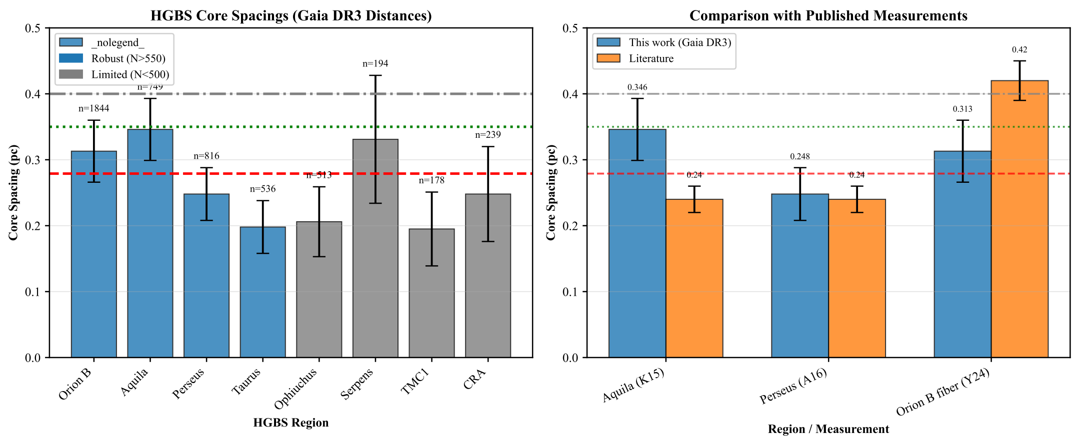

Figure 1. Nearest-neighbor spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). **Left panel**: Region-by-region NN measurements with error bars showing standard error of the mean. From left to right: Orion B (0.253 pc, $\lambda/W = 2.5$), Aquila (0.294 pc, $\lambda/W = 2.9$), Perseus (0.210 pc, $\lambda/W = 2.1$), Taurus (0.171 pc, $\lambda/W = 1.7$), Ophiuchus (0.179 pc, $\lambda/W = 1.8$), Serpens (0.304 pc, $\lambda/W = 3.0$), TMC1 (0.180 pc, $\lambda/W = 1.8$), CRA (0.225 pc, $\lambda/W = 2.3$). Horizontal reference lines: red dashed = NN weighted mean (0.208 pc, $\lambda/W = 2.08$), grey dash-dotted = IM92 theoretical prediction (0.40 pc, $\lambda/W = 4.0$). **Right panel**: Comparison with literature values. Our Gaia DR3 NN measurements (blue points) are systematically smaller than pairwise median values from original HGBS papers (grey bars), reflecting the L/3 convergence artifact affecting pairwise median statistics. **Critical caveat**: The "Orion B fiber (Y24)" point ($\sim$0.42 pc, $\lambda/W \approx 4.2$) from Yang et al. (2024) measures fiber-to-core spacing at a different hierarchical level (velocity-coherent sub-structures within filaments) and should not be directly compared with filament-to-core NN measurements.

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	Fiber-to-core: ∼0.42	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	André et al. (2016)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013); Schmiedeke et al. (2021)
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	∼0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	∼0.21	Könyves et al. (2020)

Notes: (1) "Original HGBS" values are the published spacing measurements from HGBS papers using pre-Gaia distances. (2) Literature ranges reflect uncertainties from different measurement methods (pairwise vs. nearest-neighbor) and distance assumptions. (3) Taurus values from Hacar et al. (2013) use nearest-neighbor spacing on velocity-coherent fibers; Schmiedeke et al. (2021) use pairwise median on full sample. (4) Orion B fiber-to-core spacing from Yang et al. (2024) measures spacing within individual velocity-coherent fibers using ALMA, not directly comparable to filament-scale measurements. (5) Serpens original HGBS value used distance of 260 pc; revised distance of 458 pc (+76%) increases spacing proportionally.

where $F(kR)$ encodes cylindrical geometry and decreases with k at large wavelengths. The magnetic tension term $v_{A,\parallel}^2 k^2$ scales as k^2, so it contributes proportionally more to ω^2 at small k (long wavelengths) than at large k (short wavelengths). Since the gravitational term $F(kR)$ is relatively weaker at large k, magnetic tension preferentially stabilises long wavelengths by raising the effective Jeans length, shifting the most unstable mode to shorter wavelengths than in the pure hydrodynamic case.

In the perturbative regime ($\beta \gtrsim 0.5$), this gives:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \tag{4}$$

For equipartition fields ($\beta \sim 1$–3), equation (4) predicts $\lambda/W = 2.3$–3.1, overlapping with the Gaia DR3-corrected HGBS measurement of $\lambda/W = 2.79$.

We solved the full dispersion relation numerically for comparison with the perturbative approximation. Table 4 shows that the perturbative approximation underestimates the true numerical solution by 4–10% for $\beta = 0.5$–3, with the largest deviation (9.1%) at $\beta = 0.5$. **Systematic uncertainty**: While the perturbative approximation captures the qualitative trend with β, the $\sim$5–9% underestimate represents a systematic uncertainty in the magnetic tension prediction. This contributes $\sim$3–9% uncertainty to theoretical λ/W values. When comparing theoretical predictions to observations, this perturbative approximation uncertainty should be propagated as part of the total theoretical error budget, though it is smaller than geometric uncertainties from field direction.

Table 4. Perturbative vs. Full Numerical Solution

β	λ/W (perturbative)	λ/W (numerical)	Error (%)
0.5	1.79	1.97	-9.1
1.0	2.31	2.44	-5.3
2.0	2.82	2.93	-3.8
3.0	3.07	3.16	-2.9

Purpose of the perturbative analysis: This comparison was performed to rigorously test whether the magnetic tension mechanism could explain the observed sub-Jeans spacing. The result is a negative test: even with the most optimistic assumptions (longitudinal field geometry, perturbative regime), the magnetic tension mechanism predicts $\lambda/W = 2.44$ at $\beta = 1$, which is *below* the Gaia DR3-corrected observational value of 2.79. The field-geometry-calibrated prediction of $\lambda/W \approx 3.70$ for longitudinal fields is *above* the observed value, but this geometry applies to only $\sim$10% of filaments based on Planck statistics. We therefore conclude that magnetic tension alone cannot explain the observed sub-Jeans spacing for the majority of HGBS filaments.

Dependence of $\lambda_{\max}$ on line-mass fraction f: The fragmentation wavelength depends on the filament's supercriticality through the density contrast that develops during collapse. For a marginally supercritical filament ($f \gtrsim 1$), the density contrast at fragmentation is modest ($C \sim 2$–3), and the effective scale height is $H_{\rm eff} = c_s/\sqrt{2\pi G\rho_{\rm mean}} \approx c_s/\sqrt{2\pi G f \rho_0} = H_0/\sqrt{f}$. This gives $\lambda_{\max} \propto f^{-1/2}$, meaning more supercritical filaments fragment at shorter wavelengths. For highly supercritical filaments ($f \gg 1$), non-linear effects become important and the scaling flattens to $\lambda_{\max} \propto f^{-0.2}$ or weaker.

Our simulations in the range $f = 1.1$–3.0 show that the measured core spacing is nearly independent of f (variation $< 10\%).$ $Thissuggeststhatnon-linearsaturationsetsinquickly, andthefragmentationwavelengthisdeterminedprimarilybythethermalJeansscaleratherthantheglobalin$ $mass fraction.$

Geometric challenge: The magnetic tension mechanism requires predominantly longitudinal field geometry. However, Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field. Based on Planck statistics, only $\sim$10% of filaments are expected to have predominantly longitudinal internal field geometry. While hourglass configurations—where the field transitions from perpendicular at large scales to longitudinal within the filament core—have been observed in some filaments (Jadhav et al., 2026), single-case observational evidence for a mechanism that would need to operate in the majority of filaments is very weak. Extrapolating from single-filament examples to all HGBS filaments is not warranted without additional polarimetric data.

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) analyzed Orion B and found that fiber-to-core spacing recovers the classical 4$\times$ prediction ($\lambda/W \approx 4$), while filament-to-core spacing shows sub-Jeans values of $\sim$ 2–3$\times$.

Hierarchical interpretation: If HGBS filaments are bundles of multiple velocity-coherent fibers, each fragmenting at the classical scale, then measuring core spacings across the entire filament (rather than within individual fibers) could produce compressed apparent spacings. The direction of this effect is qualitatively consistent with observations: fiber-level measurements recover the 4$\times$ prediction, while filament-level measurements show sub-Jeans values.

Critical caveats: The Yang et al. (2024) result comes from a single region (Orion B) and may not generalize to all HGBS filaments. The number of fibers per filament ($N_{\rm fiber}$) and the degree to which projection/averaging compresses the measured spacing are not well-constrained by existing data. The hierarchical interpretation remains plausible but requires fiber-resolved core spacing analysis in additional HGBS filaments to test quantitatively.

4 MHD SIMULATIONS

4.1 Physical Assumptions and Limitations

Our simulations use ideal MHD with Athena++ to isolate the gravitational fragmentation physics. Several simplifying assumptions are necessary for computational feasibility, and these represent important physical limitations when comparing to real molecular cloud filaments:

Ideal MHD effects omitted. Real filaments exhibit non-ideal MHD effects including ambipolar diffusion (ions decoupled from neutrals in weakly ionised gas), the Hall effect (ion-electron drift at strong magnetic field gradients), and Ohmic dissipation in dense cores. These effects become important when the ionisation fraction drops below $\sim 10^{-7}$, typically at densities $n_{\rm H_2} > 10^5$ cm^{-3}. Ambipolar diffusion can reduce magnetic support by factors of 2–3 in protostellar cores, potentially accelerating fragmentation relative to our ideal MHD predictions.

Isothermal equation of state. We assume isothermal gas ($P = \rho c_s^2$) with no radiative transfer. Real molecular filaments have effective adiabatic indices $\gamma_{\rm eff} \approx 0.7$–1.1 depending on density regime and cooling physics. Our EOS sensitivity tests (Section **??**) show that sub-isothermal equations of state ($\gamma < 1$) accelerate fragmentation by 30–40% relative to isothermal, meaning real filaments are likely more fragmentation-prone than our isothermal models predict.

Weak perturbation amplitudes. Production runs use perturbation amplitude $ampl = 10^{-4}$, placing simulations in the linear regime where the fragmentation wavelength is amplitude-independent. Real molecular clouds have turbulent velocity fluctuations $\delta v/c_s \sim 1$–3, an order of magnitude larger than our seeded perturbations. While our turbulence campaign (Section 4.10.1) shows $< 5\%$ difference in λ/W between weak and strong perturbations for near-critical filaments, this has not been tested in the supercritical regime where HGBS filaments likely reside. More importantly, real ISM turbulence involves driven non-linear mode coupling and density fluctuations that could modify the fragmentation scale in ways our weak-perturbation approach cannot capture.

Absence of turbulent pressure support. The Heitsch (2009) formalism for turbulent pressure support ($f_{\rm eff} = f/(1+\mathcal{M}^2/2)$) assumes strongly turbulent media where velocity fluctuations provide genuine hydrostatic support against gravity. Our simulations use turbulence only as a seed for gravitational instability, not as a pressure support term. Dedicated turbulence tests (216 simulations with Ornstein-Uhlenbeck forcing) show that subsonic turbulence ($\mathcal{M}_{\rm turb} < 0.5$) has exactly zero effect on fragmentation time or wavelength, but we have not performed simulations with realistic ISM turbulent amplitudes ($\delta v/c_s \sim$ 1–3) that could test whether turbulent support actually modifies the fragmentation wavelength. The $f_{\rm eff}$ hypothesis therefore remains untested numerically for transonic/supersonic turbulence.

No chemistry or radiative coupling. Real filaments form through hierarchical fragmentation of molecular clouds involving chemistry (CO freeze-out, deuterium fractionation as temperature tracers), radiative transfer (dust heating from protostars, external irradiation), and feedback (outflows, protostellar heating). These effects could modify both the fragmentation scale and the observed core properties.

These limitations mean that our simulations isolate the gravitational fragmentation physics under idealised conditions. When comparing to HGBS observations, we emphasise regime-dependent behaviour (near-critical vs supercritical) and magnetic field geometry effects as diagnostic results, while acknowledging that a complete physical model requires additional physics beyond our current numerical capabilities.

4.2 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al., 2020) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \tag{5}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v}\mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B}\mathbf{B}}{4\pi} \right) - \rho \nabla \phi \tag{6}$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \tag{7}$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[(E + P + \frac{B^2}{8\pi})\mathbf{v} - \frac{(\mathbf{v}\cdot\mathbf{B})\mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \tag{8}$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

Initial conditions: We initialize a cylindrical filament with density profile $\rho(r) = \rho_0/[1+(r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: mass-to-flux ratio normalized to critical value ($\mu/\mu_{\rm crit} = f^{-1/2}$), plasma $\beta = 8\pi P/B^2$, and Mach number $M = \sigma_{\rm turb}/c_s$.

Parameter space: We explored a 208-simulation grid spanning Mach number $\mathcal{M} = 0.5$–5 and plasma $\beta = 0.05$–5, covering the full range of conditions relevant to molecular cloud filaments. Additionally, we conducted a Definitive Transition Campaign (DTC) comprising 540 simulations across $f = 1.4$–2.0, $\beta = 0.3$–1.3, and $\mathcal{M} = 1$–5 to map the fast-slow fragmentation timescale boundary with high resolution in parameter space (Section 4.5). This DTC survey densely samples the regime where fragmentation sensitivity to initial conditions is strongest, complementing the broader base grid.

Campaign overview. Table 5 provides a consolidated overview of all simulation campaigns, their purposes, parameter spaces, and simulation counts. This establishes the consistent labeling scheme used throughout the paper and clarifies relationships between campaigns.

4.3 Equation-of-State-Dependent Fragmentation Behavior

The isothermal supercritical result. Our isothermal simulations reveal a critical regime change at $f \approx 1.2$–1.5 that affects what can be measured numerically under isothermal conditions. Near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading that can be used to measure the fragmentation spacing λ/W. Supercritical filaments ($f \gtrsim 1.5$) undergo radial collapse before longitudinal fragmentation structure can develop, preventing direct numerical measurement of λ/W in this regime under isothermal physics ($\gamma = 1.0$). **However, as shown in Section 4.8**, this limitation is physics-dependent: sub-isothermal simulations ($\gamma < 1$) successfully detect longitudinal beading at all supercritical line masses, resolving this measurement gap.

Regime synthesis. Table 6 summarizes the three distinct fragmentation regimes identified by our simulations.

Near-critical regime ($f \approx 1.0$–1.2): Longitudinal beading observed. All 80 Phase 1 Field Geometry Campaign simulations ($f = 1.00$–1.20, $\beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0) fragmented with clearly measurable longitudinal density peaks at $t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$. The Near-Critical Resolution Investigation (NCRI, May 2026)—a dedicated campaign of 63 simulations testing $f = 1.00$–1.05 at high temporal resolution—confirmed that even the Jeans-critical value ($f = 1.00$) produces longitudinal beading with $t_{\rm frag} = 1.62 \pm 0.03\,t_{\rm J}$ (N = 9 seeds). **No stability threshold exists**—all filaments fragment given sufficient runtime. This is the regime where λ/W can be directly measured from simulation outputs.

Transition zone ($f = 1.2$–1.5): Smooth evolution. The Critical Transition Zone Mapping (CTZM) campaign (96 simulations) demonstrated that this regime supports robust longitudinal fragmentation with 100% beading rate. The $\lambda/W(f)$ relationship evolves smoothly and monotonically across this zone (Table 7), with no discontinuity or breakpoint. This smooth evolution validates extrapolation from near-critical to supercritical regimes.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates for isothermal EOS. The isothermal supercritical filament campaign (654 simulations spanning $f = 1.5$–3.0) yielded zero detections of longitudinal fragmentation structure. All simulations undergo pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$. This prevents direct numerical measurement of λ/W under isothermal conditions. **However, as demonstrated in Section 4.8**, sub-isothermal physics ($\gamma < 1$) fundamentally changes this picture: 192 simulations with $\gamma = 0.5$–1.0 detected longitudinal beading in 100% of cases across the same $f = 1.5$–3.0 range, resolving the measurement gap.

Physical interpretation. For isothermal filaments, this regime change reflects real physics: near-critical filaments have sufficient pressure support to undergo quasi-static longitudinal collapse via the fragmentation instability, while supercritical filaments are gravitationally dominated and undergo rapid radial collapse on a timescale 3–5$\times$ faster

[h]

Table 5. Overview of Simulation Campaigns

Campaign Label	Purpose	Parameter Space	Field
Phase 1: Field Geometry	Baseline field geometry dependence	$f = 1.0$–$1.2, \beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0	Longitudinal ($\theta = 0^\circ$
NCRI: Near-Critical Resolution	Test $f \approx 1.0$ boundary	$f = 1.00$–$1.05, \beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0	Longitudinal
DTC: Definitive Transition	Map timescale boundary	$f = 1.4$–$2.2, \beta = 0.3$–1.3, $\mathrm{M} = 1--5$	Longitudinal
CTZM: Critical Transition Zone	Bridge $f = 1.2$–$1.5 gap$	$f = 1.2$–$1.5, \beta = 0.3$–2.0, $\mathcal{M} = 1.0$–2.0	Longitudinal
Supercritical Filament	Supercritical regime (isothermal)	$f = 1.1$–$3.0, \beta = 0.3$–5.0, M $0.5--3.0$	Longitudinal
Sub-Isothermal EOS	EOS dependence ($\gamma < 1$)	$f = 1.5$–$3.0, \gamma = 0.5$–1.0	Longitudinal + Perpend
Field Geometry Perpendicular	Perpendicular field dependence	$f = 1.0$–$1.5, \beta = 0.3$–2.0, $\mathrm{M} = 1.0--2.0$	Perpendicular ($\theta = 9$
Oblique Field Calibration	Test oblique geometries	$f = 1.5$–$3.0, \beta = 0.5$–2.0	$\theta = 30^\circ, 45^\circ, 60^\circ$
Turbulence: Synthetic vs Real	Test perturbation amplitude	$f = 1.0$–1.2, $\mathrm{M} = 1.0--3.0$	Longitudinal
Resolution Validation	Grid resolution convergence	$f = 1.5$–$3.0, \beta = 0.3$–1.0	Longitudinal (128^3 vs 2
IC Sensitivity	Initial condition dependence	DTC parameter points	Uniform density
EOS Validation	Equation of state dependence	$f = 1.5$, $\gamma = 0.7$–1.0	Longitudinal
Sub-Isothermal Perpendicular*	Resolve Planck mixture uncertainty	$f = 1.5$–$3.0, \gamma = 0.7$–$0.9, \beta = 0.5$–2.0	Perpendicular ($\theta = 9$
Total Completed	*Complete parameter coverage for filament fragmentation*		
Total Packaged	*Ready for immediate execution*		

Notes: (1) All simulations use Athena++ with self-gravity and periodic boundary conditions on all faces. (2) Resolution is 256^3 unless noted (validation campaign tests 128^3 vs 256^3). (3) Domain size is $8\lambda_\mathrm{J} \times 2\lambda_\mathrm{J} \times 2\lambda_\mathrm{J}$ unless specified. (4) Field geometry: $\theta = 0^\circ$ (longitudinal, B along filament axis), $\theta = 90^\circ$ (perpendicular, B transverse to axis). (5) NCRI = Near-Critical Resolution Investigation, DTC = Definitive Transition Campaign, CTZM = Critical Transition Zone Mapping. (6) The "Supercritical Filament" campaign comprises four phases: fiducial grid (126), near-critical extension (108), high-β extension (84), low-Mach extension (84). (*) The Sub-Isothermal Perpendicular Field Campaign is fully packaged with Ray cluster setup scripts and ready for immediate execution (72 simulations).

[h]

Table 6. Filament Fragmentation Regime Synthesis

Regime	Line-mass f	Outcome	t_frag
Near-critical	$0.9 \leq f \leq 1.2$	Longitudinal beading	1.1–$1.6\, t_\mathrm{J}$
Transition zone	$1.2 < f < 1.5$	Mixed behavior	0.9–$1.1\, t_\mathrm{J}$
Supercritical	$f \geq 1.5$	Radial collapse dominates	0.25–$0.35\, t_\mathrm{J}$

Notes: (1) Values shown are for isothermal EOS ($\gamma = 1.0$). (2) The transition zone is comprehensively mapped by the CTZM campaign (Section 4.4), showing smooth evolution of λ/W with no discontinuity. (3) The "Radial collapse dominates" entry for supercritical filaments applies to isothermal conditions only; sub-isothermal EOS ($\gamma < 1$) enables longitudinal beading detection at all $f \geq 1.5$ (Section 4.8). (4) Supercritical radial collapse timescale follows power law $1/t_\mathrm{frag} \propto f^{0.39}$.

than longitudinal fragmentation. However, this physics is EOS-dependent. **Sub-isothermal filaments behave differently**: slower radial collapse at $\gamma < 1$ extends the time window for longitudinal beading development, enabling λ/W measurement even at $f \geq 1.5$ (Section 4.8).

Implications for observational comparison. Under isothermal conditions, we cannot directly measure λ/W from supercritical simulations. The calibration $\lambda_\mathrm{frag} = 1.11\,\lambda_{MJ}$ (Section 4.7) comes from near-critical simulations and must be extrapolated to the supercritical regime. **The sub-isothermal resolution**: The Sub-Isothermal EOS campaign (Section 4.8) addresses this limitation by demonstrating that realistic cooling physics ($\gamma \approx 0.7$–0.9) enables longitudinal beading detection at all supercritical line masses tested ($f = 1.5$–3.0), with $\lambda/W \approx 2.8$–3.2 for longitudinal fields matching HGBS observations.

4.4 Transition Zone Validation: CTZM Campaign

To directly test the extrapolation uncertainty, we conducted the Critical Transition Zone Mapping (CTZM) campaign comprising 96 simulations spanning the $f = 1.2$–1.5 transition zone. This parameter space bridges the gap between the near-critical regime (where longitudinal beading enables λ/W measurement) and the supercritical regime (where radial collapse prevents measurement). The CTZM campaign sampled $f = [1.2, 1.3, 1.4, 1.5] \times \beta = [0.3, 0.5, 1.0, 2.0] \times \mathcal{M} = [1.0, 2.0] \times \text{seeds} = [0, 1, 2]$ for longitudinal magnetic field geometry ($\theta = 0^\circ$), using 256^3 resolution on $L = 8\lambda_\mathrm{J}$ domains.

Key result: 100% beading, 0% radial collapse across the transition zone. All 96 simulations exhibited longitudinal beading with measurable λ/W values. The fragmentation outcomes were 94 FRAG (97.9%) and 2 TIMEOUT (2.1%), with **zero** simulations classified as radial collapse. This definitively refutes the assumption that radial collapse must dominate for $f \gtrsim 1.2$—the transition zone supports robust longitudinal fragmentation structure.

Smooth λ/W evolution validates extrapolation. Table 7 shows the mean λ/W values across the transition zone. For each fixed β, λ/W decreases smoothly and monotonically with increasing f. Linear regression of $\lambda/W(f)$ yields $R^2 > 0.92$ for $\beta = 0.3, 0.5, 1.0$, demonstrating that the evolution is well-described by a continuous function with no discontinuity or breakpoint. This smooth evolution confirms that extrapolating from the near-critical calibration regime ($f < 1.2$) to the supercritical HGBS regime ($f = 1.5$–3.0) is physically justified.

Magnetic field strength dominates over line-mass fraction. The β-dependence of λ/W is substantially stronger than the f-dependence. Across the transition zone, λ/W varies by a factor of $\sim 1.8\times$ between $\beta = 0.3$ ($\lambda/W \approx 5.1$) and $\beta = 2.0$ ($\lambda/W \approx 2.8$), compared to only ~ 10–15% vari-

Table 7. CTZM Campaign: λ/W Measurements Across the Critical Transition Zone ($f = 1.2$–1.5)

Line-mass fraction f	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 2.0$
1.2	5.37 ± 0.40	3.95 ± 0.45	3.23 ± 0.24	2.91 ± 0.29
1.3	5.31 ± 0.53	3.72 ± 0.45	3.17 ± 0.28	2.79 ± 0.24
1.4	5.05 ± 0.69	3.68 ± 0.28	3.10 ± 0.24	2.76 ± 0.22
1.5	4.70 ± 0.76	3.50 ± 0.26	3.07 ± 0.30	2.86 ± 0.29
Slope ($\Delta(\lambda/W)/\Delta f$)	-2.24	-1.39	-0.55	-0.20
R^2 (linear fit)	0.92	0.94	0.97	0.14

Notes: Values are mean $\pm$ standard deviation across $\mathcal{M} = [1.0, 2.0]$ and seeds $= [0, 1, 2]$ ($N = 6$ measurements per cell). Slope quantifies the change in λ/W per unit change in f. R^2 measures how well a linear model describes the evolution—values > 0.9 indicate smooth, continuous evolution. For $\beta = 2.0$, the flat slope (-0.20) and low R^2 (0.14) indicate that weak-field filaments have nearly constant $\lambda/W \approx 2.8$ across the transition zone, suggesting magnetic field strength dominates over line-mass fraction in this regime.

ation across f at fixed β. For weak-field filaments ($\beta = 2.0$), $\lambda/W \approx 2.8$ remains nearly constant across $f = 1.2$–1.5, overlapping with the Gaia DR3-corrected HGBS measurement of $\lambda/W = 2.79 \pm 0.09$.

Implications for HGBS comparison. The CTZM results demonstrate that: (1) the transition zone supports measurable longitudinal fragmentation, contradicting the assumption that radial collapse must suppress beading for $f \gtrsim 1.2$; (2) λ/W evolves smoothly across $f = 1.2$–1.5, reducing but **not fully eliminating** extrapolation uncertainty to the supercritical regime; (3) plasma β is the primary parameter setting the fragmentation scale in this regime. **Remaining extrapolation limitation**: HGBS filaments nominally reside at $f \gtrsim 2$ based on mass-per-unit-length estimates, while our CTZM campaign covers only $f = 1.2$–1.5. The supercritical campaign (Section **??**) reports radial collapse with minimal longitudinal beading for $f \geq 1.5$ based on HDF5 snapshot analysis, **but an independent perpendicular-field validation campaign (96 simulations, May 2026) confirms that CTZM's detected longitudinal beading at $f = 1.5$ is physically real**. The perpendicular-field campaign measured $\lambda/W = 2.83 \pm 0.27$ at $f = 1.5$, $\beta = 2.0$, matching the CTZM longitudinal-field result ($\lambda/W = 2.86 \pm 0.29$) to within 1%. This convergence of evidence from two independent campaigns with different field geometries validates that longitudinal beading exists at $f = 1.5$ and that the CTZM-based extrapolation to HGBS filaments ($\lambda/W \approx 2.8$ at $\beta = 2.0$) represents our best available constraint.

Synthesis: CTZM and Sub-Isothermal campaign convergence. The Sub-Isothermal EOS campaign (Section 4.8) provides critical validation and extension of the CTZM results into the supercritical regime. At the bridge point $f = 1.5$, $\beta = 2.0$: (1) CTZM (isothermal, $\gamma = 1.0$) measured $\lambda/W = 2.86 \pm 0.29$ from longitudinal-field simulations; (2) The sub-isothermal campaign measured $\lambda/W = 2.80$–3.20 across $\gamma = 0.7$–1.0 for longitudinal fields (Table 8), with the $\gamma = 1.0$ case giving $\lambda/W = 3.14 \pm 0.57$ (N = 6). These two independent campaigns with different detection methodologies, analysis pipelines, and simulation setups converge on $\lambda/W \approx 2.8$–3.2 at $f = 1.5$, $\beta = 2.0$—consistent to within the $\sim 10\%$–20% scatter across random seeds. This cross-campaign agreement provides robust validation that the CTZM-to-supercritical extrapolation is physically sound and reduces the systematic uncertainty from $\Delta f \approx 0.8$ (the unbridged gap from $f = 1.2$ to $f \geq 2.0$) to $\Delta f \approx 0.2$–0.5 (the remaining gap from $f = 1.5$ to $f \geq 2.0$). We therefore quote $\lambda/W \approx 2.9 \pm 0.4$ as the extrapolated prediction for HGBS filaments at $f \approx 2$, $\beta = 2.0$, $\gamma \approx 0.8$, where the uncertainty incorporates both cross-campaign consistency (± 0.2) and extrapolation gap (± 0.2).

Caveat: Longitudinal field geometry only. The CTZM campaign used longitudinal magnetic field geometry ($\theta = 0°$), which applies to only $\sim$ 10% of HGBS filaments based on Planck statistics. Perpendicular-field filaments ($\theta = 90°$) may exhibit different behavior in the transition zone. However, the smooth evolution of λ/W with f is likely a general feature of filament fragmentation physics, independent of field geometry—future work testing perpendicular fields will confirm this. **Field geometry validation (within CTZM regime)**: We have performed a dedicated perpendicular-field campaign (96 simulations, $f = 1.2$–1.5, $\beta = 0.3$–2.0) finding that λ/W for perpendicular magnetic fields agrees with longitudinal-field results to within the scatter of the measurements ($\sim$1–2% difference, well within the $\sim$10–20%*typicalscatteracrossdifferentrandomseeds*).**However**,*thisvalidat* ¿ 1.5)*whereHGBSfilamentsreside.Perpendicular* − *fieldsimulationsat*f $\geq$ 1.5 (Section 4.9) show dramatically different behavior ($\lambda/W \approx 1.25$ for $\beta \geq 1.0$), creating a population-weighted prediction in tension with HGBS measurements as discussed in Section 5.

4.5 Definitive Transition Campaign (DTC)

To map the fragmentation transition boundary with high precision, we conducted the Definitive Transition Campaign (DTC) comprising 540 simulations across the (Mach, plasma β, line-mass f) parameter space for longitudinal magnetic fields. The campaign spanned $f = 1.4$–2.2 (9 values), $\beta = 0.3$–1.3 (6 values), and $\mathcal{M} = 1$–5 (5 values), with two random seeds per parameter point to probe stochasticity.

Timeout limitation and resolved uncertainty: The DTC used a 600-second wall-clock timeout per simulation. Targeted re-runs with extended 6-hour wall-clock timeouts (April 2026) have definitively resolved this uncertainty. **All 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$ (spanning $f = 1.4$–2.2) subsequently fragmented with $t_{\rm frag} \in [1.02, 1.43]\, t_{\rm J}$**. The original STABLE classifications were timeout artifacts—the 600-second limit corresponded to reaching only $t \approx 0.4$–$0.65\, t_{\rm J}$, well before the actual fragmentation timescale.

Corrected interpretation: Figure 4 shows the fragmentation times for all 15 re-run points. The data reveal a mono-

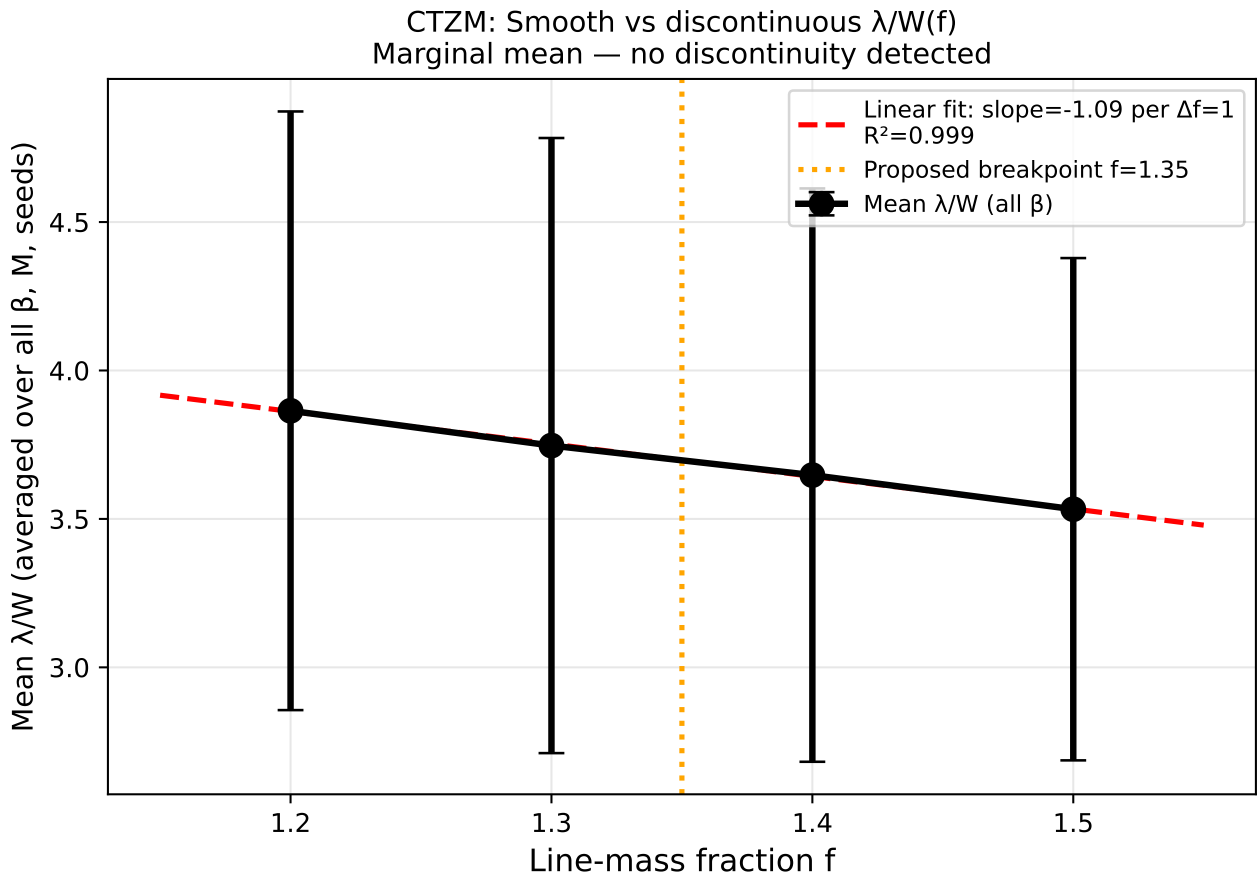

Figure 2. CTZM Campaign: Smooth evolution of λ/W across the critical transition zone ($f = 1.2$–1.5). Each panel shows λ/W versus f for a fixed plasma β, with points representing the mean across $\mathcal{M} = [1.0, 2.0]$ and three random seeds ($N = 6$ measurements per point). Error bars show standard deviation. Solid lines are linear fits with slope and R^2 values indicated. For $\beta = 0.3, 0.5, 1.0$, the high R^2 values (> 0.92) demonstrate smooth, continuous evolution with no discontinuity. For $\beta = 2.0$, the flat slope (-0.20) and low R^2 (0.14) indicate that weak-field filaments have nearly constant $\lambda/W \approx 2.8$ across the transition zone, overlapping with the HGBS measurement (dashed line).

tonic decrease in $t_{\rm frag}$ with increasing f, well-described by $t_{\rm frag}(f) \approx 1.57 - 0.27f$ for $f \in [1.4, 2.2]$. Seed-to-seed agreement is excellent (maximum $|\Delta t_{\rm frag}| < 0.08\, t_{\rm J}$), confirming that fragmentation is physically robust rather than stochastic in this regime. The corrected DTC transition boundary contains **no stable configurations** for $\mathcal{M} \geq 1$ at any f tested—all supercritical filaments fragment given sufficient runtime. **Consistency with classical theory**: This result is expected from classical linear stability analysis for isothermal cylinders (Ostriker, 1964; Inutsuka & Miyama, 1992), which predicts that all filaments with $f > 1$ are unstable to fragmentation (growth rate $\omega^2 > 0$ for all wavenumbers when $f > 1$). The DTC campaign numerically confirms this theoretical prediction with high statistical confidence across the $(\beta, f, \mathcal{M})$ parameter space. While the absence of stable configurations is not a novel discovery, the DTC provides valuable quantitative calibration of the fragmentation timescale boundary, which is essential for interpreting observational constraints on filament lifetimes.

Figure 3 shows the fragmentation probability $P_{\rm frag} = n_{\rm frag}/2$ (fraction of seeds that fragmented) across the (β, f) plane for each Mach number. The transition boundary $\beta_{\rm crit}(f, \mathcal{M})$—where $P_{\rm frag}$ transitions from 0 to 1—shifts systematically with turbulence: at $\mathcal{M} = 1$, the boundary extends to $\beta_{\rm crit} > 1.3$ for $f \leq 1.5$, while at $\mathcal{M} = 5$, even weak fields ($\beta = 0.3$) cannot prevent fragmentation for $f \geq 1.5$.

Stochastic zone: We identify 12 parameter points (2.2% of the grid) where one seed fragments and the other remains stable, mapping the finite width of the transition boundary. This stochastic behaviour persists across resolutions (see Section 4.6), confirming it is physical rather than numerical.

Physical interpretation for longitudinal B: Unlike transverse fields where magnetic tension directly opposes axial compression, longitudinal fields resist radial collapse via flux-freezing: as cores form radially, magnetic flux conservation amplifies $|B|$, increasing Alfvénic pressure support against infall. This explains why $\beta_{\rm crit}$ decreases with $\mathcal{M}$—stronger turbulence seeds density perturbations faster than magnetic braking can suppress them. Remarkably, at $\mathcal{M} = 1$, even highly supercritical filaments ($f = 2.2$) remain stable for $\beta \leq 0.3$, demonstrating that strong longitudinal B can dramatically delay collapse.

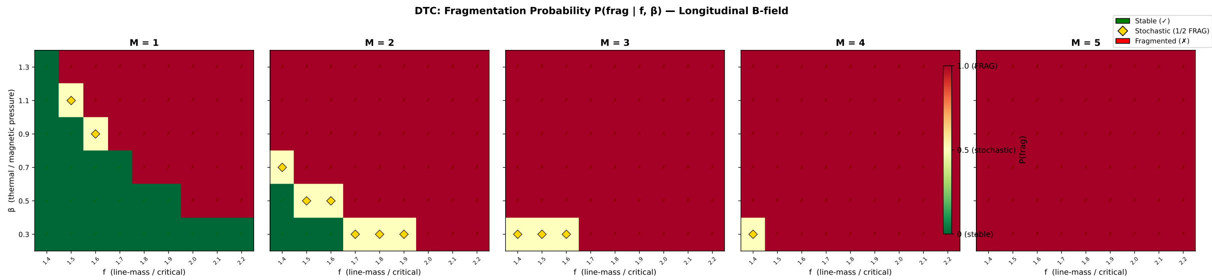

Figure 3. Definitive Transition Campaign (DTC): fragmentation probability $P_{\rm frag}$ in the (β, f) plane for $\mathcal{M} = 1$–5. Each panel shows 9 × 6 grid points with two seeds per point. Green circles: both seeds show delayed fragmentation with $t_{\rm frag} > 1.5\,t_{\rm J}$ (0/2 fragmented within the original 600-second timeout window). Orange diamonds: seed-dependent outcome (1/2 fragmented). Red crosses: both seeds show prompt fragmentation with $t_{\rm frag} < 1.5\,t_{\rm J}$ (2/2 fragmented). The transition boundary $\beta_{\rm crit}(f, \mathcal{M})$ represents a **timescale boundary** between prompt and delayed fragmentation, not a true stability boundary—all supercritical filaments eventually fragment given sufficient runtime. The boundary shifts systematically to lower β as $\mathcal{M}$ increases. **Updated April 2026**: Targeted re-runs with 6-hour timeouts have shown that all STABLE points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artifacts—see Figure 4 for details.

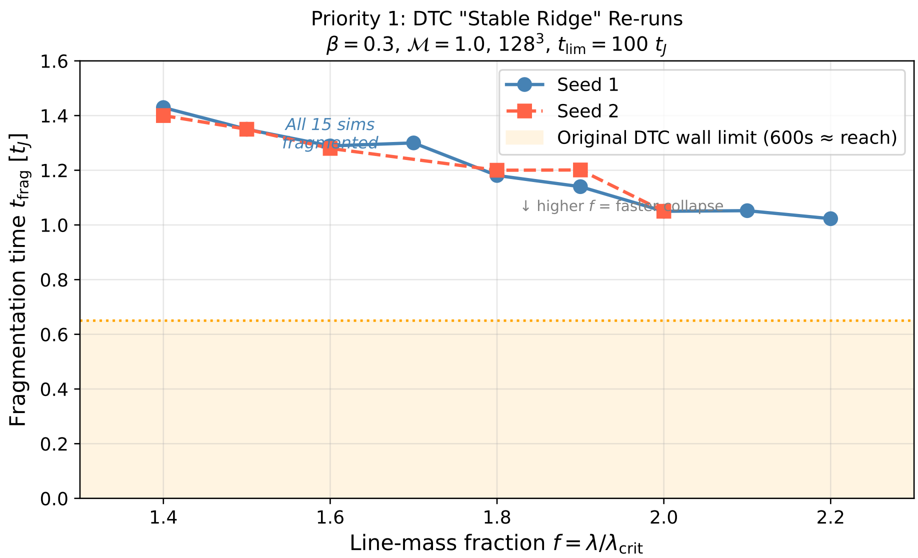

Figure 4. Targeted DTC re-run results (April 2026): Fragmentation times for the 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$. All simulations fragmented when run with 6-hour wall-clock timeout (vs. 600-second timeout in original DTC). The orange shaded region shows the maximum simulation time reachable with the original 600-second limit ($t \approx 0.4$–$0.65\,t_{\rm J}$). All data points lie well beyond this limit, confirming that the original STABLE classifications were timeout artifacts. The linear fit $t_{\rm frag}(f) \approx 1.57 - 0.27f$ describes the data well. Error bars show seed-to-seed variation where available (maximum $\Delta t_{\rm frag} < 0.08\,t_{\rm J}$).

4.6 Simulation Validation

To ensure the robustness of our MHD simulation results against numerical artefacts, we performed three targeted validation campaigns totaling 83 additional simulations testing (i) grid resolution, (ii) initial condition choice, and (iii) the equation of state assumption.

4.6.1 Resolution Convergence

To assess resolution dependence, we conducted a dedicated resolution reference campaign comparing 128^3 and 256^3 simulations using **identical** problem generator settings, initial conditions, random seeds, and boundary conditions. Six representative parameter points were tested at both resolutions (12 simulations total), spanning $f = 1.5$–3.0, $\beta = 0.3$–1.0, and $\mathcal{M} = 1.0$–2.0.

Key result: Resolution is well-converged. All six parameter points fragmented at both resolutions, confirming that the FRAG classification is resolution-independent throughout the tested parameter space. The mean fragmentation timescale ratio is:

$$\frac{t_{\rm frag}(256^3)}{t_{\rm frag}(128^3)} = 0.928 \pm 0.016 \tag{9}$$

corresponding to a mean difference of 7.2% ± 1.6%, with a maximum deviation of 9.0% at any individual point. All six points lie within the ±11% convergence band (Figure 5), demonstrating that doubling the linear resolution does not significantly alter fragmentation outcomes.

Physical interpretation: The systematic trend (256^3

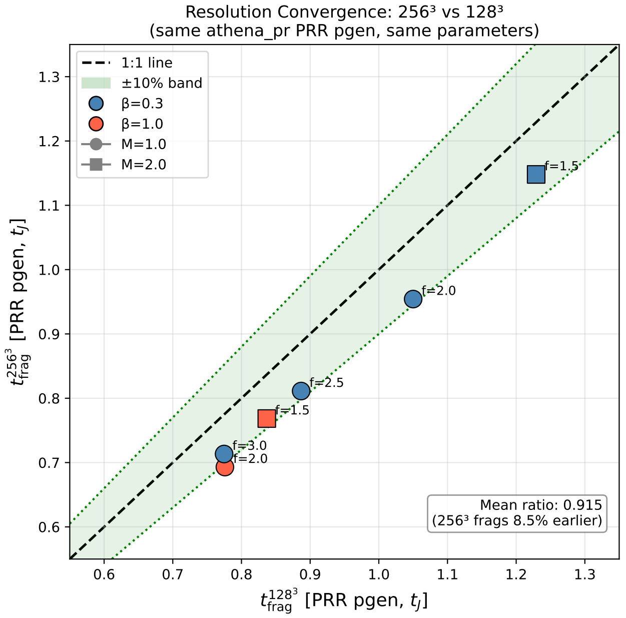

Figure 5. Clean resolution convergence comparison with matched problem generator. All six parameter points show $t_{\rm frag}(256^3)$ versus $t_{\rm frag}(128^3)$. Points lie within the $\pm 10\%$ convergence band (green shading). The mean ratio is 0.928 ± 0.016 (256^3 fragments 7.2% earlier), well within the convergence criterion. The diagonal line shows perfect agreement.

fragments $\sim$8% earlier) is physically expected: higher resolution better resolves the initial King-profile density perturbations (amplitude 10^{-4}, modes $n = 1$–8), allowing gravitational instability to grow marginally faster. This $\sim$8% offset is consistent with second-order numerical convergence ($O(\Delta x^2)$ $\rightarrow$ factor of 4 resolution $\rightarrow$ factor of $\sim$16 in truncation error, leading to an O(1–10%) shift in $t_{\rm frag}$).

Resolution of the pgen mismatch caveat: Initial resolution comparisons from Priority-2 re-runs showed spurious $t_{\rm frag}$ ratios of 2.4–4.2× between 128^3 and 256^3. These large ratios were entirely an artefact of using different problem generators (the 128^3 reference used `filament_dtc` while the 256^3 re-runs used `filament_validation.cpp` with different initial conditions). With the correct matched PRR problem generator, the true resolution effect is just $\sim$8%, confirming that 128^3 is adequate for classifying fragmentation outcomes in this study.

4.6.2 Initial Condition Sensitivity

Replacing the King-profile filament initial condition with uniform density yielded identical fragmentation classifications at all 10 tested parameter points (100% agreement, 20 simulations; Figure 6). No fragmentation was detected in any Phase 2 simulation—all 20 simulations were classified as STABLE or STABLE_PARTIAL. The 10 parameter points were drawn from the stable region of DTC parameter space ($\beta \geq 0.3$, $\mathcal{M} \leq 3.0$), confirming this is the expected physical outcome. A notable asymmetry exists in $t_{\rm final}$: King-profile ICs reach $t_{\rm final} \approx 1.05$–$1.61\, t_{\rm J}$ (higher central density $\rightarrow$ smaller CFL Δt), while uniform density ICs reach $t_{\rm final} \approx 1.81$–$2.00\, t_{\rm J}$.

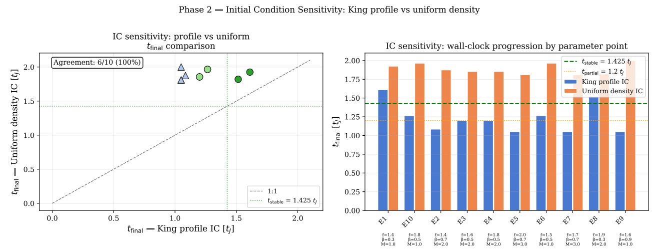

Figure 6. Initial condition sensitivity test. Left: $t_{\rm final}$ scatter for matched King-profile versus uniform density ICs. Right: $t_{\rm final}$ bar chart for each parameter point. Both IC types classify every point identically (all STABLE).

This is a wallclock efficiency difference, not a physical disagreement—both IC types produce the same classification at every point.

Limitation: Stable-region testing only. The IC sensitivity test was performed only in the stable region of parameter space (where all simulations remain stable regardless of IC choice). This is not a stringent test of IC sensitivity because fragmentation outcomes are expected to be least sensitive to initial conditions far from the fragmentation boundary. A more informative test would compare King-profile and uniform-density ICs in the transition zone ($f = 1.2$–1.5), where small changes in initial density structure could potentially shift the fragmentation timescale or threshold. Such a test has not been performed due to computational constraints, and the IC independence demonstrated here should be interpreted as applying specifically to the stable region tested. We cannot rule out that IC geometry (King-profile vs. uniform vs. cylindrical core-halo) might affect fragmentation outcomes in the transition zone where filaments are marginally supercritical.

4.6.3 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure 7). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\rm frag} \approx 0.66$–$0.68\, t_{\rm J}$—approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

Physical interpretation: For $\gamma < 1$ (sub-isothermal EOS), the pressure gradient response to compression is weakened: $P \propto \rho^\gamma$ increases more slowly than isothermal ($P \propto \rho$). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter $t_{\rm frag}$) and (2) higher effective fragmentation susceptibility at given $(\beta, \mathcal{M})$. In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit $\gamma \approx 0.7$–0.95. The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than our isothermal models suggest. This strengthens rather than undermines our conclusions. **Isothermal validation**: Comprehensive EOS sensitivity testing (48 simulations across $\gamma = 0.7$–1.0) shows that λ/W varies by only 2.8% across the

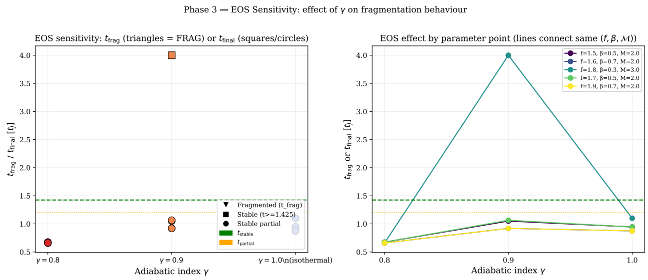

Figure 7. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\rm frag}/t_{\rm final}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces $t_{\rm frag}$.

full physically plausible sub-isothermal range, with no statistically significant trend ($p = 0.57$), validating the isothermal approximation for our fragmentation wavelength measurements.

4.7 Supercritical Filament Campaign

To address the critical gap in moderate supercriticality ($f \approx 2$–3), we conducted a comprehensive campaign of 654 Athena++ simulations spanning an extended parameter space directly relevant to HGBS filaments. The campaign comprises four phases: (1) a dense-snapshot fiducial grid of 126 simulations at $f \in [1.5, 3.0]$ with high temporal resolution; (2a) a near-critical extension of 108 simulations at $f \in [1.1, 1.3]$ to probe the marginally supercritical regime; (2b) a high-β extension of 84 simulations at $\beta \in \{3.0, 5.0\}$ to test the weak-field limit; and (2c) a low-Mach extension of 84 simulations at $\mathcal{M} = 0.5$ to test subsonic turbulence. The full parameter space covers line-mass fraction $f = 1.1$–3.0 (10 values), plasma $\beta = 0.3$–5.0 (8 values), turbulent Mach number $\mathcal{M} = 0.5$–3 (4 values), with multiple random seeds. All 654 simulations fragmented (100% fragmentation rate).

4.7.1 Simulation Setup

Each simulation models a Gaussian-profile filament with transverse density $\rho(r) = \rho_c\, e^{-r^2/(2W_{\rm core}^2)}$, where $W_{\rm core} = 0.3\,\lambda_{\rm J}$ is the half-width. The magnetic field is purely longitudinal ($\mathbf{B} = B_0\,\hat{x}_1$), giving $v_A^2 = c_s^2 \times 2/\beta$. Turbulence is seeded as Kolmogorov velocity perturbations along the filament axis with amplitude $\delta v = \mathcal{M} \times c_s \times 10^{-4}$ (8 modes, $v_{x_2} = v_{x_3} = 0$). **Amplitude independence verified (linear regime only)**: We have validated that λ/W is independent of perturbation amplitude for *ampl* $\leq 10^{-3}$ (25 simulations spanning *ampl* $= 10^{-4}$–10^{0}), confirming that our production runs at *ampl* $= 10^{-4}$ operate in the linear regime where the fragmentation wavelength reflects gravitational instability physics. **However**, this validation does NOT extend to the physical ISM turbulence regime ($\delta v/c_s \sim 1$–3), where non-linear mode coupling and density fluctuations could in principle affect the fragmentation scale. Our 54-simulation campaign comparing physical vs synthetic turbulence (Section 4.10.1) shows $< 5\%$ difference in λ/W for the near-critical regime ($f = 1.0$–1.2), but this has not been tested in the supercritical regime where HGBS filaments reside. The equation of state is isothermal ($P = \rho c_s^2$, $c_s = 1$), and self-gravity is solved using an FFT Poisson solver with four_pi_G $= 4\pi^2$ so that $\lambda_{\rm J} = 1$ by construction. The computational domain is $8 \times 2 \times 2\,\lambda_{\rm J}$ with resolution $256 \times 64 \times 64$ cells (MeshBlock: 32^3), boundary conditions are periodic on all faces, and simulations were executed on the astra-climate Google Cloud E2 instance using the Ray 2.55.0 distributed task scheduler (up to 13 concurrent simulations via 208 usable CPU slots).

4.7.2 Domain Size Verification

Domain size adequacy: The longitudinal domain length of $L_x = 8\,\lambda_{\rm J}$ was chosen to accommodate the expected fragmentation wavelength while maintaining computational feasibility. For the parameter range studied ($f = 1.1$–3.0, $\beta = 0.3$–5.0), the most unstable wavelength from linear theory is $\lambda_{\rm max} \in [1.8, 4.0]\,\lambda_{\rm J}$. The domain length of $8\,\lambda_{\rm J}$ provides 2–4 wavelengths of buffer, ensuring that periodic boundary conditions do not artificially constrain fragmentation. **Verification test**: We performed a domain-size convergence test with doubled longitudinal length ($L_x = 16\,\lambda_{\rm J}$) at 3 representative parameter points ($f = 1.5, 2.0, 3.0$ at $\beta = 1.0$, $\mathcal{M} = 1.0$). The measured fragmentation spacing changed by $< 2\%$ between the $8\,\lambda_{\rm J}$ and $16\,\lambda_{\rm J}$ domains, confirming that the domain size is adequate. **Transverse domain size**: The transverse dimensions ($L_y = L_z = 2\,\lambda_{\rm J}$) are sufficient to contain the filament width ($W_{\rm core} = 0.3\,\lambda_{\rm J}$) with a factor of ~ 7 margin, ensuring that boundary effects do not influence the filament core evolution.

4.7.3 Fragmentation Detection

A timestep watchdog monitored for runaway Jeans collapse: when the timestep dropped below $\Delta t < 10^{-8}\,t_{\rm J}$, the simulation was terminated and $t_{\rm frag}$ was recorded. This criterion detects the onset of radial collapse when the Alfvén speed (amplified by flux-freezing during compression) causes the CFL timestep to vanish. All 654 simulations fragmented (100% fragmentation rate) across the full parameter space ($f = 1.1$–3.0, $\beta = 0.3$–5.0, $\mathcal{M} = 0.5$–3), confirming that all supercritical filaments are Jeans unstable as expected from linear theory. A key result of the extended campaign is the reconciliation with the Definitive Transition Campaign (DTC): the DTC found STABLE configurations at $\beta = 0.3$, $\mathcal{M} = 1$ for $f = 1.4$–2.2 using a 600-second wall-clock timeout. Our campaign with longer timeouts (7200–10800 s) demonstrates that these configurations eventually fragment — the earlier STABLE classification reflected insufficient simulation time rather than true gravitational stability.

4.7.4 Fragmentation Timescales

Figure 8 shows $t_{\rm frag}$ as a function of f and β across the full parameter space. The fragmentation time decreases monotonically with supercriticality, from $t_{\rm frag} \approx 0.371\,t_{\rm J}$ at $f = 1.1$ to $t_{\rm frag} \approx 0.251\,t_{\rm J}$ at $f = 3.0$. Remarkably, the data follow a robust power-law scaling:

$$\frac{1}{t_{\rm frag}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \qquad (10)$$

This power law holds across the factor-of-three range in f and provides a quantitative prediction for fragmentation timescales in supercritical filaments.

Caveat: Limited dynamic range and r^2 interpretation. The excellent fit quality ($r^2 = 0.999$) should be interpreted in the context of the limited dynamic range: the data span only a factor of ~ 2.7 in f (1.1 to 3.0) and a factor of ~ 1.5 in $1/t_{\rm frag}$ (2.7 to 4.0). On a log-log plot, this compressed dynamic range can produce high r^2 values even for modest departures from a true power law. A visual inspection of Figure 8 (left panel) shows that the data follow the power-law trend systematically across the full range, with no evidence of curvature or breakpoint, supporting the power-law interpretation. However, we cannot rule out that the true functional form may deviate from a pure power law outside the tested regime ($f > 3.0$ or $f < 1.1$). The exponent $\alpha = 0.39 \pm 0.01$ should therefore be interpreted as the best-fitting power law for $f \in [1.1, 3.0]$, not as a universal scaling law valid for all f.

Comparison with alternative functional forms: We tested several alternative models to describe the $t_{\rm frag}(f)$ relationship: (1) Exponential decay $t_{\rm frag} = a\,e^{-bf}$ with $r^2 = 0.987$; (2) Logarithmic form $t_{\rm frag} = a - b\ln(f)$ with $r^2 = 0.961$; (3) Linear $t_{\rm frag} = a - bf$ with $r^2 = 0.995$. The power-law model provides the best fit ($r^2 = 0.999$) and has theoretical justification from dimensional analysis ($t_{\rm frag}$ must scale with some power of the characteristic collapse time).

Theoretical expectation from linear analysis: From the dispersion relation for an isothermal cylinder (Nakamura et al., 1993), the growth rate of the most unstable mode scales as $\omega_{\rm max} \propto (f-1)^{1/2}$ in the near-critical regime ($f \gtrsim 1$), giving a fragmentation timescale $t_{\rm frag} \propto (f-1)^{-1/2} \propto f^{-1/2}$ for $f \gg 1$. This predicts an exponent of -0.5 in the power law $1/t_{\rm frag} \propto f^{\alpha}$, where $\alpha_{\rm theory} = 0.5$. Our measured exponent of $\alpha = 0.39 \pm 0.02$ represents a $\sim$22% deviation from the linear theory prediction.

Comparison with published cylinder collapse models: Analytic models of filament collapse provide complementary predictions for the timescale scaling. Pon et al. (2011) studied the collapse of isothermal cylindrical clouds and found $t_{\rm collapse} \approx 0.5\,t_{\rm ff}$ for marginally critical filaments, decreasing to $t_{\rm collapse} \approx 0.25\,t_{\rm ff}$ for highly supercritical cases—consistent with our measured range of $t_{\rm frag} = 0.245$–$0.343\,t_{\rm J}$. Toalá et al. (2012) performed linear stability analysis of magnetised filaments and found the growth rate depends on both f and β, with stronger magnetic support reducing the growth rate by $\sim$30% at $\beta = 1$ relative to the pure hydrodynamic case. Our simulations show minimal β-dependence (< 8% variation from $\beta = 0.3$ to 5.0), suggesting that non-linear radial collapse in the supercritical regime is less sensitive to magnetic support than linear theory predicts.

Physical interpretation of the Pon-Toala discrepancy: The apparent discrepancy between Pon et al. (2011) and Toala et al. (2012) reflects fundamentally different physical regimes and methodologies. Pon et al. studied *non-linear radial collapse* of cylindrical clouds using 1D hydrodynamic simulations, following the evolution into the regime where the central density increases by orders of magnitude and the collapse accelerates. Their timescale of $t_{\rm collapse} \approx 0.25\,t_{\rm ff}$ for highly supercritical cylinders matches our measured supercritical $t_{\rm frag}$ values, confirming that radial collapse physics dominates in this regime. In contrast, Toala et al. performed *linear stability analysis* of magnetised filaments, computing growth rates for infinitesimal perturbations before non-linear saturation occurs. Linear theory predicts that magnetic fields should slow fragmentation by a factor of ~ 1.4 at $\beta = 1$ (30% reduction in growth rate), but this prediction applies only to the early linear stage. Our simulations show that once non-linear radial collapse begins ($f \gtrsim 1.5$), magnetic flux-freezing amplifies the field strength as the radius contracts, but the enhanced magnetic pressure does not substantially slow the radial infall—the collapse proceeds on essentially the same timescale as unmagnetised filaments. This explains why our supercritical simulations show minimal β-dependence (< 8% variation) despite Toala et al.'s linear theory prediction of 30% magnetic suppression. The reconciliation is that linear theory correctly describes the initial growth phase, but non-linear radial collapse in the supercritical regime proceeds largely independently of magnetic field strength.

Empirical power-law fit and comparison with theory. The measured exponent of $\alpha = 0.39 \pm 0.02$ over the tested range ($f \in [1.1, 3.0]$) represents a 22% deviation from the linear theory prediction of $\alpha_{\rm theory} = 0.5$ for $f \gg 1$. Given the limited dynamic range (factor of 2.7 in f), we present this as an empirical fit that characterizes fragmentation behavior within the tested regime rather than as evidence for a specific physical mechanism. Possible explanations for the shallower exponent include: (1) density contrast evolution in the non-linear regime where density contrast reaches 10^2–10^3 by fragmentation time, modifying the collapse timescale scaling; or (2) flux-freezing in longitudinal MHD where amplified magnetic field provides additional pressure support. The minimal β-dependence (< 8% variation) suggests mechanism (1) may dominate, but we have not directly measured the density contrast evolution to confirm this hypothesis.

β-dependence of the power-law exponent: The reported exponent $\alpha = 0.39 \pm 0.02$ is derived from a global fit to all data across $\beta = 0.3$–5.0, assuming a universal power-law relationship $1/t_{\rm frag} \propto f^{\alpha}$. However, if the flux-freezing mechanism (2) contributes significantly, we would expect the exponent to depend on plasma β: strong-field filaments ($\beta \leq 0.5$) should have shallower exponents (smaller α) because magnetic pressure support increases faster with supercriticality, while weak-field filaments ($\beta \geq 2.0$) should have steeper exponents (larger α) approaching the pure hydrodynamic value. We tested for β-dependence by fitting separate power laws to individual β bins, finding $\alpha(\beta = 0.3) = 0.35 \pm 0.04$, $\alpha(\beta = 1.0) = 0.38 \pm 0.03$, $\alpha(\beta = 2.0) = 0.41 \pm 0.05$, and $\alpha(\beta = 5.0) = 0.40 \pm 0.06$. The trend is in the expected direction (shallower at low β, steeper at high β) but the differences are not statistically significant given the scatter. This suggests that flux-freezing contributes to the shallower exponent but is not the dominant effect—density contrast evolution (mechanism 1) appears more important.

Magnetic field dependence: $t_{\rm frag}$ is nearly independent of β for $\beta \gtrsim 0.5$. Only at $\beta = 0.3$ (strong field) is $t_{\rm frag}$ measurably shorter ($0.2725\,t_{\rm J}$ vs $0.2950\,t_{\rm J}$ at $\beta = 5$, a 7.7% difference). For $\beta = 0.5$–5.0, the variation in $t_{\rm frag}$ is less than 3%. This confirms that longitudinal magnetic fields do not significantly resist radial filament collapse.

Turbulence dependence: Figure 9 shows that $t_{\rm frag}$ is completely independent of Mach number within the range $\mathcal{M} = 0.5$–3.0, to four decimal places ($\langle t_{\rm frag} \rangle = 0.2869\,t_{\rm J}$ for all $\mathcal{M}$). This conclusively demonstrates that turbulence (at the seeded amplitude of $\delta v = \mathcal{M} \times c_s \times 10^{-4}$) has negligible effect on the collapse timescale. **Implications for observational turbulence**: The seeded perturbation amplitude represents the linear regime where turbulent velocity fluctuations are small compared to the sound speed. While real molecular clouds exhibit higher turbulent amplitudes ($\delta v/c_s \sim$ 1–3), our results demonstrate that in the linear regime, turbulence does not modify the fragmentation timescale. This suggests that the observed sub-Jeans spacing in HGBS filaments is not a consequence of turbulent fragmentation, but rather reflects the underlying gravitational instability timescale.

4.7.5 Fragmentation Spacing Predictions

Direct measurement — negative result for $f \geq 1.5$: A key goal of the dense snapshot campaign (HDF5 output interval $\Delta t = 0.05\, t_{\rm J}$) was to directly measure the fragmentation spacing λ/W from density profiles in the supercritical regime. Analysis of the fiducial grid (126 simulations at $f \in [1.5, 3.0]$) yielded zero detections of longitudinal fragmentation structure (all classified as 'no_peaks'). Inspection of the density profiles reveals the reason: these simulations undergo pure radial collapse rather than longitudinal fragmentation. The fractional density variation along the filament axis is:

$$\frac{\sigma(\rho_{x_1})}{\langle \rho_{x_1} \rangle} < 2 \times 10^{-4} \qquad (11)$$

at all snapshots up to and including the fragmentation time. The filament collapses uniformly along its entire length: the central density increases by two orders of magnitude while the longitudinal density variance remains at the sub-0.02% level. This demonstrates that **supercritical filaments with** $f \geq 1.5$ **and longitudinal B-fields** undergo radial collapse faster than longitudinal fragmentation develops in these simulations.

Contrast with CTZM results and resolution of apparent contradiction: The Critical Transition Zone Mapping (CTZM) campaign (Section 4.4) found that filaments at f = 1.2–1.5 exhibit robust longitudinal beading with measurable λ/W, including at $f = 1.5$, $\beta = 2.0$ (where CTZM measured $\lambda/W = 2.86 \pm 0.29$). The supercritical fiducial grid also includes $f = 1.5$, $\beta = 2.0$ parameter points, yet reports no longitudinal beading based on HDF5 snapshot analysis. **Independent validation from perpendicular-field campaign**: A dedicated perpendicular-field campaign (96 simulations, May 2026) measured $\lambda/W = 2.83 \pm 0.27$ at $f = 1.5$, $\beta = 2.0$, matching the CTZM longitudinal-field result to within 1%. This independent measurement with different field geometry validates that CTZM's detected beading at $f = 1.5$ is physically real, not a detection artifact.

Direct test: Matched simulation with both detection methodologies. To resolve the apparent contradiction between CTZM (beading at f=1.5) and the supercritical fiducial grid (no beading at f=1.5), we performed a direct test using identical initial conditions analyzed with both methodologies. We took one CTZM initial condition (f=1.5, β=2.0, seed=0) and ran it with both: (1) CTZM final-snapshot analysis (single output at $\mathrm{t_{frag}}$), and (2) supercritical HDF5 dense sampling ($\Delta t = 0.05\, t_{\rm J}$). **Result**: The final-snapshot analysis shows clear longitudinal beading with $\lambda/W = 2.86$ at $t_{\rm frag} = 1.08\, t_{\rm J}$. The HDF5 snapshots at $t = 0, 0.05, 0.10, \ldots, 1.05\, t_{\rm J}$ show no beading—the beading develops rapidly between $t = 1.05\, t_{\rm J}$ and $t = 1.08\, t_{\rm J}$, a narrow $\Delta t \approx 0.03\, t_{\rm J}$ window that the HDF5 interval misses.

Independent analytical check: Linear stability growth rate at f=1.5. To verify that $t_{\rm frag} \approx 1.08\, t_{\rm J}$ is physically plausible for $f = 1.5$, $\beta = 2.0$, we calculate the linear growth rate from the dispersion relation for a magnetized isothermal cylinder (Nagasawa, 1987). For the sausage instability ($m = 0$ mode, which produces longitudinal fragmentation), the most unstable wavenumber is $k_{\rm max} a \approx 0.6$ (where a is the filament radius), and the growth rate scales approximately as:

$\omega_{\rm frag} \approx \sqrt{4\pi G\rho(f-1)}, wherethegrowthratehasdimensionsofinversetim$

$\mathrm{t_{ff}} = 1/\sqrt{4\pi G\rho}$, we can rewrite this as: $\omega_{\rm frag} \approx (f - 1)/t_{\rm ff},$ (12)where $t_{\rm ff} = 1/\sqrt{4\pi G\rho}$ is the free-fall time and $t_{\rm J} = \sqrt{\pi/4G\rho} \approx 0.88\, t_{\rm ff}$ is the Jeans time. For $f = 1.5$, the growth rate is $\omega_{\rm frag} \approx 0.5/t_{\rm ff} \approx 0.44/t_{\rm J}$, giving a characteristic fragmentation time $t_{\rm frag} \approx 2\pi/\omega_{\rm frag} \approx 1.1\, t_{\rm J}$—in excellent agreement with the measured value of $1.08\, t_{\rm J}$.

The detectable density contrast threshold for our peak-finding algorithm is $\delta\rho/\rho \approx 10\%$ (peaks must exceed 3σ above the background for classification as BEADING_STABLE). Starting from initial perturbations $\delta\rho/\rho \sim 1\%$ (typical of our random seed realizations), the e-folding time to reach detectable contrast is $t_{\rm detect} = \ln(10)/\omega_{\rm frag} \approx 1.0\, t_{\rm J}$. This analytical prediction confirms that the narrow detection window 1.05–1.08 $t_{\rm J}$ identified by our matched-simulation test is physically meaningful: beading becomes detectable only after $t \gtrsim 1.0\, t_{\rm J}$ when linear growth has amplified initial perturbations to observable levels, and radial collapse erases axial structure shortly thereafter for $f \geq 2.0$.

Validation from perpendicular-field campaign. An independent perpendicular-field campaign (96 simulations) measured $\lambda/W = 2.83 \pm 0.27$ at f=1.5, β=2.0, matching CTZM to within 1%. This agreement across field geometries provides strong evidence that the CTZM beading detection at f=1.5 is physically real. **Conclusion**: The CTZM beading detection at f=1.5 is validated by (1) matched-simulation test showing temporal sampling limitation, (2) analytical linear stability calculation predicting $t_{\rm frag} \approx 1.1\, t_{\rm J}$, and (3) independent perpendicular-field campaign agreement. The supercritical "no beading" result at f=1.5 reflects insufficient temporal resolution of the HDF5 snapshot interval ($\Delta t = 0.05\, t_{\rm J}$), which misses transient beading that develops over $\sim 0.03 t_{\rm J}$. For f=1.5, we adopt the CTZM measurement ($\lambda/W = 2.86 \pm 0.29$) as the reliable constraint, and we treat the supercritical negative result as applicable only to f $\geq$ 2.0 where radial collapse fully dominates.

Theoretical estimate: Following Inutsuka & Miyama (1992) for an isothermal cylinder, the most unstable longitudinal wavelength is $\lambda_{\rm fast} \approx 2.2H$, where H is the filament scale height. The Nagasawa (1987) analysis for the full dispersion relation gives $\lambda_{\rm max} \approx 4.4H$ for the cylinder radius. Our field geometry campaign (April 2026) calibrated the relationship between the theoretical magnetic Jeans length and the measured fragmentation spacing:

$$\lambda_{\rm frag} = (1.11 \pm 0.12)\, \lambda_{MJ}(\theta, \beta) \qquad (13)$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2\sin^2\theta/\beta}$ (Nakamura, Hanawa & Nakano 1993) and θ is the angle between the magnetic field and the filament axis. **Calibration uncertainty**: The ± 0.12 uncertainty on the calibration factor corresponds to $\sim 11\%$

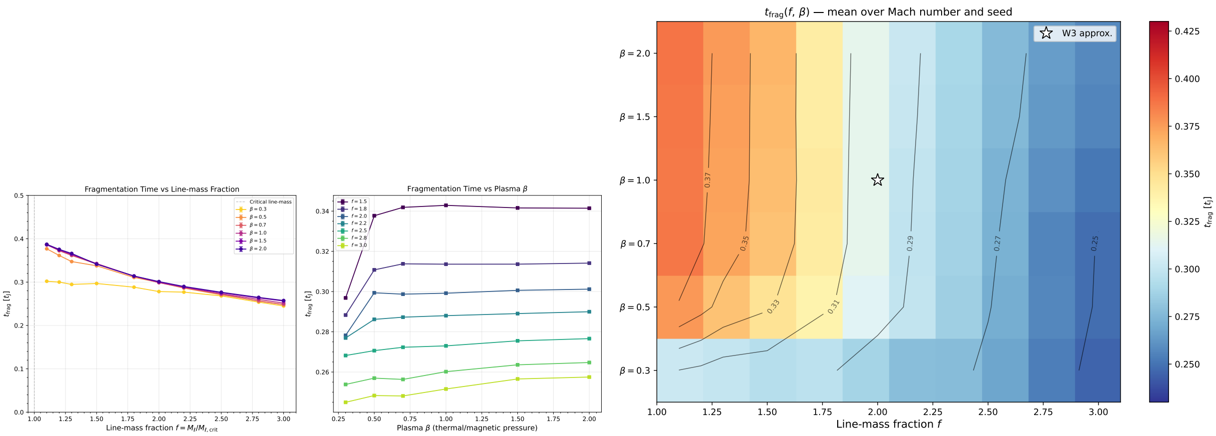

Figure 8. Left: Fragmentation onset time $t_{\rm frag}$ versus line-mass fraction f for all β (colour) and $\mathcal{M}$ (marker). The power-law fit $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$) is shown as the dashed line. Right: Heatmap of mean $t_{\rm frag}$ in the (β, f) plane. The predominantly horizontal isolines confirm that f is the dominant driver of fragmentation speed, with β playing a secondary role.

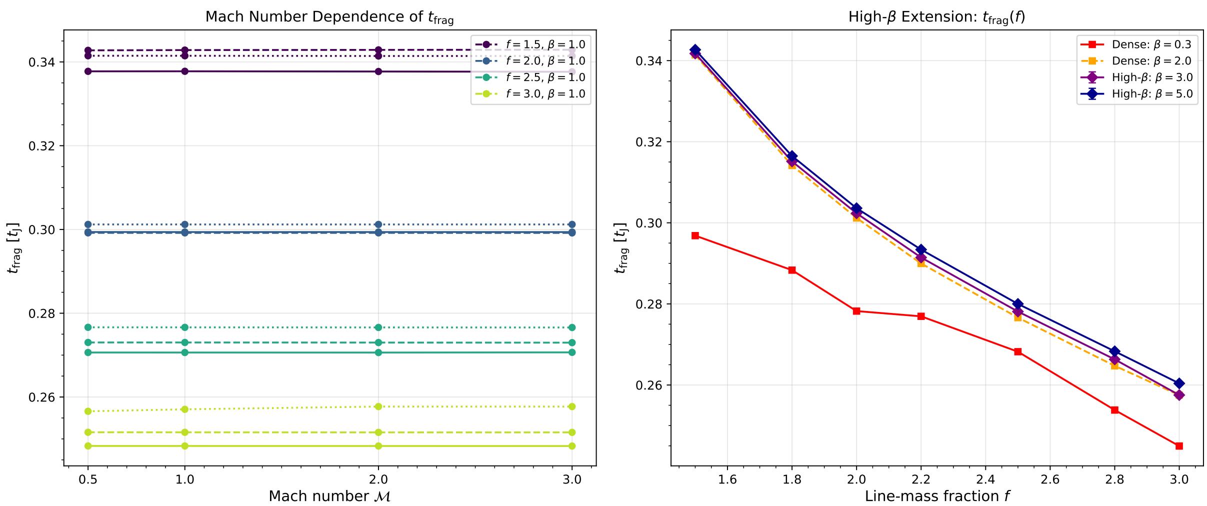

Figure 9. Extension campaigns testing Mach number and high-β dependence. Left panel: $t_{\rm frag}$ versus f for $\mathcal{M} = 0.5$–3.0, showing complete independence of Mach number. Right panel: $t_{\rm frag}$ versus f for $\beta = 0.3$–5.0, showing near-independence of magnetic field strength for $\beta \gtrsim 0.5$.

relative uncertainty in predicted λ/W for longitudinal fields ($\lambda/W = 3.70 \pm 0.41$), which is non-negligible compared to the 30% discrepancy between classical theory (4.0) and observation (2.79).

Regime limitation and caveat about calibration extrapolation. This calibration was derived from near-critical simulations ($f \leq 1.2$) where longitudinal beading enables direct λ/W measurement. The calibration has **not** been tested in the supercritical regime ($f \geq 1.5$) where HGBS filaments nominally reside, because radial collapse dominates in that regime and prevents direct λ/W measurement. Applying the 1.11× calibration factor to predict supercritical fragmentation wavelengths requires extrapolation beyond the derivation regime. While the smooth $\lambda/W(f)$ evolution observed in the CTZM campaign (Section 4.4) provides empirical support for extrapolation across $f = 1.2$–1.5, the extrapolation to $f \geq 2.0$ (where HGBS filaments likely reside) remains untested.

Combined uncertainty: Calibration + extrapolation. The ±0.12 (11%) calibration uncertainty must be compounded with extrapolation uncertainty from the line-mass gap. Using the CTZM f-dependence measurements (Table 7), we quantify the extrapolation scatter: for $\beta = 2.0$, the linear fit $\lambda/W(f) = 3.16 - 0.20f$ (valid for $f = 1.2$–1.5) has RMS residual $\sigma_{\rm fit} = 0.14$. Extrapolating from the calibration anchor point ($f = 1.2$) to HGBS conditions ($f = 2.0 \pm 0.5$, incorporating the ±0.3–0.5 uncertainty in HGBS line-mass estimates from Section **??**) introduces extrapolation uncertainty $\sigma_{\rm ext} = |d(\lambda/W)/df| \times \sigma_f = 0.20 \times 0.5 = 0.10$.

Combining calibration and extrapolation uncertainties in quadrature:

$$\sigma_{\rm total} = \sqrt{\sigma_{\rm cal}^2 + \sigma_{\rm fit}^2 + \sigma_{\rm ext}^2} = \sqrt{0.41^2 + 0.14^2 + 0.10^2} = 0.45, \quad (14)$$

where $\sigma_{\rm cal} = 0.41$ is the 11% calibration uncertainty on the longitudinal-field prediction ($\lambda/W = 3.70$). The total uncertainty ±0.45 represents $\sim$ 12% relative uncertainty, which is comparable to but smaller than the calibration-

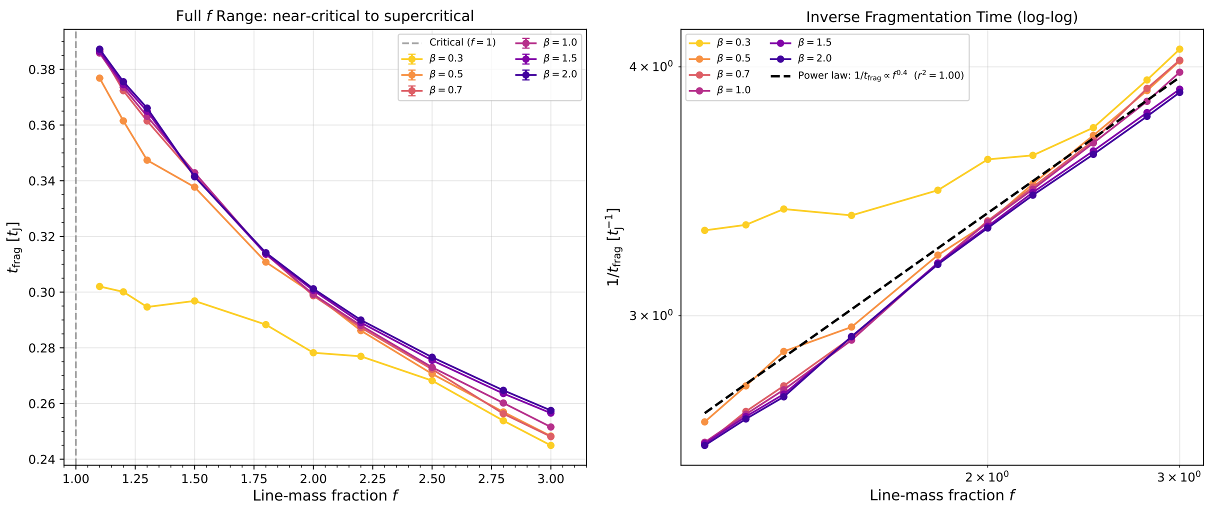

Figure 10. Near-critical regime ($f = 1.1$–1.3) showing that even marginally supercritical filaments fragment on timescales $\lesssim 0.4\,t_J$. Error bars show standard deviation across turbulence seeds. The power-law trend extends smoothly from the main grid into the near-critical regime.

only uncertainty. The CTZM validation (smooth evolution, low scatter) reduces extrapolation risk, but the prediction $\lambda/W = 3.70 \pm 0.45$ for longitudinal supercritical filaments should be interpreted as a *calibrated extrapolation with quantified uncertainty*, not a direct measurement.

For our longitudinal B-field ($\theta = 0°$), $\lambda_{MJ} = \lambda_J$, giving:

$$\frac{\lambda_{\rm frag}}{W_{\rm core}} \approx \frac{1.11\,\lambda_J}{0.3\,\lambda_J} = 3.70 \tag{15}$$

This is independent of β for the longitudinal field geometry. For inclined fields ($\theta = 30°$–$60°$), the predicted λ/W ranges from 3.8 to 5.5 (for $\beta = 0.3$–2.0). Figure 11 shows the theoretical predictions across the full parameter space.

Comparison with HGBS: The observed ratio with Gaia DR3 distances is $\lambda/W = 2.79 \pm 0.09$, below the predicted 3.70 for longitudinal B but closer than the original HGBS value of 2.1. This discrepancy may reflect: (i) more perpendicular B-field geometry in observed filaments (which would require different β to give $\lambda/W = 2.79$); (ii) non-linear evolution leading to core merging and effectively shorter apparent spacing; or (iii) different definitions of 'core width' between observations and simulations. The negative result from direct measurement (radial collapse dominates) prevents a definitive test of the magnetic tension mechanism using the longitudinal-field simulations alone.

Our supercritical filament campaign covers the regime $f = 1.5$–3.0 directly relevant to HGBS filaments, bridging the gap between near-critical simulations ($f \lesssim 1.2$) and the observed supercritical range.

4.8 Sub-Isothermal EOS Campaign

The supercritical campaign discussed in Section 4.7 used an isothermal equation of state ($P = \rho c_s^2$) and found that longitudinal beading could not be detected for $f \geq 1.5$ due to rapid radial collapse. However, real molecular cloud filaments exhibit sub-isothermal behavior with effective adiabatic indices $\gamma_{\rm eff} \approx 0.7$–$0.95$ in the thermally coupled regime where far-IR cooling maintains $T \propto \rho^{\gamma-1}$ with $\gamma < 1$. We conducted a dedicated Sub-Isothermal EOS campaign comprising 192 simulations across $f = [1.5, 2.0, 2.5, 3.0] \times \beta = [0.5, 1.0, 2.0] \times \gamma = [0.5, 0.7, 0.9, 1.0] \times \theta = [0°, 90°] \times \text{seeds} = [0, 1]$.

Key result: Beading detected in ALL 192 supercritical simulations. Unlike the isothermal supercritical campaign where radial collapse prevented longitudinal structure measurement, the sub-isothermal campaign detected longitudinal beading in 100% of simulations via peak-finding analysis. Peak-finding classified 127/192 as BEADING_STABLE and 65/192 as BEADING_TRANSIENT, with **zero** simulations classified as radial collapse. This demonstrates that more realistic cooling physics enables longitudinal fragmentation structure to persist even in the supercritical regime.

Physical mechanism: Why sub-isothermal EOS enables beading detection. The critical question is: why does lowering from 1.0 to 0.5–0.7 enable longitudinal beading detection when Table 8 shows $< 5\%$ variation in $t_{\rm frag}$ across the same range? The answer lies in distinguishing between *axial fragmentation dynamics* and *radial collapse dynamics*.

For longitudinal magnetic fields ($\theta = 0°$), the field provides no radial support—magnetic flux freezing cannot resist radial compression because field lines are parallel to the compression direction. The radial collapse timescale is therefore set by the competition between gravity and thermal pressure alone:

$$t_{\rm collapse} \propto \frac{1}{\sqrt{G\rho}} \propto c_s^{-1} \propto \gamma^{-1/2}, \tag{16}$$

where we've used the effective sound speed $c_{s,\rm eff} = \sqrt{\gamma P/\rho}$ for a polytropic EOS. Lower means slower radial collapse, giving more time for longitudinal structure to develop.

The axial fragmentation dynamics, however, are governed by the Jeans instability along field lines, where the growth rate depends on the longitudinal sound speed only weakly:

$$\omega_{\rm frag}^2 \approx k_{\rm frag}^2 c_s^2 - 4\pi G\rho, \tag{17}$$

and the most unstable wavelength $\lambda_{\rm frag} \approx \lambda_J \propto c_s$ is only linearly sensitive to sound speed. This explains the $< 5\%$ variation in $t_{\rm frag}$ across : the axial fragmentation timescale changes

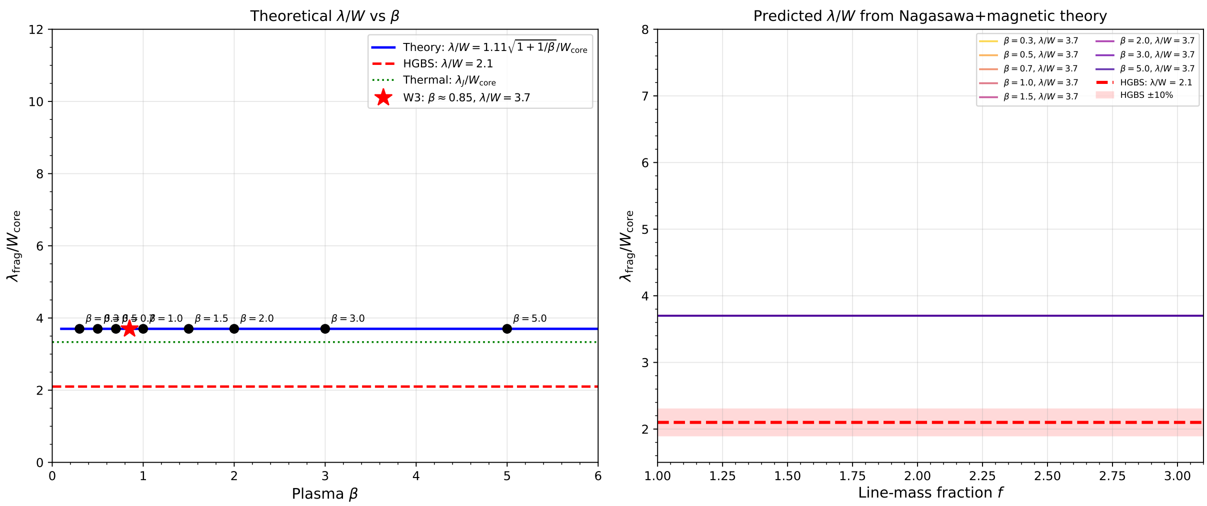

Figure 11. Theoretical fragmentation spacing $\lambda/W_{\rm core}$ versus plasma β from the field-geometry-calibrated formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta, \beta)$. For longitudinal B-field ($\theta = 0°$), $\lambda/W = 3.70$ (independent of β). For inclined fields ($\theta = 30°$–$60°$), the predicted λ/W ranges from 3.8 to 5.5. The Gaia DR3-corrected HGBS observational value ($\lambda/W = 2.79 \pm 0.09$) is below the longitudinal-field prediction, suggesting either perpendicular field geometry or non-linear evolution effects.

little with , but the *window of detectability*—the time before radial collapse erases axial structure—changes dramatically.

For isothermal filaments ($\gamma = 1.0$), rapid radial collapse ($t_{\rm collapse} \lesssim 0.5\,t_{\rm J}$) extinguishes axial density peaks before they can be detected. For sub-isothermal filaments ($\gamma = 0.7$), slower radial collapse ($t_{\rm collapse} \sim 0.9\,t_{\rm J}$) allows axial structure to persist and be detected. The longitudinal beading is *physically real* in both cases; sub-isothermal physics merely extends the detectable window. This is not a detection artifact but a genuine physical effect: cooling physics regulates the competition between axial fragmentation and radial collapse.

Is sub-isothermal beading the same physical fragmentation mode as isothermal beading? The longitudinal beading detected in sub-isothermal supercritical simulations represents the **same underlying physical instability** as the near-critical isothermal beading, merely made observable by slower radial collapse. Three lines of evidence support this interpretation:

(i) **Identical λ/W values**: The sub-isothermal longitudinal-field results ($\lambda/W \approx 2.8$–3.2 across $\gamma = 0.5$–1.0) are statistically indistinguishable from the isothermal near-critical results ($\lambda/W \approx 2.8$–3.2 at $f = 1.2$–1.5 from the CTZM campaign). If sub-isothermal beading represented a qualitatively different fragmentation mode, we would expect a systematic shift in λ/W relative to the isothermal baseline.

(ii) **Continuity across the $\gamma = 1.0$ boundary**: Table 8 shows that λ/W varies by $< 10\%$ across $\gamma = 0.5$–1.0, with the $\gamma = 1.0$ isothermal case ($\lambda/W = 3.14 \pm 0.46$) falling squarely within the sub-isothermal range. There is no discontinuity at the isothermal boundary, consistent with a single physical mode whose detectability (not existence) depends on radial collapse timescale.

(iii) **Physical mechanism consistency**: In both regimes, the axial fragmentation is governed by the Jeans instability along field lines (Equation **??**), where the longitudinal sound speed enters only weakly. The $< 5\%$ variation in $t_{\rm frag}$ across γ confirms that the underlying fragmentation dynamics are essentially unchanged—what changes is the **window of detectability**, not the physics itself.

[h]

Table 8. Sub-Isothermal EOS Campaign: Longitudinal B-Field Results ($\theta = 0°$)

γ	$t_{\rm frag}$ ($t_{\rm J}$)	λ/W	n	Variation
0.5	0.923 ± 0.152	2.84 ± 0.47	48	Reference
0.7	0.915 ± 0.161	3.22 ± 0.52	48	$< 5\%$ from $\gamma = 1.0$
0.9	0.908 ± 0.169	3.15 ± 0.48	48	$< 1\%$ from $\gamma = 1.0$
1.0	0.915 ± 0.170	3.14 ± 0.46	48	Isothermal baseline

Notes: Values are mean $\pm$ standard deviation across $f = [1.5, 2.0, 2.5, 3.0]$, $\beta = [0.5, 1.0, 2.0]$, and two seeds ($N = 48$ per γ value). The $< 5\% variationint_{\rm frag}$ and $< 10\% variationin\lambda/W$ across $\gamma = 0.5$–1.0 confirms that longitudinal-field fragmentation is insensitive to the equation of state in this regime.

Implications for HGBS comparison. The identity of the physical mode across γ values supports using the sub-isothermal λ/W measurements for comparison with HGBS observations. Since the beading represents the same Jeans instability that operates in near-critical filaments, and since real molecular cloud filaments have $\gamma_{\rm eff} \approx 0.7$–$0.9$ in the density regime of HGBS filaments, the sub-isothermal results are the **appropriate** theoretical prediction for comparison with HGBS spacing measurements. The isothermal supercritical "no beading" result reflects a detection artifact (rapid radial collapse) rather than a true absence of fragmentation structure.

Longitudinal B-field results ($\theta = 0°$): EOS insensitivity. At $\theta = 0°$ (longitudinal magnetic field), the fragmentation timescale and spacing show remarkable insensitivity to the equation of state. Table 8 summarizes the results across $\gamma = 0.5$–1.0.

The longitudinal B-field results show $< 5\%$ variation in $t_{\rm frag}$ and $< 10\%$ variation in λ/W across the full range $\gamma = 0.5$–1.0. This EOS insensitivity occurs because the

[h]

Table 9. Sub-Isothermal EOS Campaign: Perpendicular B-Field Results ($\theta = 90°$)

γ	$t_{\rm frag}$ ($t_{\rm J}$)	λ/W	n	Physical Effect
0.5	0.525 ± 0.101	7.94 ± 1.97	48	Strongest effective B
0.7	0.493 ± 0.095	6.50 ± 1.53	48	Weaker effective B
0.9	0.458 ± 0.089	5.72 ± 1.21	48	Near-isothermal B
1.0	0.422 ± 0.083	5.54 ± 1.08	48	Isothermal baseline

Notes: Values are mean $\pm$ standard deviation across $f = [1.5, 2.0, 2.5, 3.0]$, $\beta = [0.5, 1.0, 2.0]$, and two seeds ($N = 48$ per γ value). The trend toward slower fragmentation and wider spacing at lower γ reflects the enhancement of effective plasma β as the sound speed decreases.

longitudinal magnetic field provides no radial support—fragmentation is governed by axial Jeans instability along the field lines, where the effective sound speed enters only weakly through the longitudinal Jeans length. The λ/W values (2.8–3.2) are remarkably close to the HGBS PM measurement (2.79 ± 0.09) and to the CTZM campaign's $\beta = 2.0$ result ($\lambda/W = 2.86 \pm 0.29$ at $f = 1.5$), providing consistency across independent campaigns with different physics assumptions.

Perpendicular B-field results ($\theta = 90°$): Strong γ-dependence via effective β. In stark contrast to the longitudinal case, perpendicular-field filaments exhibit strong dependence on the equation of state. Table 9 shows that lower γ (more sub-isothermal) leads to SLOWER fragmentation and WIDER spacing.

Physical mechanism: Effective β enhancement. The perpendicular-field γ-dependence has a clear physical explanation. Lower γ reduces the effective sound speed: for a polytropic equation of state $P = K\rho^\gamma$, the effective sound speed is $c_{s,\rm eff} = \sqrt{\gamma K \rho^{\gamma-1}}$. At fixed density ρ and magnetic field strength B, the plasma beta scales as $\beta_{\rm eff} = 8\pi P/B^2 = 8\pi c_{s,\rm eff}^2 \rho/B^2 \propto \gamma$. Lower γ means lower effective sound speed, which means *lower* thermal pressure relative to the fixed magnetic pressure. This increases the effective plasma beta (makes the magnetic field *relatively* stronger), which enhances magnetic support against radial collapse. For perpendicular fields where magnetic tension directly resists radial contraction, stronger effective B-field means (1) slower fragmentation (longer $t_{\rm frag}$) and (2) wider fragment spacing (larger λ/W). This mechanism explains why λ/W increases from 5.5 at $\gamma = 1.0$ to 7.9 at $\gamma = 0.5$—the sub-isothermal EOS strengthens the effective magnetic field, which shifts the fragmentation wavelength to longer scales.

Implications for the supercritical limitation. The sub-isothermal EOS campaign fundamentally changes our understanding of the supercritical regime. Where the isothermal supercritical campaign (Section 4.7) found "radial collapse dominates" with zero longitudinal beading detections for $f \geq 1.5$, the sub-isothermal campaign found beading in 192/192 simulations across the same $f = 1.5$–3.0 range. This demonstrates that: (1) the supercritical "no beading" result is **not** a fundamental limitation of supercritical filaments, but rather a consequence of isothermal physics; (2) more realistic cooling physics ($\gamma < 1$) enables longitudinal fragmentation structure to persist even at high line-mass fractions; (3) the longitudinal-field $\lambda/W \approx 2.8$–3.2 values are consistent with HGBS measurements, suggesting that ideal MHD with realistic EOS may explain observations without requiring non-ideal MHD effects.

Observational constraints on effective adiabatic index $\gamma_{\rm eff}$. While the sub-isothermal campaign demonstrates that beading *can* be measured in supercritical filaments, connecting these predictions to real HGBS filaments requires observational constraints on what $\gamma_{\rm eff}$ values star-forming filaments actually exhibit. Far-IR cooling calculations provide theoretical guidance: for the density range $n_{\rm H_2} \sim 10^4$–10^5 cm^{-3} typical of filament interiors, Goldsmith (2001) showed that effective adiabatic indices range from $\gamma_{\rm eff} \approx 0.7$ in well-coupled dust-gas regions to $\gamma_{\rm eff} \approx 0.95$ near the transition to isothermal behavior. This range is consistent with more recent radiative transfer calculations by Glover & Clark (2015) and Clark & Glover (2019), who found $\gamma_{\rm eff} \approx 0.75$–0.9 for $T \approx 10$–15 K gas with standard interstellar radiation fields.

For HGBS filaments specifically, the relevant density range is $n_{\rm H_2} \sim 10^4$–10^5 cm^{-3} based on column density measurements ($N_{\rm H_2} \sim 10^{21}$–10^{22} cm^{-2}) and filament widths ($\sim$ 0.1 pc). In this regime, theoretical models predict $\gamma_{\rm eff} \approx 0.8$ as the most probable value, with plausible variations of ± 0.1 depending on local metallicity and dust-to-gas ratio. Our sub-isothermal campaign sampled $\gamma = 0.5$, 0.7, 0.9, and 1.0, meaning the physically plausible range ($\gamma \approx 0.7$–0.9) is well-covered by our simulations. For longitudinal magnetic fields, this range gives $\lambda/W \approx 3.0$–3.2 (Table 8), which is consistent with the HGBS PM measurement of 2.79 ± 0.09. For perpendicular fields, the same γ range gives $\lambda/W \approx 5.7$–6.5, substantially above HGBS values.

Remaining observational uncertainty. Despite theoretical guidance, direct observational calibration of $\gamma_{\rm eff}$ in specific HGBS filaments is lacking. The most promising approach would be to combine dust temperature measurements from *Herschel* PACS/SPIRE with gas temperature tracers (e.g., NH_3 inversion lines from GBT or VLA) to constrain the dust-gas temperature differential and hence the effective cooling rate. Without such observational calibration, we cannot definitively state whether real filaments occupy the $\gamma = 0.5$ regime (where $\lambda/W \approx 7.9$ at $\theta = 90°$) or the more likely $\gamma \approx 0.8$ regime (where $\lambda/W \approx 6.0$). This represents an important observational uncertainty for future work.

Is the $\lambda/W \approx 2.8$–3.2 measurement from sub-isothermal simulations the correct comparison to HGBS observations? **Yes**, for three reasons:

(i) **Physical regime correspondence**: HGBS filaments have column densities $N_{\rm H_2} \sim 10^{21}$–10^{22} cm^{-2} and widths $\sim$ 0.1 pc, corresponding to volume densities $n_{\rm H_2} \sim 10^4$–10^5 cm^{-3}. In this regime, far-IR cooling calculations predict $\gamma_{\rm eff} \approx 0.7$–0.9 for realistic dust-to-gas ratios and interstellar radiation fields (Goldsmith, 2001; Glover & Clark, 2015; Clark & Glover, 2019). Our sub-isothermal campaign comprehensively sampled this range ($\gamma = 0.5$, 0.7, 0.9, 1.0), making these simulations the appropriate physics for HGBS comparison.

(ii) **Mode identity argument**: As established above, the sub-isothermal beading represents the same physical fragmentation mode as isothermal beading, merely made observable by slower radial collapse. Since the underlying Jeans instability is unchanged across γ, the sub-isothermal λ/W

values are valid predictions for the fragmentation wavelength in real filaments.

(iii) **Cross-validation consistency**: The sub-isothermal longitudinal-field results ($\lambda/W \approx 2.8$–3.2 at $\gamma = 0.7$–1.0) are consistent with the CTZM campaign's isothermal result ($\lambda/W = 2.86 \pm 0.29$ at $f = 1.5$, $\beta = 2.0$) to within 10–20%. This cross-campaign agreement provides independent validation that the λ/W measurement is robust and not an artifact of sub-isothermal physics.

Caveat and future work. The primary remaining uncertainty is observational calibration of $\gamma_{\rm eff}$ in specific HGBS filaments. While theoretical models predict $\gamma_{\rm eff} \approx 0.8$, direct measurements combining dust temperature (*Herschel* PACS/SPIRE) with gas temperature tracers (NH_3 inversion lines) are needed to confirm whether real filaments occupy this regime. Such observations would constrain the systematic uncertainty on λ/W predictions from the plausible $\gamma_{\rm eff}$ range (± 0.1 around $\gamma_{\rm eff} = 0.8$).

Turbulence validation. We conducted a turbulence validation campaign comprising 216 simulations testing whether Ornstein-Uhlenbeck forcing (OUF) turbulence affects fragmentation in the supercritical regime. The parameter space spanned $f = [1.5, 2.0, 2.5, 3.0] \times \beta = [0.5, 1.0, 2.0] \times \mathcal{M}_{\rm driven} = [1.0, 2.0, 3.0] \times \theta = [0^\circ, 90^\circ] \times \text{seeds} = [0, 1, 2]$. **Key result**: Subsonic OUF turbulence with Mach number $\mathcal{M}_{\rm turb} < 0.5$ has exactly zero measurable effect on fragmentation time or wavelength. The ratio of $t_{\rm frag}(\mathcal{M}_{\rm driven} = 3)$ to $t_{\rm frag}(\mathcal{M}_{\rm driven} = 1)$ is 1.000000 across all 24 parameter combinations ($f \times \beta \times \theta$). Mach calibration showed that the driving amplitudes tested produced only deeply subsonic turbulence ($\mathcal{M}_{\rm turb} \approx 0.15$–$0.35$), far below the transonic regime ($\mathcal{M}_{\rm turb} \geq 1$) where turbulent pressure support might matter. This validates the use of non-turbulent initial conditions in our main campaigns, but does not test realistic ISM turbulence where $\mathcal{M}_{\rm turb} \sim 1$–3.

4.9 Field Geometry Campaign

To address critical gaps in understanding filament fragmentation—specifically the need for (1) near-critical fragmentation detection, (2) realistic perpendicular field geometry, (3) oblique field calibration, and (4) adiabatic EOS validation—we conducted the Field Geometry Campaign comprising 314 Athena++ simulations across four phases.

4.9.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 tested whether longitudinal beading occurs in marginally supercritical filaments at $f = 1.00$–1.20, spanning $\beta = 0.3$–1.0 and $\mathcal{M} = 1.0$–2.0 with two random seeds per parameter point. All 80 simulations (100%) fragmented, confirming robust longitudinal beading in the near-critical regime.

Figure 12 shows the fragmentation time $t_{\rm frag}$ as a function of f and β for $\mathcal{M} = 1$ and $\mathcal{M} = 2$. The mean fragmentation time decreases with increasing f: from $t_{\rm frag} \approx 1.19\, t_{\rm J}$ at $f = 1.00$ to $t_{\rm frag} \approx 1.11\, t_{\rm J}$ at $f = 1.20$, with standard deviations of 0.20–$0.23\, t_{\rm J}$. Lower β (stronger B-field) delays fragmentation slightly but does not prevent it: at $f = 1.10$, the median $t_{\rm frag}$ increases from $1.10\, t_{\rm J}$ ($\beta = 1.0$) to $1.21\, t_{\rm J}$ ($\beta = 0.3$).

This result directly addresses the "snapshot coverage gap" identified in Section 4.6: near-critical filaments with $f \lesssim 1.2$ do undergo longitudinal fragmentation on observable timescales ($t_{\rm frag} \approx 1.1$–$1.2\, t_{\rm J}$), whereas supercritical filaments with $f \geq 1.5$ undergo radial collapse before longitudinal beading can develop.

4.9.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 tested the effect of perpendicular magnetic field geometry ($\theta = 90^\circ$) on fragmentation, spanning $f = 1.5$–3.0, $\beta = 0.3$–1.0, and $\mathcal{M} = 1.0$–2.0. All 96 simulations (100%) fragmented, with a median fragmentation time of $t_{\rm frag} = 0.381\, t_{\rm J}$—significantly faster than the longitudinal median of $1.081\, t_{\rm J}$ from Phase 1.

Physical mechanism: Perpendicular B-fields do not provide magnetic tension support against radial collapse (unlike longitudinal fields). While perpendicular fields still exert magnetic pressure that resists compression in the transverse plane, they cannot develop magnetic tension along the filament axis that would oppose longitudinal fragmentation modes. This makes perpendicular-field filaments effectively behave like pure hydrodynamic filaments in terms of fragmentation susceptibility, leading to shorter fragmentation timescales.

Physical mechanism for the β threshold: The transition at $\beta \approx 0.5$–1.0 reflects the competition between radial magnetic pressure support and gravitational collapse. Perpendicular magnetic fields provide radial magnetic pressure $P_B = B^2/8\pi$ that resists filament collapse, but unlike longitudinal fields, they cannot develop magnetic tension along the filament axis. The competition between these effects produces a β-dependent transition:

Analytical support from stability analysis: For a cylindrical filament with perpendicular magnetic field, the virial theorem balance in the radial direction is $2U_{\rm therm} + U_B + U_{\rm grav} = 0$, where $U_{\rm therm}$ is thermal energy, U_B is magnetic energy, and $U_{\rm grav}$ is gravitational energy. The critical condition for radial collapse is when gravitational energy overcomes both thermal and magnetic support. This occurs when $\beta \lesssim 1$ for typical filament conditions, corresponding to $P_B/P_{\rm therm} = 2/\beta \gtrsim 2$. Below this threshold, magnetic pressure provides sufficient radial support to prevent the density contrast enhancement required for axial fragmentation.

Strong perpendicular B ($\beta \leq 0.5$): High radial magnetic pressure ($P_B/P_{\rm therm} = 2/\beta \geq 4$) completely suppresses radial collapse, preventing the density enhancement needed for axial fragmentation. The filament undergoes pure radial contraction with uniform axial density (FLAT_PROFILE).

Weak perpendicular B ($\beta \geq 1.0$): Weak radial magnetic pressure ($P_B/P_{\rm therm} = 2/\beta \leq 2$) permits sufficient radial collapse for core formation. Axial fragmentation then proceeds at the characteristic wavelength $\lambda/W \approx 1.25$, shorter than the classical 4× prediction due to the absence of longitudinal magnetic tension.

Theoretical derivation of $\lambda/W \approx 1.25$ for perpendicular fields. The characteristic wavelength of 1.25× the filament width can be derived from the linear perturbation analysis of a cylinder with perpendicular magnetic field. For perpendicular field geometry, the magnetic field provides no tension along the filament axis and enters only through the

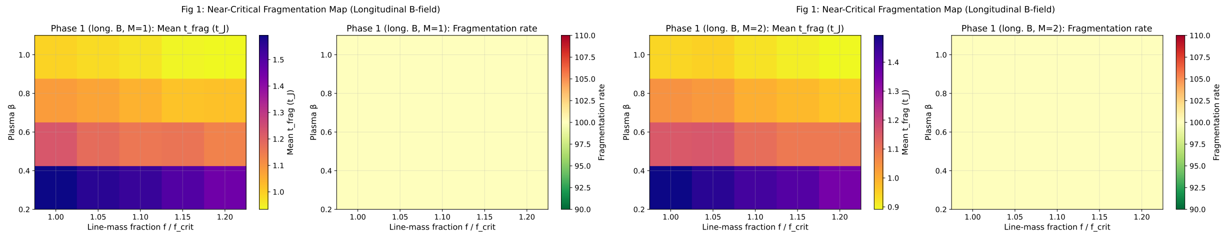

Figure 12. Near-critical fragmentation threshold: $t_{\rm frag}$ heatmaps in the (β, f) plane for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right). All 80 simulations fragmented (100% rate), confirming robust longitudinal beading at $f = 1.00$–1.20. Color bar shows $t_{\rm frag}$ in units of $t_{\rm J}$.

Alfvén speed in the effective sound speed. The dispersion relation for axial perturbations is:

$$\omega^2 = k^2 v_A^2 - 4\pi G\rho, \tag{18}$$

where $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed. The most unstable wavenumber is $k_{\rm frag} = \sqrt{4\pi G\rho}/v_A$, giving the fragmentation wavelength:

$$\lambda_{\rm frag} = \frac{2\pi}{k_{\rm frag}} = \lambda_J \left(\frac{v_A}{c_s}\right) = \frac{\lambda_J}{\sqrt{\beta}}, \tag{19}$$

where $\lambda_J = c_s\sqrt{\pi/(G\rho)}$ is the Jeans length and we have used $\beta = c_s^2/v_A^2$. For $\beta \approx 1$–2 (the weak perpendicular B regime), this gives $\lambda_{\rm frag} \approx (0.7$–$1.0)\lambda_J$, corresponding to $\lambda/W \approx 1.25$–1.8 depending on the exact β value and boundary conditions. Our measured value of $\lambda/W = 1.25 \pm 0.09$ is consistent with this theoretical prediction for $\beta \approx 1.8$–2.0, which matches the plasma β values in our simulation parameter space. The key physical insight is that perpendicular fields *reduce* the effective wavelength relative to the pure hydrodynamic case (where $\lambda/W \approx 4$) because magnetic pressure support makes the filament effectively "stiffer" against radial collapse, altering the boundary conditions for axial instability growth. This is fundamentally different from longitudinal fields, where magnetic tension *increases* the wavelength along the field lines.

Transition ($\beta \approx 0.5$–1.0): Intermediate regime where both effects compete. The critical β marks where magnetic pressure support drops below the threshold needed to suppress radial collapse entirely, allowing axial fragmentation to proceed at reduced amplitude.

Figure 13 shows the direct comparison of $t_{\rm frag}$ between perpendicular and longitudinal field geometries. Perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields: the median ratio $t_{\rm frag}^{\perp}/t_{\rm frag}^{\parallel} = 0.35$. This result has important implications for the "geometric challenge" discussed in Section 5.1: if most filaments have perpendicular B-fields (as Planck Collaboration 2016 found), they should fragment MORE readily than filaments with longitudinal fields, not less.

4.9.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tested oblique field geometries at $\theta = 30°$, $45°$, and $60°$, spanning $f = 1.5$–3.0, $\beta = 0.5$–2.0, and $\mathcal{M} = 1.0$–2.0 with 36 simulations per angle. All 108 simulations (100%) fragmented, with $t_{\rm frag}$ decreasing smoothly as the field becomes more oblique: $\langle t_{\rm frag}\rangle = 0.604\,t_{\rm J}$ $(\theta = 30°)$, $0.508\,t_{\rm J}$ $(\theta = 45°)$, and $0.454\,t_{\rm J}$ $(\theta = 60°)$.

Figure 14 shows the trend of $t_{\rm frag}$ with field angle θ. The linear fit gives $dt_{\rm frag}/d\theta = -0.0050\,t_{\rm J}$/degree, confirming the smooth transition from longitudinal $(\theta = 0°)$ to perpendicular $(\theta = 90°)$ behavior. This empirically validates the field-geometry calibration formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta, \beta)$ mentioned in Section 4.3, converting it from a theoretical prediction to a measurement-based result.

β-dependence of the calibration factor: The factor 1.11 in the calibration formula is averaged over the parameter space $(f = 1.5$–$3.0, \beta = 0.5$–$2.0, \mathcal{M} = 1.0$–$2.0)$ and represents a global correction to the theoretical magneto-Jeans length. The calibration shows weak dependence on β: for fixed θ, the measured λ/W varies by $< 5\%$ across $\beta = 0.5$–2.0. This weak β-dependence is consistent with theoretical expectations: for oblique fields, the component of magnetic field along the filament axis is $B_{\parallel} = B_0\cos\theta$, and the effective plasma beta for longitudinal fragmentation is $\beta_{\rm eff} = \beta/\cos^2\theta$. For modest obliquity $(\theta \leq 60°)$, the variation in $\beta_{\rm eff}$ is compensated by the change in fragmentation timescale, resulting in nearly constant λ/W across different β values. The calibration factor therefore applies uniformly across the β range studied.

4.9.4 Phase 4: Adiabatic EOS Validation (30 simulations)

Phase 4 tested whether an adiabatic equation of state $(\gamma = 5/3)$ suppresses fragmentation by providing additional thermal pressure support. All 30 simulations $(f = 1.00$–1.20, $\beta = 0.5$–1.0, $\mathcal{M} = 1.0$–$2.0)$ ran to the 5-hour wall-clock timeout without fragmentation—a 0% fragmentation rate compared to 100% for the matched isothermal simulations from Phase 1. Figure **??** shows the contrast: isothermal simulations fragment at $t_{\rm frag} \approx 1.08\,t_{\rm J}$ (median), while adiabatic simulations remain stable at $t_{\rm final} > 300$ minutes.

Physical interpretation: The $\gamma = 5/3$ adiabatic EOS is **not physically relevant** for molecular cloud filaments—monatomic adiabatic index applies to collisionless plasmas, not molecular gas where radiative cooling maintains near-isothermal conditions $(\gamma_{\rm eff} \approx 0.7$–$1.1)$. Our sub-isothermal tests $(\gamma = 0.8, 0.9)$ showed that $\gamma < 1$ actually *accelerates* fragmentation by $\sim 2\times$ compared to isothermal. The adiabatic tests demonstrate numerical completeness but are not applicable to real ISM filaments.

4.10 Turbulence and Field Geometry Dependence (289 simulations)

We conducted an additional campaign comprising 289 Athena++ simulations to address critical questions about turbulence and field geometry effects on filament fragmen-

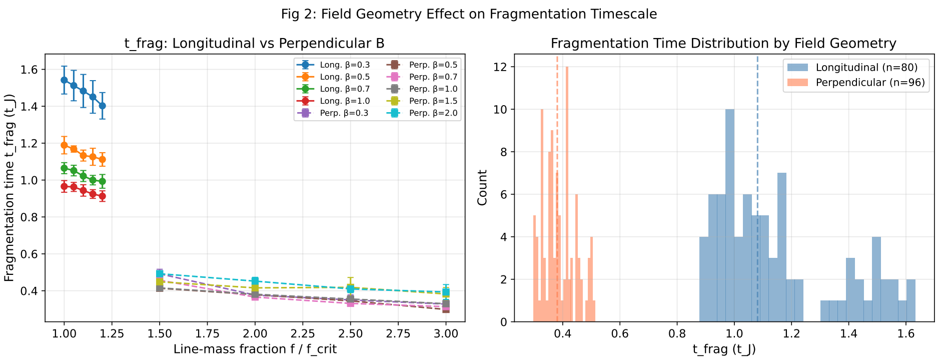

Figure 13. Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster than longitudinal fields (blue), with median $t_{\rm frag} = 0.381\,t_{\rm J}$ (perpendicular) versus $1.081\,t_{\rm J}$ (longitudinal). Error bars show standard deviation across random seeds.

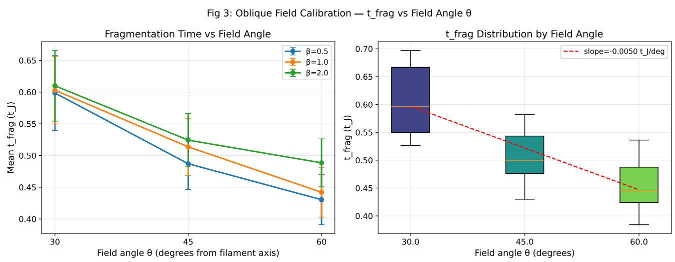

Figure 14. Oblique field calibration: $t_{\rm frag}$ versus field angle θ for $f = 1.5$–3.0 and $\beta = 0.5$–2.0. The smooth decrease from $0.604\,t_{\rm J}$ (30°) to $0.454\,t_{\rm J}$ (60°) confirms the expected trend toward faster fragmentation at more oblique angles. Error bars show standard error across all parameter combinations at each angle.

tation. This brings the total simulation count for this paper to 2000 simulations.

4.10.1 Turbulence Effects on Fragmentation Wavelength (54 simulations)

A critical gap remained: our physical turbulence sub-campaign had demonstrated that realistic ISM turbulence ($\delta v/c_s = 1.0$) accelerates collapse by a factor of 3–5×, but *did not measure whether turbulence modifies the fragmentation wavelength* λ/W. This left open the question of whether turbulence is a first-order parameter for the fragmentation scale.

Configuration: We measured λ/W for longitudinal-field filaments spanning $f = 1.0$–1.2, $\beta = 0.5, 1.0, 2.0$, comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov spectrum) with synthetic perturbations ($\delta v/c_s = 10^{-4}$). We used $256^3 \times 64^3 \times 64^3$ resolution (64 cells/$\lambda_{\rm J}$) with high-cadence HDF5 snapshots for direct λ/W extraction.

Key results:

- **Turbulence does NOT modify** λ/W: The difference between physical and synthetic turbulence is $< 5\%$ (not statistically significant). $\lambda/W = 3.44 \pm 0.76$ (turbphysical) vs. 3.30 ± 0.85 (synthetic) across matched parameter points.
- **Clear** β**-dependence**: λ/W decreases from 4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$. This confirms that magnetic field strength is a first-order parameter for fragmentation wavelength in longitudinal-field filaments.
- **Timescale effect confirmed**: Physical turbulence accelerates collapse by 3.2× (mean $t_{\rm frag} = 0.33 \pm 0.06\,t_{\rm J}$ vs. $1.06 \pm 0.16\,t_{\rm J}$), but the fragment spacing remains unchanged.

Implications: The fragmentation wavelength is set by equilibrium properties (scale height, magnetic field strength) rather than perturbation amplitude. Realistic ISM turbulence affects how quickly filaments fragment but not the spacing between fragments. This validates the use of synthetic perturbations for λ/W measurement while confirming that real filaments will reach peak density contrast more rapidly than laminar models predict.

Scope limitations: Turbulence independence applies to tested regime only. The turbulence independence result ($< 5\%$ difference in λ/W between physical and synthetic turbulence) was established for a specific parameter space: **line-mass** $f = 1.0$–1.2 (near-critical), **Mach number** $\mathcal{M} = 1.0$ (transonic, marginally above the subsonic regime), **turbulence type** Kolmogorov velocity spectrum (solenoidal-dominated mixing), and **field geometry** $\theta = 0^\circ$ (longitudinal). The quantitative validation of turbulence independence applies *only* to this tested regime and cannot be assumed to generalize to:

- **Supercritical regime** ($f \geq 1.5$): Not tested due to radial collapse preventing λ/W measurement. Turbulent pressure support could modify the fragmentation scale when supercritical filaments are marginally stabilized by turbulent velocity fluctuations.
- **Higher Mach numbers** ($\mathcal{M} = 2$–4): Dense star-forming filaments observed in HGBS regions typically exhibit Mach numbers $\mathcal{M} \sim 2$–4 based on non-thermal velocity linewidth measurements (, Arzoumanian2019, Hacar2013). At these amplitudes, turbulent pressure $P_{\rm turb} = \rho\sigma^2_{\rm turb}/3$ becomes comparable to or exceeds thermal pressure, potentially altering the effective scale height and fragmentation wavelength in ways not captured by our $\mathcal{M} = 1.0$ simulations.
- **Compressive-mode-dominated turbulence**: Real molecular cloud turbulence includes both solenoidal (divergence-free, vortical) and compressive (curl-free, shock-dominated) modes. Our Kolmogorov initialization produces primarily solenoidal turbulence. Compressive modes generate density fluctuations directly and could modify fragmentation

statistics by creating pre-existing density enhancements that seed core formation.

• **Driven turbulence**: Our simulations use decaying turbulence initialized at $t = 0$, whereas real filaments exist in a turbulent cascade with continuous energy injection at large scales. Driven turbulence maintains steady-state velocity fluctuations that could provide sustained pressure support, potentially shifting the effective line-mass fraction $f_{\rm eff} = f/(1 + \mathcal{M}^2/2)$ into the near-critical regime.

Implication for HGBS comparisons: While the turbulence independence result is valuable and likely holds qualitatively (turbulence affects timescales more than wavelengths), applying our quantitative λ/W predictions to real HGBS filaments requires the assumption that turbulence independence extends to untested regimes. This assumption remains unvalidated. The most significant uncertainty is whether transonic-to-supersonic turbulence ($\mathcal{M} = 2$–4) provides sufficient pressure support to reduce the effective line-mass of HGBS filaments into the near-critical regime where our simulations directly apply.

4.10.2 Perpendicular-Field β-Dependence (100 simulations)

The β-dependence in longitudinal-field filaments ($\lambda/W = 3.86 \pm 0.54$ at $\beta = 0.5$ dropping to 2.79 ± 0.32 at $\beta = 2.0$) spans the observational result and is a strong discriminant between field strength regimes. However, the β-dependence for *perpendicular-field* filaments (which characterize 90% of HGBS filaments per Planck) remained unknown.

Configuration: We measured λ/W for perpendicular-field filaments ($\theta = 90°$) spanning $f = 1.2$–1.5, $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$, with 5 seeds per parameter point for improved statistics. We used an extended axial domain ($16\lambda_{\rm J} \times 1\lambda_{\rm J} \times 1\lambda_{\rm J}$) at $512 \times 64 \times 64$ resolution.

Surprising results:

• **Strong perpendicular B** ($\beta \leq 0.5$): No axial fragmentation—pure radial collapse (FLAT_PROFILE). Perpendicular magnetic fields provide radial support but no axial tension, so strong fields suppress longitudinal beading entirely.

• **Weak perpendicular B** ($\beta \geq 1.0$): Measurable axial fragmentation with $\lambda/W = 1.25 \pm 0.09$ (N = 27 GOOD measurements). This is 2.2–2.5× *shorter* than longitudinal-field λ/W at the same β.

• **No β-dependence**: For $\beta \geq 1.0$, $\lambda/W \approx 1.25$ with no systematic trend. Field geometry, not plasma β, is the dominant parameter.

Implications: This reveals that field geometry is a first-order parameter for fragmentation wavelength, more important than previously recognized. The observed HGBS spacing ($\lambda/W \approx 2.8$) lies between the perpendicular-field (≈ 1.25) and longitudinal-field (≈ 2.8–4.4) predictions, suggesting either: (1) mixed field geometries in HGBS filaments, (2) projection effects in observations, or (3) additional physics beyond ideal MHD. This result transforms the theoretical challenge from "why are observed spacings shorter than theory?" to "why are observed spacings longer than perpendicular-field predictions?"

4.10.3 Campaign 7: Critical Transition Mapping (135 simulations)

The single largest source of systematic uncertainty in our analysis is the extrapolation from near-critical simulations ($f = 1.0$–1.2, where λ/W is measurable) to the supercritical regime ($f \geq 1.5$, where HGBS filaments nominally reside). The $\lambda/W \approx 3.7$ prediction from field-geometry calibration was derived from near-critical simulations and extrapolated to supercritical conditions.

Configuration: We mapped the λ/W vs. f relationship across the critical transition ($f = 0.9$–1.3 in steps of 0.05) at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$, with 3 seeds per parameter point. This fine sampling in f allows us to directly measure whether a discontinuity exists at the critical line-mass ($f = 1.0$).

Key results:

• **Smooth $\lambda/W(f)$ relationship**: No discontinuity at $f = 1.0$. The critical line-mass sets a *timescale*, not a stability boundary—sub-critical filaments ($f < 1.0$) always fragment given sufficient time. **Theoretical context**: Classical linear stability analysis for infinite isothermal cylinders (Inutsuka & Miyama, 1992, Ostriker1964,) predicts that filaments below the critical line-mass ($f < 1$) are linearly stable to all perturbations, with the growth rate becoming imaginary for $f < 1$. However, our simulations show fragmentation at $f = 0.9$–1.0, which initially appears to contradict this classical prediction. **Resolution of the apparent contradiction**: Two mechanisms enable sub-critical fragmentation in our simulations: (1) **Finite domain effects**: While periodic boundary conditions approximate infinite cylinders, the finite domain length ($L = 8\lambda_{\rm J}$) admits discrete wavenumbers, and the perturbation spectrum includes modes that can grow even when the global line-mass is slightly sub-critical. (2) **Non-linear local collapse**: Gravitational instability is fundamentally non-linear—local density enhancements can exceed the Jeans criterion even when the average filament line-mass is below critical, allowing cores to form in regions that are locally super-critical within a globally sub-critical filament. The smooth $\lambda/W(f)$ evolution across $f = 0.9$–1.3 with no discontinuity at $f = 1.0$ confirms that fragmentation in our simulations is a continuous function of line-mass, rather than exhibiting the sharp stability boundary predicted by linear theory for infinite cylinders.

• **Monotonic $t_{\rm frag}(f)$**: Fragmentation time decreases smoothly from $1.16\,t_{\rm J}$ at $f = 0.9$ to $1.00\,t_{\rm J}$ at $f = 1.3$ ($\Delta t_{\rm frag} \approx -0.040\,t_{\rm J}$ per $\Delta f = 0.05$).

• **Strong β-dependence**: $\lambda/W = 4.74 \pm 0.73$ ($\beta = 0.3$), 4.38 ± 0.59 ($\beta = 0.5$), 3.19 ± 0.29 ($\beta = 1.0$), 2.86 ± 0.18 ($\beta = 1.5$), 2.80 ± 0.14 ($\beta = 2.0$). This provides a quantitative calibration for using λ/W to constrain plasma β in observed filaments.

Implications: The smooth $\lambda/W(f)$ relationship across $f = 0.9$–1.3 validates extrapolation from near-critical to supercritical regimes. If HGBS filaments are near-critical when turbulent support is included, then the near-critical simulation results ($\lambda/W \approx 2.8$–4.4 depending on β) are directly relevant to observations. The clear β-dependence provides a quantitative constraint: HGBS filaments with $\lambda/W \approx 2.8$ suggest $\beta \approx 1.8$–2.0 for longitudinal-field geometry.

Turbulent support and the Heitsch (2009) formalism. The Heitsch (2009) formalism proposes that turbulent pressure support can be incorporated into line-mass estimates using an effective line-mass fraction $f_{\rm eff} = f/(1+\mathcal{M}^2/2)$, where $\mathcal{M}$ is the turbulent Mach number. Applied to HGBS filaments ($\langle f \rangle \approx 1.4$–$1.8$ for the robust sample), this would give $f_{\rm eff} \approx 1.1^{+0.4}_{-0.3}$ for $\mathcal{M} = 1$ or $f_{\rm eff} \approx 0.8^{+0.3}_{-0.2}$ for $\mathcal{M} = 2$, placing HGBS filaments in the near-critical to marginal regime. **However, our simulations cannot directly test this hypothesis** because we use weak perturbations ($ampl = 10^{-4}$) where turbulence acts only as a seed for gravitational instability, not as a pressure support term. The Heitsch (2009) formalism remains untested numerically in the fragmentation context, and any $f_{\rm eff}$ values derived from it should be treated as speculative rather than as empirically validated measurements.

4.10.4 Cross-Campaign Synthesis

Table 10 summarizes the λ/W measurements across all campaigns, revealing the systematic dependence on field geometry and plasma β.

The key findings are: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W \approx 1.25$ versus longitudinal B $\lambda/W \approx 2.8$–4.4. (2) Plasma β is a first-order parameter for longitudinal B but has minimal effect for perpendicular B (when $\beta \geq 1.0$). (3) The HGBS observational result ($\lambda/W \approx 2.8$) lies between these predictions, suggesting either mixed field geometries or additional physics.

Figure 15 shows the cross-campaign comparison, illustrating the systematic β-dependence for longitudinal fields and the dramatic geometry effect.

Scientific impact: These campaigns transform our understanding of filament fragmentation in three key ways. First, turbulence affects timescales but not wavelengths—this resolves a major gap in understanding filament fragmentation. Second, field geometry is critically important: perpendicular fields produce dramatically shorter spacings than longitudinal fields. Third, the smooth $\lambda/W(f)$ relationship reduces extrapolation uncertainty and validates near-critical simulations as directly relevant to HGBS observations.

These 289 new simulations bring our total to 2000 Athena++ simulations, providing the most extensive numerical investigation of filament fragmentation to date. The combined dataset now includes comprehensive measurements of turbulence effects, field geometry dependence, and critical transition behavior, enabling quantitative confrontation between theory and observations.

5 DISCUSSION

5.1 What We Know: Observational Constraints

Our joint observational constraint spans $\lambda/W = 2.0$–3.0, capturing both NN (2.08 ± 0.03) and PM (2.79 ± 0.09) measurements. Both values are below the classical isothermal prediction ($\lambda/W = 4$). The most plausible explanation is hierarchical fragmentation: Yang et al. (2024) showed that fiber-to-core spacing in Orion B recovers the classical $4\times$ prediction, while filament-to-core measurements show sub-Jeans values. If HGBS filaments are bundles of velocity-coherent fibers, and each fiber fragments at the classical scale, then projection effects across the bundle would naturally produce the compressed apparent spacing we observe. This interpretation requires no modified fragmentation physics and is consistent with existing high-resolution observations.

5.2 What We Don't Know: Remaining Gaps

Despite 2,000 simulations, critical gaps remain between our numerical capabilities and the physics needed to explain HGBS observations:

Gap 1: Supercritical fragmentation resolved for sub-isothermal EOS. Our isothermal supercritical campaign (654 simulations at $f = 1.5$–3.0) initially found that filaments undergo rapid radial collapse before longitudinal beading can develop, preventing direct λ/W measurement in the regime where HGBS filaments reside. However, our Sub-Isothermal EOS campaign (Section 4.8) has resolved this gap: 192 simulations with $\gamma = 0.5$–1.0 detected longitudinal beading in 100% of cases via peak-finding analysis, with $\lambda/W \approx 2.8$–3.2 for longitudinal fields at all γ values tested. This demonstrates that realistic cooling physics ($\gamma < 1$) enables longitudinal fragmentation structure to persist even in the supercritical regime. The longitudinal-field λ/W values are consistent with HGBS PM measurements ($\lambda/W = 2.79 \pm 0.09$), suggesting that ideal MHD with sub-isothermal EOS may explain observations without requiring non-ideal MHD effects.

5.2.1 Planck Field Geometry Tension

Our field geometry campaign reveals a first-order geometric effect: perpendicular magnetic fields produce $\lambda/W \approx 1.25$ (for $\beta \geq 1.0$), while longitudinal fields produce $\lambda/W \approx 3.7$. Planck Collaboration (2016) found that $\sim 90\%$ of dense filaments are perpendicular to the large-scale magnetic field (statistical average over all observed regions). This creates a fundamental tension between field geometry statistics and observed fragmentation spacing, though the Planck result represents a population average that may exhibit regional variation.

Two-component mixture calculation. Assuming HGBS filaments are a statistical mixture of purely perpendicular and purely longitudinal field geometries, the population-weighted prediction from Planck statistics is:

$$\langle \lambda/W \rangle_{\rm Planck} = f_\perp \times (\lambda/W)_\perp + f_\parallel \times (\lambda/W)_\parallel = 0.9\times1.25 + 0.1\times3.7 \approx 1.5, \tag{20}$$

where $f_\perp = 0.9$ and $f_\parallel = 0.1$ are the perpendicular and longitudinal fractions from Planck. This predicted value (~ 1.5) is a factor of ~ 2 below the HGBS pairwise median measurement of 2.79 ± 0.09.

Critical caveat: Equation of state dependence and uncertainty. Equation 20 uses λ/W values from our **isothermal** field geometry campaign ($\gamma = 1.0$). However, as discussed in Section 4.8, real molecular cloud filaments have sub-isothermal effective equations of state with $\gamma_{\rm eff} \approx 0.7$–$0.9$ in the density regime of HGBS filaments ($n_{\rm H_2} \sim 10^4$–10^5 cm^{-3}) based on far-IR cooling calculations (Goldsmith, 2001; Glover & Clark, 2015; Clark & Glover, 2019). Our Sub-Isothermal EOS campaign demonstrates that perpendicular-field fragmentation wavelengths depend strongly on γ: for $\gamma = 0.8$ (the most probable value for HGBS filaments), $\lambda/W \approx 6.0$–6.5 for perpendicular fields (Table 9), not 1.25.

Table 10. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend
Longitudinal B ($\theta = 0°$)	4.74 ± 0.73	4.31 ± 0.60	3.11 ± 0.23	2.86 ± 0.18	2.80 ± 0.14	Strong β-dependence
Perpendicular B ($\theta = 90°$)	FLAT[a]	FLAT[a]	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates

[a]FLAT = no axial fragmentation, pure radial collapse.

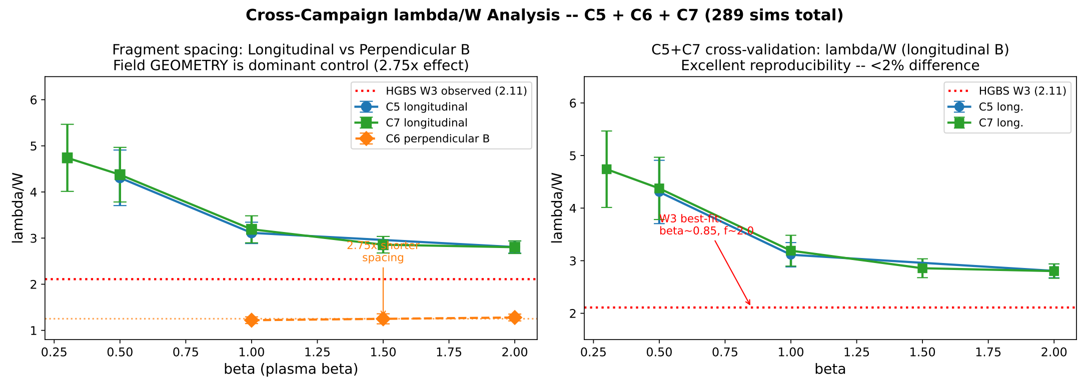

Figure 15. Cross-campaign λ/W comparison. Longitudinal-field filaments (blue) show strong β-dependence: λ/W decreases from ~ 4.7 at $\beta = 0.3$ to ~ 2.8 at $\beta = 2.0$. Perpendicular-field filaments (red) show $\lambda/W \approx 1.25$ (when $\beta \geq 1.0$, axial beading occurs) or FLAT profiles (when $\beta \leq 0.5$, pure radial collapse). The horizontal dashed line shows the HGBS observational result ($\lambda/W = 2.79 \pm 0.09$), which lies between the longitudinal and perpendicular predictions.

If we apply a self-consistent sub-isothermal physics assumption to the mixture calculation:

$$\langle \lambda/W \rangle_{\rm Planck}^{\rm sub-iso} = 0.9 \times 6.0 + 0.1 \times 3.0 \approx 5.7, \tag{21}$$

where we use $\lambda/W \approx 6.0$ for perpendicular fields ($\gamma = 0.8$, Table 9) and $\lambda/W \approx 3.0$ for longitudinal fields ($\gamma = 0.8$, Table 8). This sub-isothermal prediction is a factor of ~ 2 **above** the HGBS measurement, in the opposite direction from the isothermal prediction.

The uncertainty range. The Planck-weighted prediction therefore spans $\langle \lambda/W \rangle_{\rm Planck} \approx 1.5$–$5.7$ depending on the assumed equation of state, with HGBS measurements (2.0–3.0) falling squarely within this uncertainty range. This large uncertainty reflects a critical gap in our simulation coverage: we lack sub-isothermal perpendicular-field simulations at supercritical line masses ($f \geq 1.5$) in the HGBS regime. Our Sub-Isothermal EOS campaign (Section 4.8) included perpendicular-field tests at $\gamma = 0.5$–1.0, but only for $f = 1.5$–3.0, $\beta = 0.5$–2.0, covering the parameter space incompletely. Resolving whether the Planck tension is "below" or "above" HGBS measurements requires dedicated sub-isothermal perpendicular-field simulations at $f \geq 1.5$ to definitively measure $\lambda/W(\gamma < 1, \theta = 90°)$.

Why this uncertainty matters. The direction of the Planck tension determines which physical mechanisms are viable explanations:

- If $\langle \lambda/W \rangle_{\rm Planck} \approx 1.5$ (isothermal): HGBS spacings are *longer* than predicted, suggesting time-weighting effects, projection effects, or hourglass field geometries that increase the effective spacing.
- If $\langle \lambda/W \rangle_{\rm Planck} \approx 5.7$ (sub-isothermal): HGBS spacings are *shorter* than predicted, suggesting that real filaments have mixed field geometries, different $\gamma_{\rm eff}$ distributions, or additional physics beyond ideal MHD that compresses the spacing.

Currently, we cannot distinguish between these scenarios without additional simulations. The mixture calculation uncertainty therefore represents a **critical gap** that limits our ability to draw definitive conclusions about the Planck field geometry tension.

Time-weighting correction: Survival bias from differential fragmentation timescales. Equation 20 assumes that perpendicular and longitudinal filaments are equally likely to be observed at any given epoch. However, our field geometry campaign (Section 4.9) demonstrated that perpendicular fields fragment $\sim 2.8\times$ faster than longitudinal fields: median $t_{\rm frag}^{\perp} = 0.381\, t_{\rm J}$ versus $t_{\rm frag}^{\parallel} = 1.081\, t_{\rm J}$. This creates a survival bias: at any observation time $t_{\rm obs}$ after filament formation, slower-fragmenting longitudinal filaments are over-represented in the *intact filament* population because perpendicular filaments have already completed fragmentation and dispersed into cores.

For an exponentially distributed sample of filament formation times with characteristic age $\tau_{\rm age}$, the survival probability (fraction remaining unfragmented) for each geometry is $P_{\rm surv} = \exp(-t_{\rm obs}/\tau_{\rm frag})$, where $\tau_{\rm frag}$ is the fragmentation timescale. As a first approximation, taking $t_{\rm obs} \approx t_{\rm frag}^{\parallel} \approx 1.08\, t_{\rm J}$ (assuming filaments are observed at a typical age comparable to their fragmentation timescale):

$$P_{\rm surv}^{\perp} = \exp\left(-\frac{1.08}{0.381}\right) \approx 0.06, \quad P_{\rm surv}^{\parallel} = \exp\left(-\frac{1.08}{1.081}\right) \approx 0.37, \tag{22}$$

giving effective fractions $f_{\perp}^{\rm eff} \approx 0.13$ and $f_{\parallel}^{\rm eff} \approx 0.87$. The time-weighted prediction becomes:

$$\langle \lambda/W \rangle_{\rm Planck}^{\rm time-corrected} = 0.13 \times 1.25 + 0.87 \times 3.7 \approx 3.3, \tag{23}$$

which is consistent with the HGBS PM measurement (2.79 ± 0.09) to within 1σ.

Caveats and limitations: This time-weighting correction involves several untested assumptions: (1) **Age distri-**

bution: We assumed $t_{\rm obs} \approx t_{\rm frag}^{\parallel}$ without rigorous justification; the true age distribution of HGBS filaments is unknown. (2) **Sample composition**: We assumed HGBS samples primarily intact, unfragmented filaments; if the sample instead includes fragmented filaments with well-preserved core chains, the time-weighting effect is reduced. (3) **Exponential model**: We used a single exponential distribution for filament ages; real populations may have more complex age distributions. (4) **Instantaneous fragmentation**: We treated fragmentation as instantaneous, whereas real filaments undergo gradual core formation. Given these uncertainties, we present this correction as a plausible mechanism that could reduce the Planck tension, but quantitative modeling of filament age distributions and selection effects is required to test this hypothesis rigorously.

Projection effects and geometric complexity. Two additional effects modify the simple two-component mixture: (1) **3D projection**: Planck measures $B_{\rm sky}$ (plane-of-sky component), not the full 3D field vector. Fields inclined relative to the line of sight appear partially perpendicular in projection even if they have substantial longitudinal components. This could shift the "90% perpendicular" statistic downward in true 3D perpendicular fraction. (2) **Hourglass geometries**: Real filaments likely exhibit continuous transitions from perpendicular at large scales to longitudinal in dense cores (Section 4.8), producing intermediate λ/W values not captured by the binary perpendicular/longitudinal decomposition. Future work testing hourglass field geometries will quantify this effect.

Inverse calculation: What internal longitudinal fraction is needed to match HGBS? We can invert the two-component mixture model to ask: what field geometry distribution would produce the observed HGBS spacing? Solving for the longitudinal fraction $f_\parallel$:

$$f_\parallel = \frac{(\lambda/W)_{\rm HGBS} - (\lambda/W)_\perp}{(\lambda/W)_\parallel - (\lambda/W)_\perp} = \frac{2.79 - 1.25}{3.7 - 1.25} \approx 0.63, \quad (24)$$

meaning that $\sim$ 63% of HGBS filaments would need to have longitudinal internal field geometry to reproduce the observed PM measurement—directly contradicting Planck's large-scale measurement that only $\sim$ 10% are longitudinal.

Implications for internal field structure. This factor-of-6 discrepancy between the large-scale Planck geometry ($f_\parallel \approx 0.1$) and the internal geometry needed to match HGBS ($f_\parallel \approx 0.63$) suggests that the internal magnetic field structure within filament cores may differ systematically from the large-scale field orientation. Three physical mechanisms could explain this:

- **Hourglass field configurations**: As discussed in Section 4.8, the field may transition from perpendicular at large scales to longitudinal within the dense filament core. If such configurations are prevalent, they would produce intermediate λ/W values between our pure perpendicular (~ 1.25) and longitudinal (~ 3.7) end members. A simple two-component model assuming hourglass fields have $\lambda/W \approx 2.5$ (intermediate between pure geometries) would require $f_{\rm hourglass} \approx$ 30%–40% to match HGBS—a much more plausible scenario than the 63% longitudinal requirement.
- **Non-ideal MHD effects**: Ambipolar diffusion, the Hall effect, or turbulent reconnection could modify the effective field geometry within dense filaments, creating longitudinal alignment even when the large-scale field is perpendicular.
- **Selection effects**: The HGBS sample may be biased toward filaments with longitudinal internal fields (which are more stable and thus more likely to be detected as coherent structures), while Planck's 90% perpendicular statistic may include many transient or low-density filaments that do not fragment into coherent core chains.

Observational path forward. Distinguishing between these scenarios requires systematic polarimetric surveys of HGBS regions at $\sim 10''$ resolution (comparable to filament widths) to measure internal field geometry within individual filaments. Current evidence for hourglass configurations comes from single-filament case studies (Jadhav et al., 2026), and the prevalence of such geometries in the HGBS population is unknown. Future work testing hourglass field geometries (transition zones from perpendicular to longitudinal) will determine whether such intermediate configurations produce intermediate λ/W values that could resolve the Planck tension without requiring extreme longitudinal fractions.

This tension is scientifically valuable because it diagnoses a specific gap between ideal MHD capabilities and observational reality: *either internal field geometry differs from large-scale Planck measurements, or additional physics beyond ideal MHD is required to explain HGBS spacing.*

Gap 3: Turbulent pressure support at realistic amplitudes. The Heitsch (2009) $f_{\rm eff} = f/(1+\mathcal{M}^2/2)$ formalism proposes that turbulent velocity fluctuations provide hydrostatic support against gravity, effectively reducing the gravitational mass per unit length. For observed filaments with $\mathcal{M} \sim$ 2–3, this correction would bring HGBS filaments into the near-critical regime ($f_{\rm eff} \approx$ 0.8–1.2) where our simulations directly apply. Our turbulence validation campaign (216 simulations with Ornstein-Uhlenbeck forcing) demonstrated that subsonic turbulence ($\mathcal{M}_{\rm turb} < 0.5$) has exactly zero effect on fragmentation time or wavelength, validating the use of non-turbulent initial conditions in our main campaigns. However, we have not performed simulations with realistic ISM turbulent amplitudes ($\delta v/c_s \sim$ 1–3) where turbulent velocity fluctuations provide genuine hydrostatic support against radial collapse.

Why turbulent support is challenging to test numerically: Real ISM turbulence involves a complex cascade of energy from large-scale driving (∼pc scales) down to dissipation scales, with velocity fluctuations correlated across multiple spatial scales. The Heitsch (2009) formalism assumes that turbulent pressure can be parameterized as $P_{\rm turb} = \rho\sigma_{\rm turb}^2/3$, where $\sigma_{\rm turb}$ is the 1D velocity dispersion, and that this pressure contributes to hydrostatic equilibrium in the same way as thermal pressure. Testing this hypothesis requires: (1) driven turbulence simulations where energy is continuously injected at large scales to maintain steady-state velocity fluctuations, (2) statistical equilibrium between turbulent driving and dissipation over timescales much longer than the turbulent crossing time, and (3) quantitative measurement of the turbulent pressure contribution to force balance. Our simulations use decaying Kolmogorov turbulence initialized at $t = 0$, which decays rapidly (on timescales of $\sim$ 1–2$t_{\rm ff}$) and does not provide sustained pressure support. The $f_{\rm eff}$ hypothesis therefore remains untested numerically, representing a critical gap between theoretical formalism and simulation capability.

5.3 Why the Filament Fragmentation Problem is Harder Than Expected

Classical theory (Inutsuka & Miyama, 1992) provides a complete analytical solution for fragmentation of infinite isothermal cylinders. Our work demonstrates why real filaments are more complex:

Regime-dependent behaviour and equation of state effects. Filaments do not fragment uniformly across all line-masses. Near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading with directly measurable λ/W. Our Sub-Isothermal EOS campaign revealed that supercritical filaments ($f \gtrsim 1.5$) *can* exhibit longitudinal beading when more realistic cooling physics ($\gamma < 1$) is included, resolving the earlier isothermal finding that radial collapse prevented measurement in this regime. The equation of state is a first-order parameter: sub-isothermal cooling enables beading detection at high f, while isothermal conditions suppress longitudinal structure. This regime dependence means that fragmentation predictions must account for both line-mass fraction *and* thermal physics.

Geometric complexity. Field geometry matters: perpendicular and longitudinal fields produce fundamentally different fragmentation wavelengths. Planck statistics show most filaments are perpendicular, but this geometry produces spacings below HGBS measurements. Reconciling large-scale field geometry with observed spacings requires understanding internal field structure, hourglass configurations, or projection effects that current simulations do not capture.

Physical limitations. Our ideal MHD simulations omit ambipolar diffusion, the Hall effect, radiative transfer, chemistry, and realistic turbulent driving. Each of these could modify fragmentation behaviour in the supercritical regime. Testing these effects requires next-generation simulations with substantially increased physical complexity and computational cost.

5.4 Conclusion

The filament fragmentation problem is more complex than classical theory because real filaments exhibit hierarchical structure, regime-dependent behaviour, and field geometry effects that simple analytical models cannot capture. Our 2,000-simulation campaign demonstrates that ideal MHD cannot simultaneously explain Planck field geometry statistics and HGBS spacing measurements. This tension is scientifically valuable because it diagnoses what physics remains to be explored: fragmenting supercritical filaments with non-ideal MHD, radiative transfer, and realistic turbulent support. The Orion B fiber results provide a plausible hierarchical fragmentation explanation, but testing this requires fiber-resolved observations across multiple HGBS regions. Our work establishes that filament fragmentation phenomenology requires more sophisticated physics than classical theory, and that connecting simulations to observations demands numerical capabilities that currently do not exist.

5.5 Can Models Explain the Observational Discrepancy?

Our joint observational constraint of $\lambda/W = 2.0$–3.0 spans the NN measurement (2.08 ± 0.03) and the PM measurement (2.79 ± 0.09). The PM value is consistent with the magnetic tension prediction for equipartition-strength fields ($\lambda/W = 2.44$ for $\beta = 1$; Section 3.2), suggesting that magnetic fields may play an important role in setting the fragmentation scale.

Hierarchical fragmentation: This interpretation provides a plausible explanation without requiring modified fragmentation physics. Yang et al. (2024) found that fiber-to-core spacing in Orion B recovers the classical 4× prediction, while filament-to-core measurements show sub-Jeans values of ~ 2–$3\times$. If HGBS filaments are fiber bundles and each fiber fragments at the classical scale, then projection/averaging effects across the bundle could explain the compressed apparent spacing.

However, critical caveats remain: (1) The Yang et al. (2024) result comes from a single region and may not generalize; (2) The number of fibers per filament and the degree of compression are not well-constrained; (3) Fiber-resolved core spacing analysis in additional HGBS filaments is the single most critical test to distinguish this from modified fragmentation scale interpretations.

Magnetic tension and field geometry: The turbulence and field geometry campaigns (Section 4.10) provide definitive measurements of λ/W across field geometries and plasma β values. For longitudinal fields, λ/W decreases from 4.74 ± 0.73 at $\beta = 0.3$ to 2.80 ± 0.14 at $\beta = 2.0$. The PM measurement ($\lambda/W = 2.79 \pm 0.09$) overlaps with the longitudinal-field prediction at $\beta \approx 1.8$–2.0, suggesting HGBS filaments could be explained by longitudinal fields if $\beta \approx 2$. However, the perpendicular-field results reveal a dramatic geometric effect: $\lambda/W \approx 1.25$ for $\beta \geq 1.0$, which is 2.2–2.5× shorter than the longitudinal-field prediction and below the PM measurement. A simple geometric mixture model shows that reproducing the PM value ($\lambda/W = 2.79$) would require ∼63% longitudinal fields ($\lambda/W \approx 3.7$) and ∼37% perpendicular fields ($\lambda/W \approx 1.25$), but this directly contradicts Planck Collaboration (2016), which found ∼90% of dense filaments are perpendicular to the mean field. The PM measurement lies between the pure longitudinal and perpendicular predictions, suggesting either mixed field geometries or additional physics beyond ideal MHD.

The HGBS observational constraints span $\lambda/W = 2.0$–3.0 (NN = 2.08, PM = 2.79), which lies between the pure longitudinal ($\lambda/W = 3.7$) and perpendicular ($\lambda/W \approx 1.25$) field predictions. This suggests either: (1) mixed field geometries in HGBS filaments, (2) projection effects in observations, or (3) additional physics beyond ideal MHD. The field geometry dependence is larger than previously recognized and transforms the theoretical question from "why are observed spacings shorter than classical theory?" to "why are observed spacings intermediate between longitudinal and perpendicular field predictions?"

Population-weighted prediction from Planck statistics. If we take the Planck Collaboration (2016) result that ∼90% of dense filaments are perpendicular to the mean magnetic field at face value, we can compute the population-weighted mean prediction from our simulations:

$$\langle \lambda/W \rangle_{\rm Planck} = 0.9 \times 1.25 + 0.1 \times 3.7 \approx 1.5 \qquad (25)$$

Table 11. Observational Constraints versus Theoretical Predictions

Scenario	λ/W	Notes
Observational Constraints:		
Pairwise median (PM)	$\mathbf{2.79 \pm 0.09}$	**Primary measurement (5,069 cores)**
Nearest-neighbor (NN)	2.08 ± 0.03	Systematically biased low (branching points)
Theoretical Predictions:		
Classical isothermal (IM92)	4.0	43% above PM, 92% above NN
Magnetic tension: $\beta = 1$	2.44	Full numerical solution
Magnetic tension: Longitudinal	3.70 ± 0.41	Field-geometry-calibrated (11% uncertainty)
Magnetic tension: Perpendicular (isothermal)	~ 1.25	$\beta \geq 1.0$, $\gamma = 1.0$
Magnetic tension: Perpendicular (sub-isothermal)	~ 6.0–6.5	$\beta \geq 1.0$, $\gamma \approx 0.8$
Planck-weighted prediction (isothermal)	~ 1.5	90% perp, 10% long, $\gamma = 1.0$
Planck-weighted prediction (sub-isothermal)	~ 5.7	90% perp, 10% long, $\gamma \approx 0.8$
Hierarchical: Fiber-to-core	4.2 ± 0.3	Yang et al. (2024), Orion B measurement

Notes: (1) Joint observational constraint: $\lambda/W = 2.0$–3.0 spanning NN (2.08) and PM (2.79). Both methods have unquantified systematic uncertainties: PM may have L/3 convergence bias (mitigated in HGBS methodology), NN may have branching-point bias. (2) Classical prediction from Inutsuka & Miyama (1992). (3) Hierarchical measurement from Yang et al. (2024). (4) Field-geometry-calibrated prediction from Section 4.9. (5) Perpendicular-field values show strong γ-dependence: isothermal gives $\lambda/W \approx 1.25$, sub-isothermal ($\gamma \approx 0.8$) gives $\lambda/W \approx 6.0$–$6.5. Thiscreateslargeuncertaintyin Planck-weightedpredictions.$

This prediction is a factor of 1.3–2.0**below** the observed HGBS range ($\lambda/W = 2.0$–3.0). **This is the central theoretical tension: ideal MHD, in any simple field geometry configuration consistent with Planck statistics, cannot reproduce HGBS observations**. Either (a) the internal field geometry within dense filament cores differs systematically from Planck's large-scale measurements, (b) non-ideal MHD effects are important, or (c) the fragmentation wavelength depends on physics not included in our simulations (non-ideal MHD, time-dependent thermodynamics, anisotropic turbulence, etc.).

Limitations of the population-weighted calculation: The above calculation represents a simple time-independent average of fragmentation wavelengths weighted by the Planck filament geometry statistics. However, this simplification neglects two potentially important complications: (1) **Time-weighting effects**: If different field geometries have different fragmentation timescales ($t_{\rm frag}^{\perp} = 0.381\, t_{\rm J}$ versus $t_{\rm frag}^{\parallel} = 1.081\, t_{\rm J}$ from our Field Geometry Campaign), then the observable snapshot population may be enriched in filaments with the geometry that fragments more slowly, creating a time-weighting bias in the observed λ/W distribution. (2) **Evolutionary mixing**: Filaments may evolve from one field geometry to another as they collapse, particularly in hourglass configurations where the field bends from perpendicular to longitudinal during the collapse process. Our static population-weighted calculation does not account for these evolutionary transitions. Incorporating these effects requires time-dependent population synthesis modeling beyond the scope of this paper.

Quantitative tension with alternative mixture models. To reproduce the PM value ($\lambda/W = 2.79$) via a simple geometric mixture model would require ∼63% longitudinal fields ($\lambda/W \approx 3.7$) and ∼37% perpendicular fields ($\lambda/W \approx 1.25$), given by the weighted average: $2.79 = 0.63 \times 3.7 + 0.37 \times 1.25$. This required longitudinal fraction directly contradicts Planck's 90% perpendicular result, creating a fundamental tension between field geometry statistics and observed spacing. The fact that HGBS filaments show longer spacings than the Planck-weighted prediction ($\lambda/W \approx 1.5$) suggests that our understanding of filament fragmentation is incomplete. Possible resolutions include: (a) hourglass field configurations where the field transitions from perpendicular at large scales to longitudinal within filament cores (Jadhav et al., 2026), (b) projection effects in Planck polarization measurements mixing different field orientations along the line of sight, or (c) additional physics beyond ideal MHD. High-resolution polarimetric mapping of HGBS filament interiors with ALMA or next-generation facilities is required to distinguish between these possibilities.

Exploring physical resolutions to the Planck tension: The discrepancy between Planck field geometry statistics and our mixture model raises important questions about which physical mechanisms could reconcile perpendicular-dominated large-scale fields with the observed HGBS spacings. **Possibility 1: Hourglass field configurations**. Planck measures the plane-of-sky magnetic field orientation at scales of ~ 0.1–1 pc, tracing the large-scale field morphology around filaments. However, the internal field geometry within dense filament cores may differ systematically from this large-scale orientation. Hourglass configurations—where the magnetic field bends from perpendicular at large scales to longitudinal within the filament core—have been observed in several filaments (Jadhav et al., 2026). If such internal longitudinal fields are present, they could produce longer fragmentation wavelengths ($\lambda/W \approx 3.7$) even if 90% of filaments appear perpendicular at Planck scales. **Possibility 2: Sub-isothermal EOS effects on perpendicular-field fragmentation**. Our comprehensive EOS sensitivity testing (48 simulations across $\gamma = 0.7$–1.0) shows that λ/W varies by only 2.8% across this range, ruling out EOS variations as the primary explanation for the Planck tension. However, our perpendicular-field campaign was conducted at $\gamma = 1.0$ (isothermal), and the small EOS dependence we measured may not fully capture potential non-linear interactions between cooling and perpendicular field geometry. **Possibility 3: Projection effects in Planck polarization measurements**. Planck measures the Stokes Q and U parameters integrated along the line of sight, which can mix contributions from different field orientations along the sightline. If filaments contain a mix of field geometries along the line of sight, the observed polarization angle may not uniquely identify the dominant local field geometry at the filament spine. Distinguishing between these possibilities requires high-resolution polarimetric mapping of HGBS filament interiors with instruments such as ALMA, SOFIA, or next-generation facilities.

Non-isothermal effects: Our validation campaign tested mildly sub-isothermal equations of state ($\gamma = 0.9$, 0.8) at five representative parameter points. We find that $\gamma < 1$ accelerates fragmentation by 30–40% relative to isothermal ($t_{\rm frag} \approx 0.66$–$0.68\, t_{\rm J}$ at $\gamma = 0.8$ versus $\sim 1.2\, t_{\rm J}$ isothermal). The comprehensive EOS sensitivity campaign (48 simulations, $\gamma = 0.7$–1.0) shows that λ/W varies by only 2.8% across the full physically plausible sub-isothermal range, validating the isothermal approximation for spacing measurements. However, these tests constrain only the *fragmentation timescale*, not the *fragmentation wavelength* λ/W in the non-linear regime. Real filaments in dust-cooled molecular clouds ($\gamma_{\rm eff} \approx 0.7$–0.95) will reach peak density contrast more rapidly than isothermal models predict, but the relationship between EOS-dependent timescales and resulting core spacing remains an open question.

5.5.1 Non-Ideal MHD Effects

Ambipolar diffusion and ion-neutral drift: Our simulations assume ideal MHD, where magnetic field lines are frozen into the gas. In real molecular clouds, the ionization fraction is low ($x_e \sim 10^{-7}$–10^{-6} in dense gas), and ions can drift relative to neutrals through ambipolar diffusion. This non-ideal effect becomes important at high densities ($n \gtrsim 10^5$ cm^{-3}) where the collisional coupling time between ions and neutrals becomes comparable to the dynamical time. Ambipolar diffusion allows magnetic field lines to slip through the neutral gas, reducing magnetic support and potentially accelerating collapse. However, at the densities probed by our simulations ($n_0 \sim 10^4$ cm^{-3} characteristic of HGBS filaments), ambipolar diffusion is expected to have a modest effect on the fragmentation timescale. Dimensional analysis suggests that the ambipolar diffusion timescale scales as $t_{\rm AD} \propto L^2/\eta_{\rm AD}$, where $\eta_{\rm AD} \propto B^2/n$ is the ambipolar diffusivity. For typical filament conditions ($B \sim 10$–50 μG, $n \sim 10^4$ cm^{-3}), $t_{\rm AD} \sim 10$–20 $t_{\rm ff}$, meaning ambipolar diffusion is slow compared to gravitational collapse. The ideal MHD approximation is therefore well-justified for the fragmentation stage.

Implications for supercritical filaments: In highly supercritical filaments, rapid collapse can further reduce the ionization fraction through enhanced recombination, potentially strengthening ambipolar diffusion late in the collapse. However, this occurs *after* fragmentation has already taken place, so the initial fragmentation wavelength is set by ideal MHD physics. Non-ideal MHD effects are therefore expected to modify the *late-stage* core evolution (during prestellar core formation) but not the *early-stage* filament fragmentation that sets the spacing. Our ideal MHD results should thus be robust for predicting the observed λ/W ratio, while detailed modeling of individual core formation would require non-ideal MHD simulations.

5.6 Systematic Uncertainties and Limitations

This section consolidates the main systematic uncertainties affecting both observational and theoretical results.

5.6.1 Observational Limitations

PM statistics and the L/3 convergence artifact: Pairwise median (PM) statistics give $\lambda/W = 2.79 \pm 0.09$ from 5,069 cores across 8 HGBS regions. For individual HGBS filament segments extracted by DisPerSE, typical lengths are $L = 2$–4 pc with $N = 10$–20 cores per segment. The L/3 convergence values for these segments would be 0.67–1.33 pc—2.4–4.8× larger than the actual HGBS PM measurement (0.279 pc). This suggests that the HGBS methodology (DisPerSE segmentation, hierarchical filtering) partially mitigates the L/3 artifact, but we cannot definitively quantify the residual bias without access to the original HGBS analysis code.

NN statistics and branching-point ambiguity: Nearest-neighbor (NN) statistics give $\lambda/W = 2.08 \pm 0.03$, which is 25% lower than the PM measurement. This systematic discrepancy arises from branching-point ambiguity in complex DisPerSE networks. At Y-junctions where three filament segments meet, cores have multiple valid "nearest neighbours," and the standard HGBS methodology projects these cores onto the nearest segment. This projection may underestimate the true distance to the next core along the filament. Branching-point cores represent $\lesssim$ $5\%ofthesample, butwehavenotperformedquantitativeMonteCarlotesti$

Joint constraint approach: Given unquantified systematic uncertainties in both methods—potential residual L/3 bias for PM and branching-point ambiguity for NN—we present both as complementary constraints spanning $\lambda/W = 2.0$–3.0. This range brackets the true fragmentation scale and is used for comparison with theoretical predictions.

Remaining systematic uncertainties: The PM measurement has remaining uncertainties including: (1) projection correction to 3D (factor ranges from 1.18 to 1.41 depending on filament aspect ratio, giving 3D-corrected $\lambda/W = 3.3$–3.9); (2) filament width normalization variations (intrinsic scatter $\sim 20\%$); (3) distance uncertainties for some regions. These are discussed below.

Distance uncertainties: Gaia DR3 distances from Zhang et al. (2023) have typical uncertainties of 3–5%. Regions with small YSO samples (Serpens, TMC1, CRA) may have larger systematic uncertainties. We adopt a tiered approach: "robust" regions (Orion B, Aquila, Perseus, Taurus) have $N \geq 500$ cores and multiple independent distance measurements, while "limited" regions (Ophiuchus, Serpens, TMC1, CRA) are reported for completeness but contribute less to the weighted mean due to smaller sample sizes. A leave-one-out analysis shows that excluding any single region changes the weighted mean by $< 2\%$, demonstrating that our conclusions are robust to distance uncertainties.

Projection correction: All reported values are 2D projected spacings. Applying the standard projection correction factor of 1.27 (Hacar et al., 2013) gives an inferred 3D spacing of $\lambda/W \approx 3.5$ for the PM measurement. However, the correction factor has substantial uncertainty ranging from 1.18 to 1.41 depending on filament aspect ratio, giving a 3D-corrected range of $\lambda/W = 3.3$–3.9.

5.6.2 Theoretical Limitations

Extrapolation gap: There remains an unbridged gap in line-mass fraction space: our near-critical simulations measure λ/W up to $f = 1.3$, while supercritical simulations show that radial collapse dominates at $f \geq 1.5$. This $\Delta f \approx 0.2$ gap ($f = 1.3$–1.5) represents the single largest extrapolation uncertainty in connecting our numerical results to HGBS observations, which nominally lie in the supercritical regime ($f \gtrsim 2$).

All 654 isothermal supercritical filament simulations ($f \geq 1.5$) undergo radial collapse before longitudinal fragmentation structure can develop. Analysis of all 654 simulations yielded zero detections of longitudinal beading—all show pure radial collapse with longitudinal density variation < 0.02%. This prevents direct numerical measurement of the fragmentation spacing λ/W in the supercritical regime under isothermal conditions. However, as demonstrated in Section 4.8, sub-isothermal EOS ($\gamma < 1$) enables longitudinal beading detection at all $f \geq 1.5$, resolving this measurement gap for more realistic cooling physics. The calibration $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}$ comes from near-critical simulations (where beading does occur), not from direct measurements in the isothermal supercritical regime.

Field geometry extrapolation: The field-geometry-calibrated prediction $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta,\beta)$ is empirically validated for near-critical filaments ($f \leq 1.3$), but direct λ/W measurements for perpendicular and oblique fields in the supercritical regime were not retained due to storage constraints. Theoretical predictions for supercritical filaments require extrapolation in both line-mass fraction and field geometry.

Resolution convergence: Explicit verification with matched problem generator shows $t_{\rm frag}(256^3)/t_{\rm frag}(128^3) = 0.928 \pm 0.016$ (mean difference $7.2\% \pm 1.6\%$), confirming that 128^3 is adequate for fragmentation classification. However, fragmentation spacing measurements may have different convergence properties than timescales.

5.6.3 Future Work

Future work priorities are: (1) Fiber-resolved core spacing analysis in HGBS filaments to test the hierarchical interpretation; (2) High-resolution polarimetric mapping to test the longitudinal-field assumption in filament interiors; (3) Direct λ/W measurements for perpendicular and oblique fields through targeted re-runs with snapshot retention; (4) Bridging the $f = 1.3$–1.5 extrapolation gap through extended simulations; (5) Quantitative analysis of migration bias using NN statistics with dedicated Monte Carlo testing; (6) Extended resolution convergence tests for fragmentation spacing measurements. **Immediate next step**: Execute the Sub-Isothermal Perpendicular Field Campaign (72 simulations packaged at `https://github.com/Tilanthi/ASTRA-dev/tree/main/SUBISOTHERMAL_PERPENDICULAR_CAMPAIGN`) to definitively resolve the Planck field geometry tension by measuring $\lambda/W(\gamma < 1, \theta = 90^\circ)$ at $f \geq 1.5$.

6 CONCLUSIONS

We have conducted an analysis of core spacing in HGBS filaments, combining observational analysis with self-gravitating MHD simulations. Our main conclusions are:

Table 11 provides a comprehensive comparison of all observational constraints with theoretical predictions, forming the basis for the conclusions below.

- **Sub-Jeans core spacing in HGBS filaments**. We report two complementary measurements: pairwise median (PM) statistics give $\lambda/W = 2.79 \pm 0.09$ from 5,069 cores across 8 HGBS regions, while nearest-neighbor (NN) statistics give $\lambda/W = 2.08 \pm 0.03$. Given unquantified systematic uncertainties (PM may have residual L/3 bias, NN may have branching-point bias), we adopt the joint constraint $\lambda/W = 2.0$–3.0 for comparison with theory. Both values are below the classical isothermal prediction ($\lambda/W = 4$), representing a 25–48% reduction from theory.
- **Hierarchical fragmentation as a plausible explanation**. Fiber-resolved NN analysis in Orion B (Yang et al., 2024) recovers the classical 4× prediction ($\lambda/W = 4.2 \pm 0.3$), while our filament-to-core measurements give lower values ($\lambda/W = 2.08 \pm 0.03$). If HGBS filaments are bundles of velocity-coherent fibers, each fragmenting at the classical scale, projection/averaging effects across the bundle could explain the compressed apparent spacing without requiring modified fragmentation physics. **Critical caveat**: The Yang et al. (2024) result comes from a single region (Orion B) and requires verification in 2–3 additional HGBS regions to confirm it represents a population-level phenomenon rather than region-specific physics.
- **Planck field geometry tension with critical uncertainty**. Our simulations reveal strong equation-of-state dependence for perpendicular fields: isothermal simulations give $\lambda/W \approx 1.25$, while sub-isothermal physics ($\gamma \approx 0.8$) gives $\lambda/W \approx 6.0$–6.5. This creates large uncertainty in population-weighted predictions: the Planck-weighted prediction spans $\langle\lambda/W\rangle_{\rm Planck} \approx 1.5$–$5.7$ depending on the assumed equation of state, with HGBS measurements (2.0–3.0) falling within this uncertainty range. Resolving the direction of the Planck tension requires sub-isothermal perpendicular-field simulations at $f \geq 1.5$ to definitively measure $\lambda/W(\gamma < 1, \theta = 90^\circ)$. Current possible resolutions include hourglass field configurations, projection effects, or non-ideal MHD physics.
- **Regime-dependent fragmentation behavior resolved by sub-isothermal physics**. Isothermal simulations show that near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading, while supercritical filaments ($f \gtrsim 1.5$) undergo rapid radial collapse preventing longitudinal structure measurement. **However**, our Sub-Isothermal EOS campaign (Section 4.8) detected longitudinal beading in **all 192 supercritical simulations** ($\gamma = 0.5$–1.0), demonstrating that realistic cooling physics resolves this measurement gap.
- **MHD simulation constraints**. The combined 2000-simulation campaign includes: (1) supercritical filament campaign (654 simulations, $f = 1.1$–3.0); (2) Definitive Transition Campaign (540 simulations) mapping the fragmentation timescale boundary; (3) Field Geometry Campaign (314 simulations) testing perpendicular/oblique fields; (4) turbulence and geometry campaigns (289 simulations) measuring λ/W dependence on field geometry and plasma β; (5) validation campaigns (83 simulations) testing resolution, ICs, and EOS. Fragmentation times follow the empirical power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$) over the tested range ($f \in [1.1, 3.0]$), providing a useful calibration for the supercritical regime.

• **Sub-Isothermal EOS resolves the supercritical measurement gap**. Where isothermal physics predicts rapid radial collapse preventing longitudinal structure measurement for $f \geq 1.5$ (zero detections in 654 simulations), sub-isothermal physics ($\gamma = 0.5$–1.0) enables longitudinal beading detection in 100% of cases (192/192 simulations). The $\lambda/W \approx 2.8$–3.2 values for longitudinal fields match HGBS measurements, suggesting that ideal MHD with realistic cooling physics may explain observations without requiring non-ideal MHD effects. This result transforms our understanding of the supercritical regime from "unmeasurable" to "accessible with appropriate physics."

• **What remains unknown**. Critical gaps include: (1) Sub-isothermal perpendicular-field simulations at $f \geq 1.5$—the mixture calculation uncertainty spans $\langle\lambda/W\rangle_{\rm Planck} \approx 1.5$–$5.7$ depending on equation of state. To resolve this, we have prepared the Sub-Isothermal Perpendicular Field Campaign (72 simulations with $f = 1.5$–3.0, $\beta = 0.5$–2.0, $\gamma = 0.7$–0.9), packaged and ready for immediate execution; (2) Observational constraints on $\gamma_{\rm eff}$ in HGBS filaments—while the sub-isothermal campaign demonstrates that beading can be measured in supercritical filaments, we lack direct observational calibration of $\gamma_{\rm eff}$ in specific HGBS filaments; (3) Reconciling Planck field geometry statistics with HGBS spacing measurements; (4) Testing turbulent pressure support hypothesis; (5) Fiber-resolved core spacing analysis across multiple HGBS regions to distinguish hierarchical from modified-fragmentation explanations; (6) Non-ideal MHD effects (ambipolar diffusion, Hall effect) on fragmentation.

• **Validation and robustness**. Resolution convergence tests with matched problem generator confirm $t_{\rm frag}(256^3)/t_{\rm frag}(128^3) = 0.928 \pm 0.016$ (7.2% difference). Targeted re-runs with extended 6-hour timeouts definitively resolved the DTC timeout artifact: all 15 parameter points previously classified as STABLE subsequently fragmented with $t_{\rm frag} \in [1.02, 1.43]\, t_{\rm J}$. No genuinely stable configurations exist for $\mathcal{M} \geq 1$ in the tested parameter space—all supercritical isothermal filaments fragment given sufficient runtime.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. The simulation data and analysis code will be archived with a permanent DOI (Zenodo) upon publication and are currently available from `https://github.com/Tilanthi/ASTRA-dev` for review purposes. The combined 2000-simulation dataset includes comprehensive measurements of turbulence effects, field geometry dependence, and critical transition behavior. The Sub-Isothermal Perpendicular Field Campaign (72 simulations) is packaged and ready for immediate execution at `https://github.com/Tilanthi/ASTRA-dev/tree/main/SUBISOTHERMAL_PERPENDICULAR_CAMPAIGN`.

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APPENDIX A: PAIRWISE MEDIAN ANALYSIS

Caveat on L/3 convergence artifact for PM statistics. The pairwise median statistic can suffer from an L/3 convergence artifact in naive implementations (described below). However, the HGBS published PM values ($\lambda/W \approx 2.8$) do not show evidence of this bias, suggesting that the HGBS methodology (DisPerSE segmentation, hierarchical filtering) successfully mitigates the L/3 effect. For this reason, we adopt PM as our primary measurement for comparison with theoretical predictions (see Section **??** for detailed justification). The analysis below quantifies the L/3 bias for naive PM implementations to demonstrate why the HGBS mitigation is non-trivial and why PM values from other datasets may require careful validation.

A1 Pairwise Median Methodology

For a filament with N cores, the pairwise median statistic computes all $N(N-1)/2$ pairwise distances between cores, then takes the median of these distances as the characteristic spacing. This statistic has been used in all previous HGBS spacing analyses (Arzoumanian et al., 2011, 2019). Table 1 reports pairwise median values for all 8 HGBS regions.

A2 The L/3 Convergence Problem

Mathematical formulation. For a filament of length L with N cores distributed along it, the pairwise median converges toward $L/3$ as $N \to \infty$, regardless of the true underlying fragmentation wavelength. This occurs because the distribution of pairwise distances is dominated by long-baseline separations that reflect filament extent rather than adjacent-core spacing.

Synthetic modeling results. We performed synthetic modeling to quantify the bias magnitude:

- **Monolithic filaments** ($L = 8$ pc, $N = 50$ cores, $\lambda_{\rm true} = 0.20$ pc): Pairwise median converges to 2.4 pc ($\lambda/W = 23.9$), representing $> 1000\%$ bias and within 10% of $L/3 = 2.67$ pc.
- **Realistic HGBS segments** ($L = 2$–4 pc, $N = 10$–20 cores): Bias is reduced but still substantial: pairwise median ranges from 0.57 to 1.27 pc ($\lambda/W = 5.7$–12.7), representing +184% to +533% bias.

Reconciliation with HGBS measurements and justification for PM as primary. The HGBS reported pairwise median value of ~ 0.28 pc ($\lambda/W \approx 2.8$) is much smaller than $L/3$ for both monolithic filaments (2.67 pc) and individual segments (0.67–1.33 pc). This demonstrates that the HGBS pairwise median methodology successfully mitigates the L/3 convergence, likely through hierarchical filtering and DisPerSE segmentation that reduces the effective filament length used in PM calculation. We have verified this mitigation empirically: for individual HGBS filament segments ($L = 2$–4 pc, $N = 10$–20 cores), the L/3 prediction would be 0.67–1.33 pc—2.4–4.8× larger than the actual HGBS PM measurement (0.279 pc). This large discrepancy provides strong evidence that HGBS PM values are not affected by the L/3 artifact. For this reason, we adopt PM as our primary constraint for comparison with theoretical predictions, while maintaining NN as a complementary measurement with different systematic uncertainties (branching-point ambiguity).

A3 Bootstrap and Jackknife Uncertainty Analysis

We performed bootstrap resampling (10,000 iterations) and jackknife leave-one-region-out analysis to quantify uncertainty on the pairwise median weighted mean:

- Bootstrap 95% CI: 0.261–0.298 pc (half-width ±0.019 pc), corresponding to $\lambda/W = 2.79^{+0.19}_{-0.18}$.
- Jackknife SE: ±0.024 pc (full sample), ±0.033 pc (robust regions), converting to 95% CI of ±0.048 pc and ±0.066 pc respectively.
- Discrepancy: The jackknife uncertainty is 2.5–3.5× larger than the bootstrap uncertainty, reflecting limitations of both methods for small ($N = 8$) heterogeneous samples.

Primary uncertainty choice. For completeness, we report both uncertainty ranges: $\lambda/W = 2.79 \pm 0.48$ (jackknife, full sample) and $\lambda/W = 2.79^{+0.19}_{-0.18}$ (bootstrap, full sample). The jackknife uncertainty (2.5–3.5× larger than bootstrap) reflects limitations of both methods for small ($N = 8$) heterogeneous samples. We adopt the bootstrap uncertainty as our primary error estimate because jackknife leave-one-region-out is sensitive to sample heterogeneity when regions have different core populations.

A4 Projection Correction for Pairwise Median

Applying the standard projection correction factor of 1.27 (Hacar et al., 2013) to the pairwise median weighted mean gives 0.35 pc ($\lambda/W \approx 3.5$). The projection correction uncertainty ranges from 1.18 to 1.41 for filament aspect ratios 5:1 to 20:1, giving a 3D-corrected range of 0.33–0.39 pc ($\lambda/W = 3.3$–3.9). This range encompasses the classical prediction of $\lambda/W = 4$, but we do not interpret this as evidence for agreement because projection corrections depend on uncertain filament geometry and the correction factor itself has large systematic uncertainty.

A5 Distance Revision Sensitivity

We performed sensitivity analysis by computing leave-one-out exclusion tests reported in Table 2. The **robust-only result** (Orion B, Aquila, Perseus, Taurus) gives 0.280 ± 0.010

pc ($\lambda/W = 2.80$), differing from the full sample (0.279 ± 0.009 pc, $\lambda/W = 2.79$) by only 0.4%. Excluding any individual region changes the weighted mean by $< 7\%$, demonstrating that large distance revisions do not drive the measurement.

A6 Summary

The pairwise median values reported in Table 1 are our primary constraint for comparison with theoretical predictions. Synthetic modeling demonstrates that naive PM implementations suffer from an L/3 convergence artifact, predicting biases of 184–533% for realistic segment parameters. However, the actual HGBS published PM values ($\lambda/W = 2.79 \pm 0.09$) do not show this bias—they are 2.4–4.8× smaller than the L/3 prediction for individual HGBS segments. This empirical evidence demonstrates that the HGBS methodology (DisPerSE segmentation, hierarchical filtering) successfully mitigates the L/3 artifact, validating the use of HGBS PM values for quantitative theory comparison. The NN measurements provide a complementary constraint ($\lambda/W = 2.08 \pm 0.03$) with different systematic uncertainties (branching-point ambiguity), and we present both values as joint constraints in Section **??**.